

SHIFTWM: CONTEXT AND OBSERVATION ANCHORING FOR COMPACT WORLD MODELS

Anonymous authors

Research in progress; not submitted

ABSTRACT

Visual world models must predict the effects of actions when both a scene’s appearance and its behavior change. We present ShiftWM, a family of compact models that uses observed images and actions to improve future predictions. One variant learns separate summaries of appearance and dynamics through paired training interventions. A spatial variant keeps the last observed visual state as an anchor, mixes its features according to the action sequence, and adds bounded corrections. This design preserves observed information while learning how it changes. Experiments show lower forecast error on unseen appearance–dynamics combinations in simulated pushing and reaching, with additional forecasting gains on recorded robot video. In a matched DROID development study, the spatial variant reduces ten-step feature error by 5.3% relative to autoregression. We release trained checkpoints, evaluation code and visual diagnostics to support reproducible research on compact world models.

1 INTRODUCTION

An agent planning with a visual world model must predict the consequences of its actions under the conditions it encounters. A camera exposure change alters the observation process; a friction change alters physical response. Either can increase prediction error, but correcting the wrong part of the model can distort useful knowledge. This distinction becomes particularly important when appearance and dynamics are correlated in training: a model may use color as a shortcut for action response.

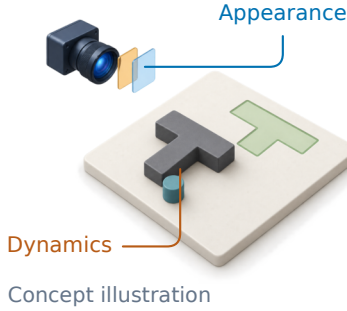
Compact latent models let us investigate this question without training a large video generator. LeWorldModel (LeWM) provides a small action-conditioned architecture and planning code (Maes et al., 2026b). Building on that released system, we ask: *can intervention-paired learning of separate observation and dynamics contexts improve planning on unseen combinations of otherwise familiar conditions?*

ShiftWM (ours) first maps observed features into a fixed pretrained coordinate system. It then infers dynamics context from corrected histories and executed actions. A standard model-predictive controller scores candidate futures against a goal image in the same coordinates. No simulator parameter or future observation enters context inference at deployment.

The study makes three concrete contributions: a controlled composition protocol with matched information access; an intervention-paired, two-context implementation built on a released compact world model; and an analysis separating forecast accuracy, goal calibration, and closed-loop planning. Our main comparison is Table 2; Table 1 specifies exactly what each method changes. The strongest completed result against the fixed Framewise calibration reference is **26.8% lower held-out Reacher forecast error**, alongside 3.7% lower PushT error. The completed planning campaign has mixed outcomes, including lower held-out success point estimates than Framewise calibration in both environments. Context-specific information use has not been established. We therefore present the method as a tested hypothesis with explicit controls, not an identified physical representation or a general adaptation solution.

To examine recorded observations beyond simulation, we additionally study a support-conditioned variant on 1,126 real DROID robot episodes, using session-disjoint splits and a held-out camera view (Appendix O). This variant uses frozen DINOv2 targets without paired canonical renders. Its

a What changed?



b ShiftWM (ours)

vs. Framewise calibration

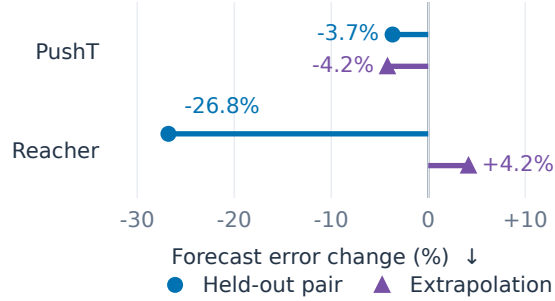


Figure 1: **Separating what is seen from how the world responds.** Left: a generated concept illustration distinguishes appearance from dynamics; it is not simulator output. Evaluated appearance changes are pointwise RGB transforms, not camera-viewpoint changes. Right: relative forecast MSE@5 change of ShiftWM (ours) versus Framewise calibration, computed from the ratio of three-training-seed means. Negative values indicate lower error. Gains on held-out pairs do not extend to Reacher extrapolation. These point summaries are not planning results or significance estimates; Figure 4 reports raw errors, all aligned controls, and seed variability.

offline forecasting results are reported separately from the controlled composition and planning experiments. A subsequent comparison freezes all methods before inspecting 65 additional recordings from 52 new sessions (Appendix P). With identical train-fitted residual calibration, ShiftWM reduces the prespecified five-block endpoint error by 0.742% versus Framewise; its paired interval excludes zero. This provides independent evidence for a small forecasting benefit, while longer-horizon comparisons remain inconclusive.

The observation-anchored spatial variant studies a complementary way to use observed evidence: retain the actual last visual grid, mix its features using an action-conditioned state, and add a bounded correction. Its matched DROID development study improves native ten-step feature error by 5.30% relative to autoregression and 3.55% relative to anchoring alone (Appendix T). This variant does not retain the original two-context factorization. We distinguish the original context model as **Ours-1**, spatial anchoring as **Ours-2**, bounded anchoring as **Ours-3**, unbounded spatial mixing as **Ours-4**, and bounded spatial mixing as **Ours-5**. Ours-3 and Ours-4 form the registered component follow-up. These identifiers name mechanisms, not a performance ranking; comparisons always use the same representation and information budget.

2 RELATED WORK

Predictive models and context. We reuse LeWM’s encoder, action embedding, causal predictor, and planning interface (Maes et al., 2026b), with `stable-worldmodel` infrastructure (Maes et al., 2026a); compactness is an upstream property. ContextWM separates scene context from dynamics (Wu et al., 2023), ICWM infers system information from interaction histories (Wang et al., 2026b), and work on habit, physics, and nuisance separates action selection, transition, and observation effects (Hang & Cai, 2026). In surgical simulation, SWoMo separates rule-based tool motion from diffusion-based appearance generation (Sivakumar et al., 2026). Our scope is support-conditioned latent prediction and matched-budget control; separation of appearance and dynamics itself is not a new claim. Our candidate distinction is intervention-paired training and held-out composition when both appearance and physical response vary. Two context vectors alone do not establish novelty.

Adaptation and physical conditioning. MoViE adapts visual model-based policies across views (Yang et al., 2023); AdaJEPA updates a latent model in the control loop (Wang et al., 2026c). ReDRAW adapts latent dynamics through learned residual corrections (Lanier et al., 2025); RLA-WM

learns residual latent actions and predicts them with flow matching (Zhang et al., 2026b). Residual prediction alone therefore does not distinguish our method. SG-JEPA studies supplied physical conditioning and multi-step latent consistency (Liu et al., 2026). ShiftWM uses inferred context without online gradients or true physical parameters. Our separate AdaJEPA reproduction retains its official checkpoint, proprioceptive inputs and planning budget; it is not a matched baseline for ShiftWM. MoVie and SG-JEPA remain unreproduced comparisons. Identifiability depends on representation assumptions and action excitation (Zhang et al., 2026a); our prediction loss does not uniquely identify physical quantities. A final literature audit and stronger direct comparisons remain necessary to substantiate a novelty claim.

Decision geometry. DA-LeWM distinguishes informative representations from planning-cost alignment and studies predicted–executed plan rankings (Wang et al., 2026a). Monotone cost ranking combines frozen visual features, recursive forecasting and action-conditioned cost shaping (Chahe et al., 2026). Our branch-ranking diagnostic acknowledges this precedent; neither that diagnostic nor a larger feature-calibration range is claimed as a new method.

3 PROBLEM AND INFORMATION ACCESS

Let x_t be hidden physical state, a_t an executed action, p a dynamics setting, and v an observation setting. Within a stationary segment,

$$x_{t+1} = F_p(x_t, a_t, \epsilon_t), \quad o_t = R_v(x_t, \eta_t). \quad (1)$$

The deployed agent receives images and actions, not x_t, p, v . Training covers selected pairs (v, p) . Compositional testing holds out a pair while retaining each factor individually in training; extrapolation to new values is a separate evaluation. Appearance transforms preserve the action coordinate system. Camera geometry changes and noninvertible occlusion require different protocols.

An earlier support segment S contains H observations and $H - 1$ actions; later query images Q are disjoint from support. At deployment, support is real past history, and the controller receives a goal image. Imagined futures never become evidence for context inference. Complete trajectory families, including appearance-paired renders, stay within one data split.

Training supervision and fairness. A canonical render depicts the *same physical state* under a reference appearance; changing dynamics requires a new rollout. All trained controls receive the same trajectories, canonical targets, training schedule, and validation selection. ShiftWM additionally uses pairing identities for context consistency. Shared versus unpaired contexts tests architecture; unpaired contexts versus ShiftWM tests the extra consistency supervision. A comparison against the frozen model alone cannot attribute a gain to factorization.

4 SHIFTWM: OBSERVATION AND DYNAMICS CONTEXTS

The visual encoder and projector E_0 are frozen. We train context networks, adapters, the action encoder, predictor, and prediction projector offline. Weights remain fixed online. Figure 2 foregrounds the shared observation calibration (A), corrected-transition dynamics inference (B), and additional paired supervision (C); the LeWM backbone and CEM planner are reused. Appendix B gives dimensions, initialization, and parameter counts.

4.1 CONTEXT INFERENCE AND PREDICTION

For $u_t = E_0(o_t) \in \mathbb{R}^d$, the observation pathway computes

$$c_o = C_o(u_S), \quad z_t = A_o(u_t, c_o). \quad (2)$$

The residual adapter A_o calibrates shifted features to reference coordinates. Dynamics context uses chronological corrected support and executed actions:

$$\begin{aligned} c_d &= C_d((z_i, a_i, z_{i+1})_{i \in S}), \\ \tilde{a}_t &= A_d(E_a(a_t), c_d), \quad \hat{z}_{t+1} = P_\theta(z_{t-L+1:t}, \tilde{a}_{t-L+1:t}). \end{aligned} \quad (3)$$

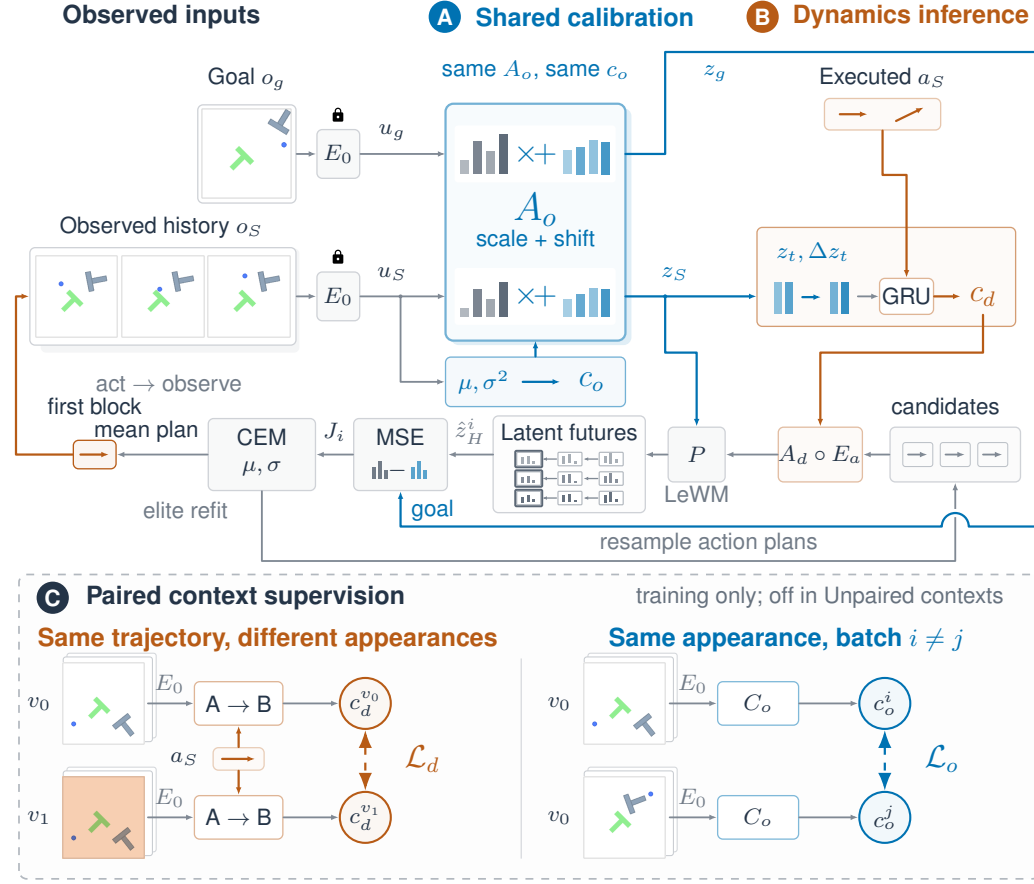


Figure 2: **ShiftWM (ours): separate contexts with paired supervision.** **A** infers observation context from past-feature statistics and shares one calibration map between history and goal. **B** infers dynamics context from corrected transitions and executed actions, then conditions action embeddings in the reused LeWM predictor. **C** adds offline penalties: $\mathcal{L}_d$ compares different appearances of the same trajectory and actions, each through its own A–B computation; $\mathcal{L}_o$ compares distinct batch entries with the same appearance. Dashed links denote penalties, not guaranteed equal or identified contexts. Training frames in C represent support clips; the upper frames illustrate deployment inputs. The encoder is frozen; contexts, adapters, and reused prediction modules train offline. Online, weights and contexts stay fixed within CEM search. Each candidate action sequence produces a latent rollout, scored by terminal MSE to the calibrated goal. CEM refits its action distribution to low-cost elites; the controller executes the first block of the final mean plan and replans. Feature glyphs and the rollout/cost example are schematic, not measurements. Canonical query losses and the exact planning budget are given in the text.

Here A_d is an action-embedding adapter and L is predictor history. Correcting support before dynamics inference is an inductive bias, not proof of independence; uninformative actions can leave dynamics ambiguous. Contexts are inferred once per replan and held fixed over candidate rollouts.

4.2 LEARNING IN FIXED COORDINATES

For a canonical training render $o_t^* = R_{v_0}(x_t)$, the target is $z_t^* = \text{sg}(E_0(o_t^*))$. The prediction loss $\mathcal{L}_{\text{pred}}$ is teacher-forced one-step query MSE, and $\mathcal{L}_{\text{align}}$ is corrected-feature MSE to the same frozen reference. Appearance variants of the same trajectory encourage equal dynamics contexts through $\mathcal{L}_d$. Different batch examples sharing an appearance label encourage equal observation contexts through $\mathcal{L}_o$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_d \mathcal{L}_d + \lambda_o \mathcal{L}_o, \quad (4)$$

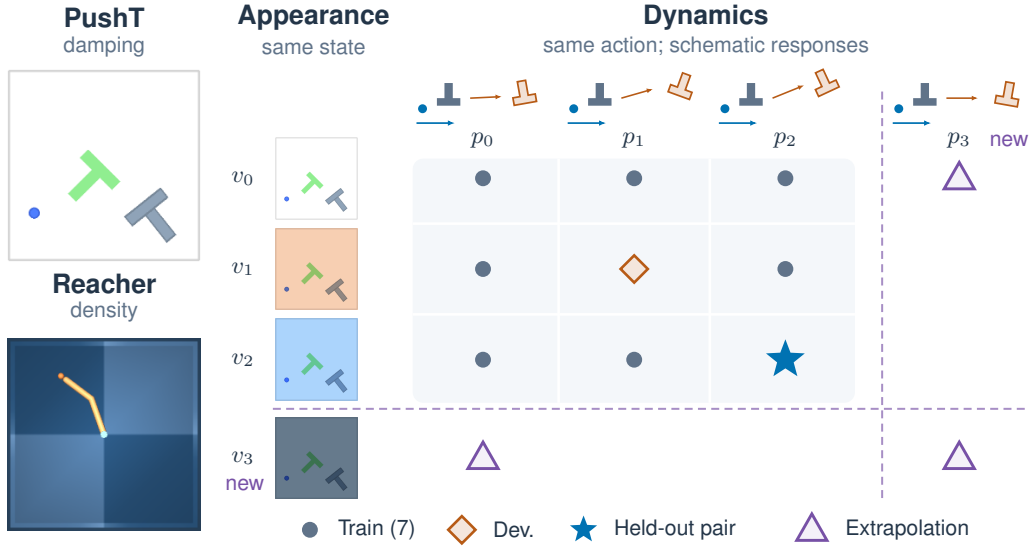


Figure 3: **Composition protocol.** Real task frames and exact RGB transforms illustrate observation changes; response glyphs are schematic. PushT varies damping and Reacher varies density. Seven pairs train the model, (v_1, p_1) is development, and (v_2, p_2) is held-out composition. Three extrapolation pairs introduce new values. Independent test trajectories cover all nine in-range pairs. Split identities precede paired rendering; factor labels are unavailable to the deployed model.

with weights $(1, 0.1, 0.01)$. Unpaired contexts disable the two consistency terms. Targets have fixed coordinates, but the contexts themselves may collapse or rescale with their adapters. Constant-context and shuffled-context controls are therefore required before interpreting the contexts as useful factor-specific information. Appendix B.2 states the exact index sets and batch-pairing rules, including the mismatch between one-step training and recursive evaluation.

4.3 PLANNING

The goal embedding is $z_g = A_o(E_0(o_g), c_o)$, assuming the goal shares the current observation setting. A common CEM/MPC solver minimizes

$$J(a_{t:t+K-1}) = \text{mse}(\hat{z}_{t+K}, z_g), \quad (5)$$

executes the first action chunk, and replans from new real observations. All methods share candidate count, iterations, action bounds, horizon, and paid interaction budget. Different-view goals and online gradient adaptation are outside the completed primary comparison.

5 EXPERIMENTS

Tasks and composition. We use visual PushT and Reacher in `stable-worldmodel`. Appearance factors are canonical, warm, and cool RGB transforms. PushT varies damping; Reacher varies arm and finger density together. Seven factor pairs train the model, (v_1, p_1) is development, and (v_2, p_2) is held-out composition (Figure 3). Separate extrapolation tests introduce dim appearance and a new physical value. Each environment has 256 training, 32 validation, 32 development, and 64 test initial-state seeds. Test trajectories are independent and cover all nine in-range pairs; extrapolation covers three additional pairs. Appendix A gives exact values, episode counts, and action interfaces.

Compared methods. *Frozen LeWM* is the released pretrained reference. *Framewise calibration*, *Shared context*, and *Unpaired contexts* are our matched controls on that same backbone, not independent reproductions of published methods (Table 1). Framewise calibration tests whether per-image

Table 1: **Methods in the primary comparison.** Frozen LeWM is the released reference; the other rows are matched in-house implementations on its backbone. Every learned visual calibration path uses canonical alignment. Only ShiftWM uses paired context consistency. None uses online gradients.

Method	Visual calibration	Context	Consistency
Frozen LeWM	None	None	None
Framewise calibration	Per frame	None	None
Shared context	Contextual	Shared	None
Unpaired contexts	Contextual	Separate	None
ShiftWM (ours)	Contextual	Separate	Paired

Method	PushT		Reacher	
	Held-out $\uparrow$	Extrap. $\uparrow$	Held-out $\uparrow$	Extrap. $\uparrow$
Frozen LeWM	4.7	7.3	35.9	17.7
Framewise calibration	3.6 ± 0.9	5.0 ± 0.8	42.7 ± 3.3	21.2 ± 5.9
Shared context	2.1 ± 0.9	7.6 ± 1.1	36.5 ± 3.9	16.8 ± 3.3
Unpaired contexts	2.6 ± 0.9	5.2 ± 1.4	34.9 ± 9.0	20.8 ± 2.6
ShiftWM (ours)	2.6 ± 0.9	6.8 ± 0.5	40.6 ± 3.1	19.6 ± 1.3
Δ vs Shared context (pp)	+0.52	-0.87	+4.17	+2.78
95% paired CI	[-1.04, +2.08]	[-2.95, +1.22]	[-4.17, +12.50]	[-1.91, +7.64]

Table 2: **Main comparison: closed-loop planning success (%)**. Higher is better. Trained entries require all three seeds and show mean $\pm$ sample SD; Frozen LeWM uses one released checkpoint per environment. The final rows give ShiftWM minus Shared context in percentage points and conditional 95% paired task-cluster intervals, not SD. **Bold green** marks positive mean differences, not statistical significance. Dashes denote incomplete evidence, not zero success. Held-out is (v_2, p_2) ; extrapolation averages three prespecified conditions. All methods share the paid support and planning budget. Framewise calibration, Shared context and Unpaired contexts are in-house controls on the same LeWM backbone. Full metrics and the unaligned diagnostic appear in Table 8.

correction suffices. Shared context tests whether one inferred context is sufficient. Unpaired contexts retains the two-branch architecture while removing context-consistency losses. *ShiftWM (ours)* combines separate contexts with those paired losses. Every trained variant has three seeds and 30 completed epochs; ordinary-validation prediction error selects checkpoints. The *Unaligned predictor* retains mismatched observed/goal coordinates and is reported only as a diagnostic in Appendix J.

Metrics and evaluation budget. Closed-loop success is primary. The main comparison separates the held-out pair from extrapolation; canonical and all-condition results remain in the appendix. Planning uses 300 CEM candidates, 30 iterations, 30 elites, a five-group horizon, and five native controls per group. The 50-native-step budget includes ten support-acquisition steps. Methods use matched initial-state/goal identities and common support. Raw success includes support-only successes; policy-eligible success excludes them. Table entries require all three trained seeds and report mean $\pm$ sample standard deviation, while paired task-cluster intervals appear separately in Appendix K. Forecast MSE@5 uses five recursive transitions with recorded actions in immutable reference coordinates. Hyperparameters are fixed without test-based selection; elapsed inference time is not compared without dedicated hardware measurements.

6 RESULTS AND ANALYSIS

All 30 training runs, main forecast evaluations, and 64 prescribed checkpoint-split planning evaluations are complete. Table 2 is the main closed-loop comparison; every trained-method entry now includes all three prescribed seeds and passes source and protocol validation. Full condition results, eligibility counts, and exact forecast values are in Appendix J.

Does composition improve planning? The complete matched comparisons show mixed effects. Relative to Framewise calibration, held-out raw-success differences are -1.04 percentage points on PushT (conditional 95% interval $[-3.65, 1.56]$) and -2.08 on Reacher ($[-11.46, 7.29]$). On extrapolation, PushT improves by $+1.74$ points ($[0.00, 3.65]$), while Reacher declines by -1.56 ($[-6.25, 3.13]$). These intervals condition on the three observed training seeds.

Positive held-out point estimates occur against Shared context ($+0.52$ points on PushT and $+4.17$ on Reacher) and against Unpaired contexts on Reacher ($+5.73$); PushT ties Unpaired contexts. Across the 12 raw-success contrasts in Table 10, six means are positive, five negative, and one zero. Every conditional 95% interval includes zero; the PushT extrapolation interval against Framewise touches it exactly. The intervals are not multiplicity-adjusted and do not capture uncertainty over the population of training runs. The completed evidence therefore does not establish consistent planning superiority or isolate a benefit from context pairing. Policy-eligible comparisons, with support-only successes excluded, are reported alongside the raw contrasts. All prescribed three-seed planning comparisons with Framewise calibration and Unpaired contexts are available in Table 2, with paired uncertainty reported in Table 10. Differences only against Frozen LeWM would conflate the proposed structure with additional training.

Where ShiftWM improves forecasting. Against the fixed Framewise calibration reference, ShiftWM achieves **26.78% lower held-out Reacher MSE@5** and 3.68% lower PushT MSE@5, using ratios of three-seed means (Figure 4). The closest aligned held-out comparator is Unpaired contexts: adding paired consistency reduces its error by 9.21% on Reacher and 2.30% on PushT. These are relative forecast-error reductions, not changes in planning success.

Relative MSE reduction (%) $\uparrow$

Reference	PushT		Reacher	
	Held-out	Extrap.	Held-out	Extrap.
Framewise calibration	+3.68	+4.20	+26.78	-4.15
Unpaired contexts	+2.30	+0.70	+9.21	-0.35

The summary defines reduction as $100(\text{baseline} - \text{ours}) / \text{baseline}$, so positive means lower error and negative means a regression. **Bold green** marks positive mean reductions, not statistical significance. On extrapolation, PushT improves over Framewise calibration by 4.20%, while Reacher worsens by 4.15%; Frozen LeWM has the lowest Reacher extrapolation error among the aligned models. The unaligned diagnostic has lower held-out PushT error than ShiftWM and remains visible in the full results. Thus, the held-out gains do not establish uniform superiority. Unpaired contexts also improves on Shared context in both held-out environments, separating architectural changes from pairing. Mean ordering and three-seed dispersion do not identify the factors encoded by either context or constitute a significance test.

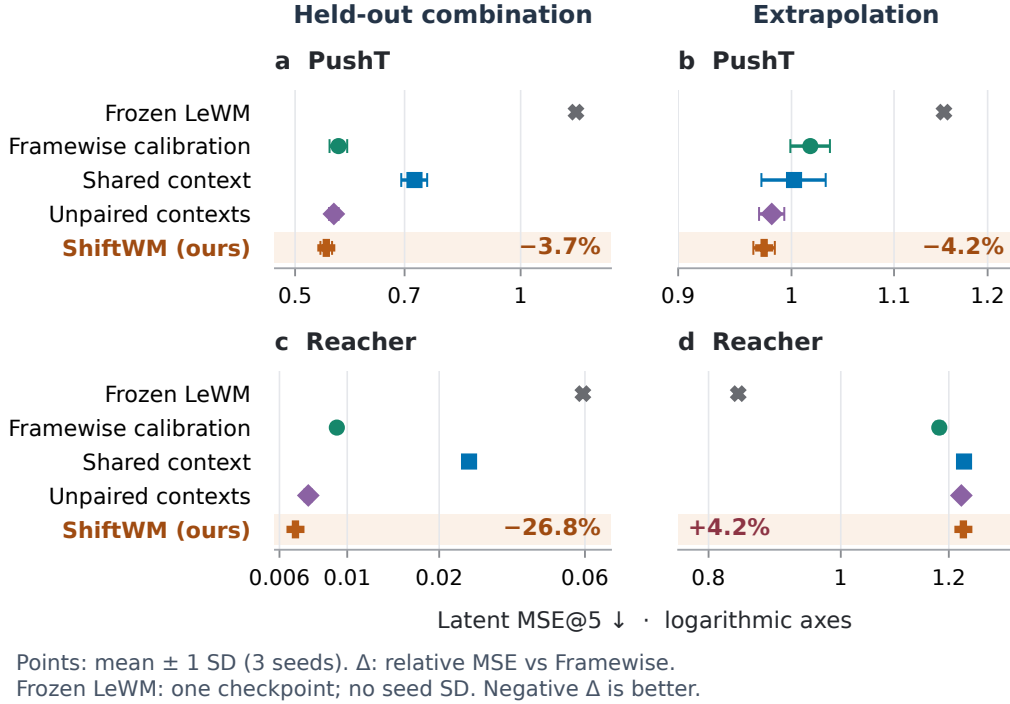


Figure 4: **Forecasting on held-out composition and extrapolation.** Terminal latent MSE after five recursive transitions (25 native actions) using recorded future actions; lower is better. Points and ranges show three-training-seed means and sample standard deviations, not confidence intervals. Frozen LeWM has one checkpoint and no training-seed error bar. Extrapolation averages the three prespecified new-value conditions. Horizontal axes are logarithmic with panel-specific limits; latent coordinates differ by environment. Gain annotations use signed error change relative to Framewise calibration: negative means lower MSE. These ratios of three-seed means are point summaries, not significance estimates. Forecast accuracy alone does not establish planning success.

Why better forecasts need not yield better control. In the completed seed-0 development comparison, ShiftWM and Shared context have equal success rates; Framewise calibration has the highest Reacher point estimate. A subsequent goal-only intervention keeps the ShiftWM predictor fixed but substitutes framewise goal correction. Eligible Reacher success increases from 6/30 to 12/30, compared with 11/30 for the complete framewise controller; PushT falls from 1/31 to 0/31 (Appendix H). This post-hoc intervention motivates separate checks of prediction and goal alignment, not a general repair. CEM also selects its own candidate actions, so mean forecast error on recorded trajectories does not control error on those selected candidates. Mechanism tests, the completed dynamics-residual revision, and the completed controlled rollout-objective study are described in Appendices G, L, and M. Matched observed rollouts in Appendix F show successes, failures, and goals reached during support before planning.

7 LIMITATIONS AND CONCLUSION

Canonical re-renderings and simulator interventions provide stronger supervision than uncontrolled real-world video. Frozen coordinates can omit dynamics-relevant information, and contexts may be ambiguous or collapse without affecting the reported losses. RGB transformations do not establish robustness to new camera geometry. The simulated tasks do not establish arbitrary model/task transfer, real-robot performance, or medical usefulness. The separate DROID study adds recorded real observations and a camera-transfer evaluation, but does not measure physical closed-loop control. On PushT, held-out and extrapolation success point estimates are below the random-action control; this descriptive comparison does not provide a matched policy-seed significance test (Ta-

ble 9). Matched online-adaptation comparisons remain unfinished. The completed eight-arm rollout-objective/context study does not satisfy its registered promotion criterion and is reported separately (Appendix M). Development-driven revisions are kept separate from the completed primary campaign; their extra optimization is not a matched-budget result. The original test outcomes have informed this follow-up work, so any confirmatory revision claim requires a fresh final holdout and additional training seeds.

ShiftWM makes the observation–dynamics composition hypothesis concrete in a compact world model. Completed forecasts favor it over framewise calibration on held-out combinations, while Reacher extrapolation regresses. The completed planning campaign shows mixed point estimates with conditional intervals that include zero. Validation-selected local checkpoints, normalization and history interfaces, and provenance-linked evaluation records are prepared; public release is pending. The evidence needed for a stronger claim is improved matched planning together with controls showing that the separate contexts provide useful information.

AI ASSISTANCE DISCLOSURE

AI assistance was used for drafting, public-source inspection, code development, and editable figures. Human authors must verify the final text, sources, implementation, and results and remain responsible for the submission. This statement must reflect the completed workflow.

REFERENCES

- Amirhosein Chahe, Siwei Cai, and Lifeng Zhou. Latent world models with monotone planning costs for image-goal navigation. *arXiv preprint arXiv:2608.09073*, 2026. URL <https://arxiv.org/abs/2608.09073>.
- Jinting Hang and Zhenhui Cai. Identifying habit, physics, and nuisance in robot world models. *arXiv preprint arXiv:2609.09210*, 2026. URL <https://arxiv.org/abs/2609.09210>.
- Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. URL <https://arxiv.org/abs/2403.12945>.
- JB Lanier, Kyungmin Kim, Armin Karamzade, Yifei Liu, Ankita Sinha, Kat He, Davide Corsi, and Roy Fox. Adapting world models with latent-state dynamics residuals. *arXiv preprint arXiv:2504.02252*, 2025. URL <https://arxiv.org/abs/2504.02252>.
- Andy Zeyi Liu, Haoran Sun, Lucas Baker, Randall Balestriero, and John Sous. Semigroup-JEPA: Latent dynamics consistency for zero-shot physics generalization. *arXiv preprint arXiv:2609.10464*, 2026. URL <https://arxiv.org/abs/2609.10464>.
- Lucas Maes, Quentin Le Lidec, Luiz Facury, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, and Randall Balestriero. stable-worldmodel: A platform for reproducible world modeling research and evaluation. *arXiv preprint arXiv:2605.21800*, 2026a. URL <https://arxiv.org/abs/2605.21800>.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026b. URL <https://arxiv.org/abs/2603.19312>.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. URL <https://arxiv.org/abs/2304.07193>.
- Jacopo Panerati, Hehui Zheng, SiQi Zhou, James Xu, Amanda Prorok, and Angela P. Schoellig. Learning to fly—a gym environment with pybullet physics for reinforcement learning of multi-agent quadcopter control. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7512–7519, 2021. doi: 10.1109/IROS51168.2021.9635857.

- Paul Maria Scheikl, Balázs Gyenes, Rayan Younis, Christoph Haas, Gerhard Neumann, Martin Wagner, and Franziska Mathis-Ullrich. Lapgym—an open source framework for reinforcement learning in robot-assisted laparoscopic surgery. *Journal of Machine Learning Research*, 24(368): 1–42, 2023. URL <https://jmlr.org/papers/v24/23-0207.html>.
- Ssharvien Kumar Sivakumar, Akwele Johnson, Anirudh Dhingra, Yannik Frisch, Ghazal Ghazaei, and Anirban Mukhopadhyay. SWoMo: Neuro-symbolic world model for cataract surgery simulation. *arXiv preprint arXiv:2605.16530*, 2026. URL <https://arxiv.org/abs/2605.16530>.
- Jiawei Wang, Yushen Zuo, Ke Rui, Yichun Feng, and Minglei Li. Decision-metric alignment in latent world models: Diagnostics and action-conditioned objectives for mpc planning. *arXiv preprint arXiv:2608.18746*, 2026a. URL <https://arxiv.org/abs/2608.18746>.
- Siyin Wang, Junhao Shi, Senyu Fei, Zhaoyang Fu, Li Ji, Jingjing Gong, and Xipeng Qiu. In-context world modeling for robotic control. *arXiv preprint arXiv:2606.26025*, 2026b. URL <https://arxiv.org/abs/2606.26025>.
- Ying Wang, Oumayma Bounou, Yann LeCun, and Mengye Ren. AdaJEPa: An adaptive latent world model. *arXiv preprint arXiv:2606.32026*, 2026c. URL <https://arxiv.org/abs/2606.32026>.
- Jialong Wu, Haoyu Ma, Chaoyi Deng, and Mingsheng Long. Pre-training contextualized world models with in-the-wild videos for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://papers.neurips.cc/paper_files/paper/2023/hash/7ce1cbdedb4b0d6202847ac1b484ee8-Abstract-Conference.html.
- Sizhe Yang, Yanjie Ze, and Huazhe Xu. MoVie: Visual model-based policy adaptation for view generalization. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://arxiv.org/abs/2307.00972>.
- Xiangteng Zhang, Yang Guan, Bo Zhang, Hongyang Li, Ya-Qin Zhang, and Shengbo Eben Li. On the identifiability of controlled world models. *arXiv preprint arXiv:2607.22430*, 2026a. URL <https://arxiv.org/abs/2607.22430>.
- Xinyu Zhang, Zhengtong Xu, Yutian Tao, Yeping Wang, Yu She, and Abdeslam Boularias. Learning visual feature-based world models via residual latent action. *arXiv preprint arXiv:2605.07079*, 2026b. URL <https://arxiv.org/abs/2605.07079>.

A COMPLETE EXPERIMENTAL PROTOCOL

Environments and splits. The core tasks are visual PushT and Reacher in `stable-worldmodel`. They have different control geometries but share infrastructure. Initial observation factors are canonical, warm, and cool pointwise RGB transforms. PushT changes space damping (0, 0.15, 0.65); Reacher changes arm and finger density together (1000, 650, 1350 in simulator units). We train on seven combinations, reserve (v_1, p_1) for development, and hold out (v_2, p_2) for the main compositional test (Figure 3). Test trajectories are independent across the full grid. Additional dim and physical conditions test extrapolation separately: dim appearance v_3 , PushT damping 0.95, and Reacher density 1500, evaluated at $(v_3, p_0), (v_0, p_3), (v_3, p_3)$. Each environment has 256 training, 32 validation, 32 development, and 64 test initial-state seeds. Three physical conditions in the first three splits and four in test produce 1,216 physical episodes per environment, each with 64 grouped transitions. Appearance versions reuse those physical trajectories. The full test planning grid has $64 \times 9 = 576$ tasks per checkpoint; the separate extrapolation grid has $64 \times 3 = 192$. The all-condition test average includes familiar factor pairs and is not a wholly unseen-composition score. This controlled extension is not a reproduction of the original paper’s success table.

A.1 PLANNING, ELIGIBILITY, AND RESOURCE MEASUREMENT

Primary outcomes and uncertainty. Closed-loop success under each environment’s declared criterion is primary. We also report steps to goal and worst-condition success under fixed budgets. Methods use matched initial-state/goal IDs. Core comparisons target three training seeds; actual seed and episode counts accompany each result. Episode-level outcomes support paired comparisons; uncertainty over tasks and variability over training seeds are reported separately. Correlated frames are not independent trials. Failed runs and timeouts remain in the record, with a stated policy for their inclusion in denominators. The main planner uses 300 candidates, 30 CEM iterations, and 30 elites, matching the released LeWM CEM sampling configuration. This budget is locked for all model comparisons before full-test learned-policy planning. It evaluates 9,000 candidate sequences per replan, versus 640 in the earlier development-only setting; this $14.0625\times$ count ratio is not a measured wall-time ratio. Both settings use five grouped actions of prediction horizon, and one group executed per replan. A group contains five native controls. The 50-step native budget includes ten steps acquiring initial support. CEM optimizes in the checkpoint’s training action z-score coordinates; it converts candidates to native controls and clips only physical bounds $[-1, 1]$. It does not clip z-scores to that range. PushT uses the relative-displacement interface from the official training archive; the earlier absolute-control corpus is quarantined. Success during support is recorded separately; policy-eligible success excludes tasks already solved by that common history. Final-state success is also retained. Per-replan latency divides actual solve time by actual replanning calls, excluding nonexistent calls on already-solved tasks. These are configuration facts, not measured costs. Main efficiency results additionally require dedicated execution with no concurrent training on the measured GPU; shared-GPU development timing is retained in raw records but excluded from efficiency tables.

A.2 ADDITIONAL DIAGNOSTICS AND TASK SCOPE

Diagnostics. Completed forecast evaluations measure recursive error in frozen reference coordinates. Action shuffling, context replacement/removal, and support-length dependence are additional mechanism tests; their outcomes are not yet reported. A low one-step error can arise from ignoring actions on predictable trajectories; action interventions and planning test that failure. Prediction diagnostics therefore include persistence, zero-action rollouts, and the change induced by actions. Zero-action predictions are compared with recorded-action targets only as a sensitivity check, not as errors against observed counterfactual trajectories. PushT additionally reports a prespecified subset requiring block translation beyond 20 pixels or rotation beyond $\pi/9$, so tasks solved by moving the agent alone do not hide manipulation failures. Both the full set and this subset remain visible. Within-episode switches are a secondary evaluation after stationary conditions. Test combinations are never used for hyperparameter selection.

Action-conditioned phantom endoscopy is an optional offline extension, outside the minimum control claim. Forecasting alone cannot demonstrate navigation or clinical benefit. Static diagnostic images do not provide the executed actions required by this study.

B IMPLEMENTATION AND TRAINING DETAILS

B.1 ARCHITECTURE AND PARAMETER ACCOUNTING

The initial implementation freezes a pretrained LeWM visual encoder and projector, denoted E_0 . Context networks, adapters, action encoder, predictor, and prediction projector are trained. Full visual fine-tuning is a distinct configuration. The frozen path permits feature caching without changing inputs or targets. Figure 2 shows the computation. The upstream architecture contains 18,034,478 parameters. The ShiftWM *active inference path* uses 18,319,598, with 12,025,454 trainable offline. Shared context has 12,025,046 trainable parameters. The current loaded model object contains 24,873,110 parameters including the duplicated frozen reference encoder and inactive ablation modules; the current package serializes that full state. Active-path counts are not claims about package size or runtime memory, which must be measured on the actual released artifact.

For $d = 192$ and $H = L = 3$, both contexts have width 32; the dynamics network uses a 128-unit GRU. The observation network pools support-feature mean and population variance. Both adapters use bounded-scale residual FiLM. Inference holds contexts fixed within a search, so imagined futures cannot modify the information used to infer dynamics.

Framewise calibration uses the per-image residual $u + \text{MLP}(\text{LN}(u))$ with dimensions $192 \rightarrow 128 \rightarrow 192$ and an initially zero output layer. It corrects history and goal images identically, receives no temporal context, and uses the prediction and canonical-alignment losses. It adds 49,856 active parameters; its active path has 18,084,334 parameters, of which 11,790,190 are trained offline. Shared context matches the context-network parameter count approximately; architecture attribution must still account for computation differences.

B.2 EXACT LOSSES AND TEMPORAL INDEXING

Let $o_t^* = R_{v_0}(x_t)$ be a paired canonical training render and $z_t^* = \text{sg}(E_0(o_t^*))$ its frozen target. For a sequence indexed $0, \dots, T - 1$, context uses only indices $0, \dots, H - 1$. Define $\mathcal{T}_{\text{pred}} = \{H, \dots, T - 2\}$ and $\mathcal{T}_{\text{align}} = \{H, \dots, T - 1\}$. The implemented query losses are

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{T}_{\text{pred}}|} \sum_{t \in \mathcal{T}_{\text{pred}}} \text{mse}(\hat{z}_{t+1}, z_{t+1}^*), \quad (6)$$

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{T}_{\text{align}}|} \sum_{t \in \mathcal{T}_{\text{align}}} \text{mse}(z_t, z_t^*). \quad (7)$$

Here $\text{mse}(u, v) = \|u - v\|_2^2 / \dim(u)$ averages over feature dimensions, matching the implementation and stated loss weights. Training uses teacher-forced one-step query prediction. With $T = 8$ and $H = 3$, context sees frames 0–2, alignment supervises frames 3–7, and prediction supervises targets 4–7 from observed history windows. Frame 3 is not a prediction target in this campaign. Evaluation instead starts immediately after support and recursively predicts frames 3–7. This boundary and teacher-forcing mismatch can affect planning even when validation prediction error is low; the completed runs are not multi-step rollout training. Frozen targets prevent an encoder from lowering the reported loss solely by shrinking or changing its coordinates; they do not guarantee that those coordinates retain every necessary physical quantity.

Appearance variants $S^{(v,p)}$ and $S^{(v',p')}$ of the same trajectory give

$$\mathcal{L}_d = \mathbb{E} \text{mse}(C_d(S^{(v,p)}), C_d(S^{(v',p')})), \quad (8)$$

where $C_d(S)$ abbreviates the corrected-support pathway. Distinct batch examples with the same observation label give the regularizer

$$\mathcal{L}_o = \mathbb{E}_{v,p,p',S,S'} \text{mse}(C_o(S^{(v,p)}), C_o(S^{(v',p')})). \quad (9)$$

The implementation averages over off-diagonal batch pairs with matching appearance IDs; it does not require their dynamics values or trajectory IDs to differ. Grouping establishes a shared transformation, not a shared state. The term does not force state latents of different trajectories to agree. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_d \mathcal{L}_d + \lambda_o \mathcal{L}_o. \quad (10)$$

The initial configuration uses $\lambda_{\text{align}} = 1$, $\lambda_d = 0.1$, and $\lambda_o = 0.01$; unpaired controls disable both context consistency terms. No additional SIGReg term is used because the target coordinates are frozen. These choices are configuration facts, not tuned results. Raw context consistency also has a scale ambiguity: shrinking a context and increasing downstream sensitivity can lower its penalty without improving factor separation. Consistency can collapse contexts to constants. Constant-context, context-shuffle, and removed-pathway controls therefore test whether both contexts carry useful information. Context swapping across trajectories is not part of the initial implementation: it requires valid matching targets.

B.3 TRAINING AND CHECKPOINT SELECTION

The reported campaign uses the fixed encoder and feature-cache path, with RGB inputs in $[0, 1]$, ImageNet channel normalization applied once, and antialiased bilinear resizing to the released encoder input size. The visual projector and prediction projector retain pretrained BatchNorm running statistics. The prediction projector’s affine parameters remain trainable; freezing running statistics prevents query-frame batch statistics from entering support features and permits batch-size-one planning. The full campaign trains five variants (Unaligned predictor, Framewise calibration, Shared context, Unpaired contexts, and ShiftWM), three seeds, and two environments: 30 runs of 30 epochs, batch size 128, stride one, and initial learning rate 5×10^{-5} . All 30 configured runs have completed their 30 epochs; the generated ledger below records the completion evidence. AdamW uses weight decay 10^{-3} , gradient clipping at norm 1, and a cosine schedule ending at 10^{-6} . Training and validation use BF16 autocast when supported; evaluation uses FP32. Checkpoints are selected by ordinary-validation prediction MSE, not last-epoch loss or total loss, whose alignment and consistency terms differ across controls. Full training completion is distinct from selecting an early validation-best checkpoint. Main planning evaluation remains separate.

Both training and CEM retain the released-data action normalization. In the new PushT training corpus, the per-coordinate native-action standard deviations are approximately 0.504 and 0.504, versus 0.208 and 0.207 in the released archive (245,760 collected training actions). This distribution difference is retained in the protocol, not retrospectively removed by renormalizing the completed campaign. Reacher’s corresponding collected and released deviations are both approximately 0.577. These statistics describe action distributions; they do not establish that normalization explains the planning failures.

Table 3: Fields required for every released checkpoint.

Field	Required record
Base architecture	LeWM source commit, checkpoint revision and weight hash
Visual input	Shape, channel order, normalization, transformation version
Action interface	Meaning, units, bounds, statistics, frame skip
Context	H , dimensions, pooling/sequence modules, reset behavior
Prediction	L , predictor configuration, recursive loss horizon, goal protocol
Loss	All weights; appearance pairing and batch grouping rules
Optimization	Optimizer, schedule, batch, steps, precision, trained modules
Data	Episodes, transitions, seeds, generation and factor/split manifests
Planning	Candidates, iterations, elite fraction, horizon, action chunk
Resources	GPU, peak memory, training hours, prediction/planning latency

Completion ledger. The configured campaign contains 30 runs. At this artifact snapshot, 30 have recorded at least one complete epoch and 30 have a verified completion summary and final checkpoint. Missing epoch logs do not establish whether a scheduler job is queued or running. Training curves show observed validation prediction loss only; they do not establish closed-loop performance.

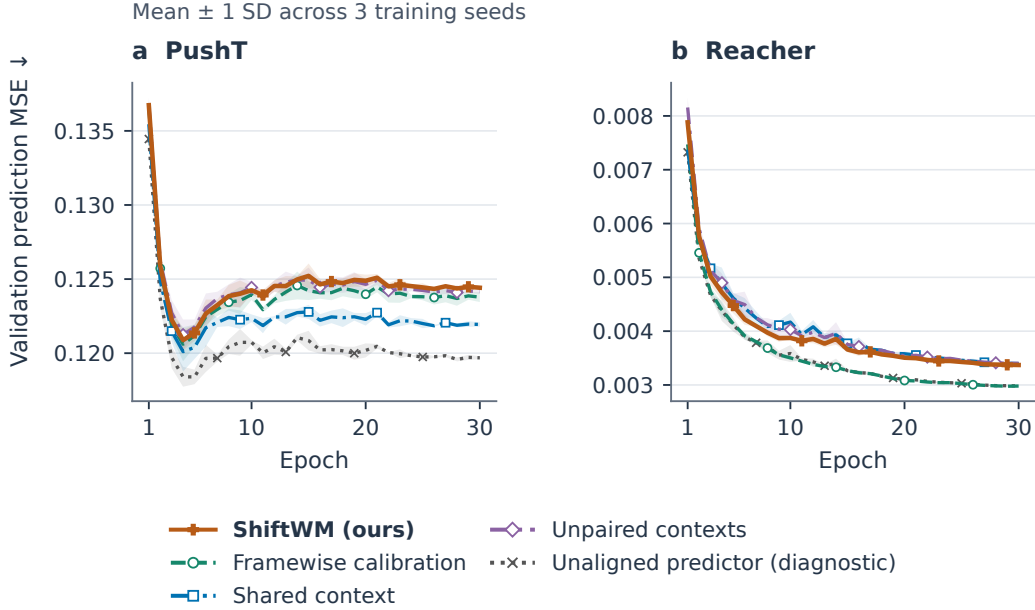


Figure 5: **Validation prediction loss over the completed training campaign.** Curves and shaded bands show means and sample standard deviations over three training seeds at common logged epochs. Colors and line styles identify model variants. No future epochs are extrapolated, and bands are not confidence intervals. The two panels have separate vertical scales. These are validation records, not test-set or planning-performance results.

C INFORMATION ACCESS AND CORRECTNESS

The support must precede query targets. No paired variant of an underlying trajectory can cross a split. Batch grouping can use factor IDs but model inputs cannot. Goal images convey target states, not hidden future trajectories. Future realized actions and rewards cannot enter context used to select an earlier action.

Checks must establish that saved models reload identically in evaluation mode; removing adapters recovers the specified base computation; prediction responds to action changes; support cannot replace query inputs; cached features match online features under identical transforms; checkpoint resumes preserve optimizer/scheduler and random state; and paired rendering does not mutate physical state. These are correctness requirements, not replacements for training and closed-loop evaluation.

What is not identifiable. For $x_{t+1} = x_t + pa_t$, $o_t = vx_t$, and $x_0 = 0$, observations depend on the product vp . For every $q > 0$, (v, p) and $(qv, p/q)$ generate the same observations under every action sequence. Separate physical interpretation therefore needs additional assumptions or information. Our contexts are functional prediction variables, not identified friction or camera parameters. This elementary example is a limitation, not a theoretical novelty claim.

D STATISTICS AND REPORTING

Every episode records training seed, condition, initial-state and goal IDs, success, length, and failure reason. A paired bootstrap resamples independent task IDs while retaining method pairing; training-seed dispersion is reported separately. The final interval level, calculation, resample count, and actual sample counts accompany each completed figure. Frame count must not inflate the number of independent trials.

The reported campaign fixes support length, loss weights, and planning settings in its versioned configurations; no hyperparameter search over these choices is claimed. Ordinary validation prediction

error selects checkpoints. Development diagnostics inform separately labeled follow-up hypotheses. Framewise calibration and Unpaired contexts were added after an implementation/validation audit, before the main learned-policy test campaign; the original runs remain unchanged. Test evaluation follows the frozen protocol. A documented implementation error can trigger a complete matched rerun; disappointing performance cannot be relabeled as an installation failure. Aborted and numerically unstable runs remain in the audit trail.

E DEVELOPMENT COMPARISONS AND BUDGET HISTORY

E.1 MATCHED OFFICIAL-BUDGET COMPARISON

Paired-support audit. PushT: the same 32 seed/trajectory identities, goal indices, support-success masks, eligibility masks and post-support distances agree across 4 completed modes. Reacher: the same 32 seed/trajectory identities, goal indices, support-success masks, eligibility masks and post-support distances agree across 4 completed modes. The comparison currently contains 8/8 completed model/environment records. Unfinished records do not contribute outcomes. Comparative conclusions remain pending; the main multi-seed test and its reporting gate are unchanged.

The seed-0 rates do not establish a multi-seed ordering. In Reacher, ShiftWM and Shared context each solve six eligible tasks but share only one success; equal aggregate rates therefore do not imply equal decisions or solved tasks.

E.2 EARLIER DEVELOPMENT-BUDGET CHECK

The first complete development check used the validation-selected checkpoint from a completed 30-epoch seed-zero factorized run in each environment, on all 32 prespecified development tasks at (v_1, p_1) , with 128 candidates, five iterations, and 16 elites. PushT achieved 1/32 total successes; 1 occurred during common support, leaving 0/31 policy-eligible successes. Reacher achieved 11/32 total successes, including 2 during support, with 9/30 policy-eligible successes. These are development diagnostics from one trained seed, not full-test results or evidence of factorization benefit. Baseline comparisons and a matched rerun at the main sampling budget are needed to interpret them. The official sampling-budget discrepancy was corrected before learned-policy full-test planning began. Increasing the search budget is not itself evidence that planning improves. Shared-GPU timings from this check are excluded from efficiency claims. This historical note records the pre-comparison budget discrepancy. The matched official-budget comparison above has since completed; the earlier check is retained without merging its outcomes into the main comparison.

Environment	Model	Raw success	During support	Policy-eligible
PushT	Frozen LeWM	2/32	1/32	1/31
PushT	Shared context	2/32	1/32	1/31
PushT	ShiftWM (ours)	2/32	1/32	1/31
PushT	Framewise calibration	2/32	1/32	1/31
Reacher	Frozen LeWM	5/32	2/32	3/30
Reacher	Shared context	8/32	2/32	6/30
Reacher	ShiftWM (ours)	8/32	2/32	6/30
Reacher	Framewise calibration	13/32	2/32	11/30

Table 4: **Official-budget development comparison, separate from the main test.** Each completed row uses all 32 prespecified development tasks at (v_1, p_1) , with 300 CEM candidates, 30 iterations and 30 elites. Raw and support rates use 32 tasks; policy-eligible rates exclude tasks already solved during common support. Trained models use seed 0; frozen LeWM has no new training seed. Dashes denote unfinished rows. These counts do not establish a multi-seed or factorization benefit; changing the CEM budget is not itself evidence of improvement.

F MATCHED QUALITATIVE EXECUTIONS

Figures 11–14 show actual simulator observations from the completed seed-zero development evaluations at (v_1, p_1) . We select the lexicographically first task in each outcome stratum relative to Framewise calibration: ShiftWM-only success, Framewise-only success, and joint failure. The fourth example is joint success for Reacher; PushT has no eligible joint success, so we explicitly show a support-only success instead. Selection is by this stated rule, not visual quality; these outcome-stratified examples do not estimate success frequency. The accompanying local gallery contains all 64 tasks and adds Shared context.

Both methods receive pixel-identical initial observations and available goals, with the same planner seed and budget: 300 candidates, 30 CEM iterations, 30 elites, horizon five action blocks, and at most 50 native calls including support. The first saved frame follows support acquisition; subsequent frames follow executed blocks of up to five actions. Labels reconstruct exact native times, including early stopping. Repeated terminal observations do not imply additional interactions. The world model predicts latents; none of the displayed images is a decoded prediction, interpolated state, or generated illustration.

F.1 WHERE SHIFTWM REACHES THE GOAL

Figures 6 and 7 show *all four* policy-eligible development tasks on which ShiftWM (ours) succeeds and Framewise fails. Shared context also fails on each. These are outcome-selected explanations, not four independent benchmark improvements: the complete eligible development comparison has one ShiftWM-only and one Framewise-only success on PushT, and three versus eight on Reacher. Counterexamples follow, and the local gallery retains every task.

The dashed goal silhouette makes the physical objective visible in each observation. Both middle-column images are recorded at native call 20; terminal images retain their actual stopping times. Reacher details use the same crop within each case for every goal, time, and method, containing all displayed arm poses with padding and marked in a full-scene overview. Endpoint errors explain why an execution satisfies or misses the scoring rule. This supports a behavioral explanation—reaching the requested block/pusher or joint configuration within the budget. It does not identify the responsible internal component. Attribution to useful dynamics context, factorization, or improved candidate ranking requires controlled interventions; the objective/context study in Appendix M tests part of this question separately.

A PushT · block and pusher reach the target

Episode 2031024 · same goal/support · 50-action allowance

ShiftWM (ours)

Goal reached

t = 21 / 50

Goal image

Matched time: t=20

Actual end: t=21



Block 6.9 px · angle 0.159 rad · pusher 14.7 px

Goal image

Matched time: t=20

Actual end: t=50



Framewise

Goal missed

t = 50 / 50

Block 317.9 px · angle 1.384 rad · pusher 328.4 px

Actions used



B Reacher · the folded pose is reached

Episode 2031004 · same goal/support · 50-action allowance

ShiftWM (ours)

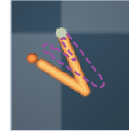
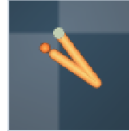
Goal reached

t = 23 / 50

Goal image

Matched time: t=20

Actual end: t=23

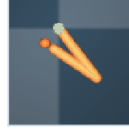


Joint-angle distance 0.048 rad

Goal image

Matched time: t=20

Actual end: t=50



Framewise

Goal missed

t = 50 / 50

Joint-angle distance 0.571 rad

Actions used



--- Goal silhouette from input pixels · observed development episodes · positive subset

Figure 6: **Positive executions made visible: reaching the requested configuration.** A: ShiftWM (ours) reaches the PushT block–pusher target at call 21; Framewise continues to call 50 and moves most of the block outside the view. The recorded terminal block/pusher errors are 6.9/14.7 versus 317.9/328.4 native pixels. B: ShiftWM attains the folded Reacher pose at call 23; Framewise ends with a different link orientation at call 50. Dashed purple outlines are extracted from the common goal pixels for display only; they are not predictions or scoring inputs. PushT’s gray block and blue pusher define the requested configuration; the green T is an unscored persistent renderer marker. The rulers count native actions within the common allowance, including ten support actions. Both methods face the same goal and recorded initial support.

C Reacher · a close-looking pose can still miss

Episode 2031008 · same goal/support · 50-action allowance

ShiftWM (ours)

Goal reached

$t = 21 / 50$

Joint-angle distance 0.026 rad

Framewise

Goal missed

$t = 50 / 50$

Joint-angle distance 0.213 rad

Actions used

0 10: support ends 50: limit

D Reacher · the forearm matches the goal

Episode 2031011 · same goal/support · 50-action allowance

ShiftWM (ours)

Goal reached

$t = 23 / 50$

Joint-angle distance 0.060 rad

Framewise

Goal missed

$t = 50 / 50$

Joint-angle distance 0.365 rad

Actions used

0 10: support ends 50: limit

--- Goal silhouette from input pixels · observed development episodes · positive subset

Figure 7: **Two further positive Reacher cases: goal-pose agreement matters.** C: Framewise’s final arm looks close to the goal, but retains a scored joint mismatch; the terminal joint-angle distances are 0.026 radians for ShiftWM (ours) and 0.213 for Framewise. D: the goal outline exposes the residual forearm-orientation mismatch, with distances 0.060 and 0.365 radians. ShiftWM succeeds at calls 21 and 23; Framewise fails after 50 in both cases. Success requires each unwrapped joint error below 0.05 radians, not an L2 distance below 0.05; case D therefore legitimately succeeds with $L2 = 0.060$. Both methods’ middle frames are recorded at call 20; endpoint frames retain their respective stopping times. Together with Figure 6, these exhaust the four positive discordant cases against Framewise in the saved development comparison; Shared context also fails on all four tasks.

F.2 FROM THE INFERENCE PATH TO MEASURED EXECUTION

Figure 8 isolates the implementation difference. Framewise calibration still uses a temporal, action-conditioned predictor; its image residual is applied independently to each frame. ShiftWM instead infers an observation context from support statistics, calibrates history and goal with that shared context, and infers a dynamics context from corrected transitions and executed actions. The second context modulates action embeddings. Contexts change after observed action blocks but remain fixed inside a CEM search; model weights remain fixed during deployment.

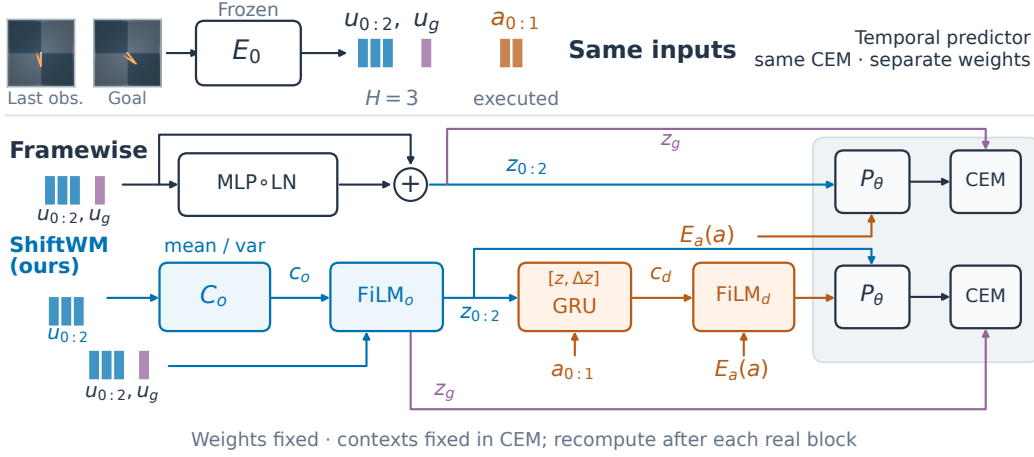


Figure 8: **What changes in the computation.** Both methods use the same frozen encoder, predictor architecture, history/action interface and CEM, with separately trained predictor weights. Framewise applies a per-image residual MLP; ShiftWM (ours) adds observation and dynamics contexts. The goal is calibrated but never enters context inference. GRU inputs are $[z_i, z_{i+1} - z_i, \tilde{a}_i]$; $E_a(a)$ embeds executed past and candidate future actions. This code-derived schematic describes operations and does not establish component causality.

Figures 9 and 10 examine the first positive discordant task in each environment under the previously declared lexicographic rule. We replay saved actions for *all thirteen* discordant tasks across three methods, including the nine Framewise-only successes. The diagnostic requires exact agreement with every archived goal and observed frame, as well as support/terminal errors and stopping behavior. Intermediate physical states are reconstructed by this verified replay; they were not policy inputs or newly selected control actions.

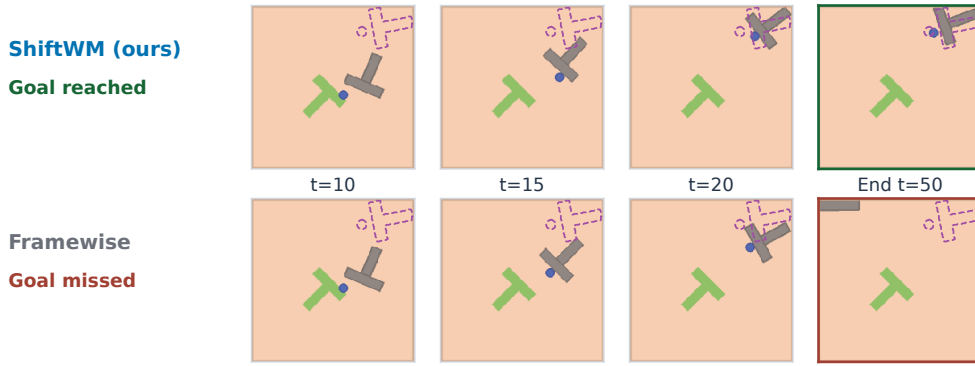
Panel B separates the two conditions actually required for success. Its green regions represent the original task tolerances, and trajectories end at their real stopping times. Panel C is a distinct, conditional prediction test: both trained checkpoints receive the same three observed frames, two past action blocks, and next executed five-action block. Error is measured against the same canonical next-observation feature from the frozen encoder. Canonical images are diagnostic targets only; they do not enter context inference. Partial terminal blocks have no full-block forecast measurement. After the common start, the two behavior policies visit different states, so predictions are compared within each observed transition. Panel C uses logarithmic MSE axes. Percentage labels show relative change against Framewise: green $\downarrow$ is lower error and red $\uparrow$ is higher error. Missing terminal measurements indicate an incomplete five-action block, never zero error. These diagnostics do not recover CEM candidate rankings or identify which component caused a successful action sequence.

Across the four positive cases, ShiftWM has lower next-observation prediction MSE on 6 of 20 matched transitions and lower observed-history calibration MSE on 0 of 20. These outcome-selected, temporally dependent transitions are descriptive counts, not independent benchmark wins or significance estimates. Success therefore cannot be equated with improvement in every intermediate model measurement.

PushT · where the successful execution differs

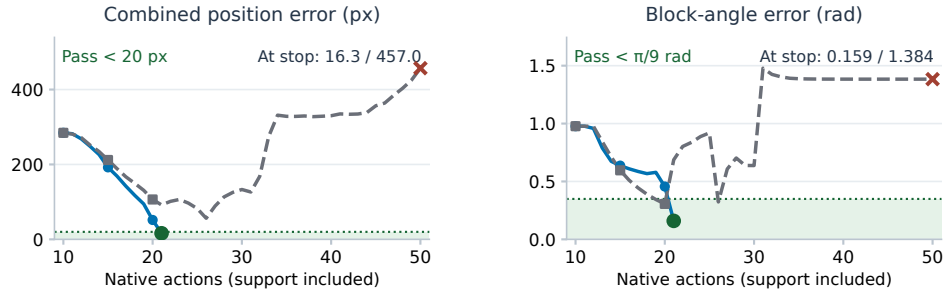
Development episode 2031024 · matched goal/support · positive case

A Observe the decisions along the executed path



Dashed outline: common goal · green T is unscored

B Both goal conditions must pass at the same time



C Same observed transition, two prediction models

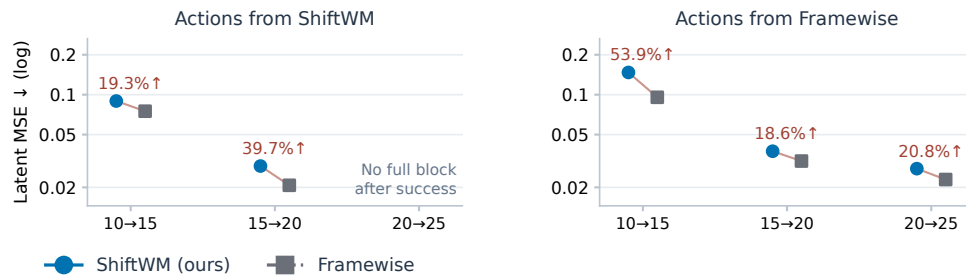


Figure 9: **PushT: success does not require winning every diagnostic.** Episode 2031024. A: real observations and a display-only goal outline. The green T is an unscored renderer marker; dashed contours show the task goal. B: at call 20, ShiftWM (ours) has approximately half the remaining combined position error, while Framewise has the smaller angle error. ShiftWM meets *both* tolerances at call 21; Framewise moves away and fails at call 50. C: each pair tests two models on the same realized transition, grouped by the behavior policy. ShiftWM has higher one-block prediction error on all five pairs despite its successful execution. The conditional diagnostic therefore rejects a uniform prediction-improvement explanation for this case.

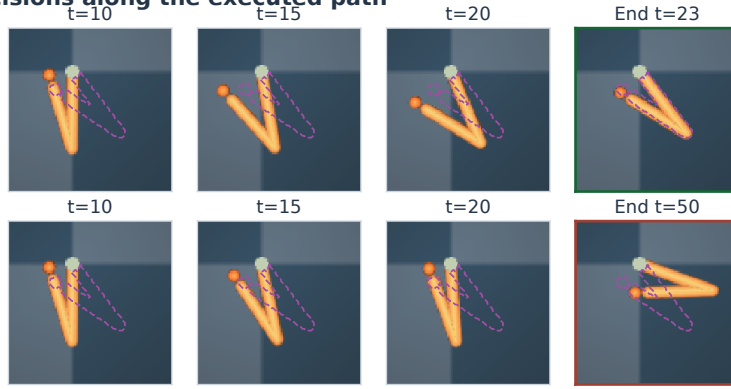
Reacher · where the successful execution differs

Development episode 2031004 · matched goal/support · positive case

A Observe the decisions along the executed path

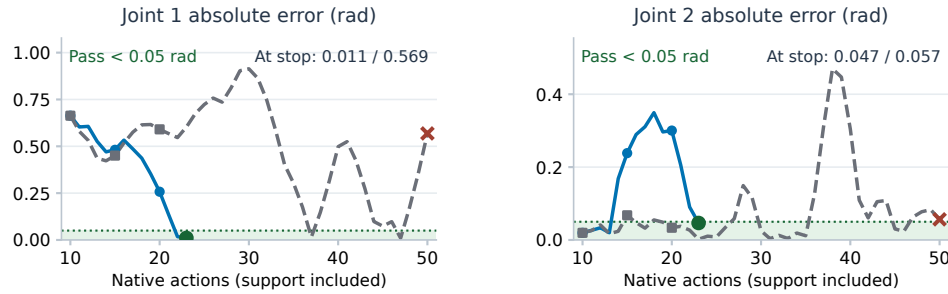
ShiftWM (ours)

Goal reached



Dashed outline: common goal · same arm crop in every cell

B Both goal conditions must pass at the same time



C Same observed transition, two prediction models

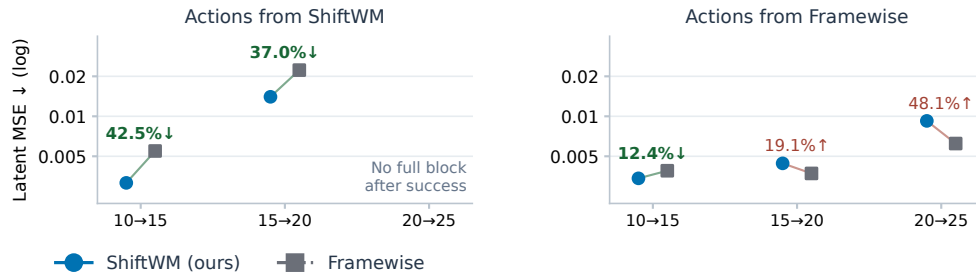
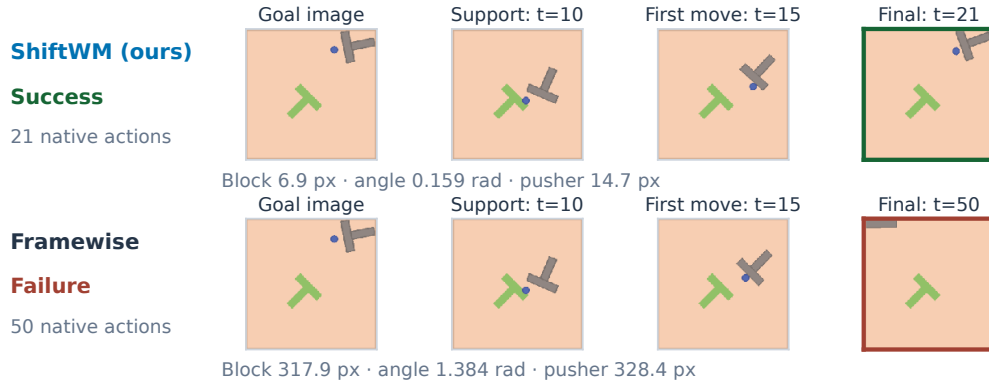


Figure 10: **Reacher: two accurate forecasts accompany coordinated success.** Episode 2031004. A: observed poses use a common crop and goal outline. B: both unwrapped joint errors fall below 0.05 radians at call 23 for ShiftWM (ours); Framewise fails at call 50. A smaller error in one joint at an earlier time does not satisfy the joint criterion. C: ShiftWM’s next-observation MSE is 42.5% and 37.0% lower on the two full transitions along its own execution, but lower on only one of three transitions along Framewise’s execution. These paired measurements show a local predictive advantage, without establishing component causality or average planning gains.

1134 F.3 CONTRASTING SUCCESSES, FAILURES, AND SUPPORT-ONLY OUTCOMES
1135
1136
1137
1138
1139
1140
1141
1142
1143
1144
1145
1146
1147
1148
1149
1150
1151
1152
1153
1154
1155
1156
1157
1158
1159
1160
1161
1162
1163
1164
1165
1166
1167
1168
1169
1170
1171
1172
1173
1174
1175
1176
1177
1178
1179
1180
1181
1182
1183
1184
1185
1186
1187

PushT | ShiftWM succeeds; Framewise fails

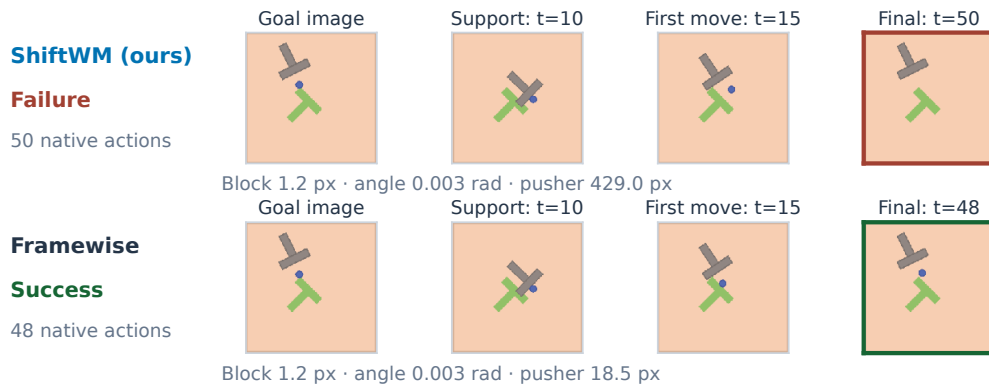
Development · episode 2031024 · training seed 0 · 50-action limit



Observed simulator frames · shared start and goal · no interpolated or predicted images

PushT | Framewise succeeds; ShiftWM fails

Development · episode 2031000 · training seed 0 · 50-action limit

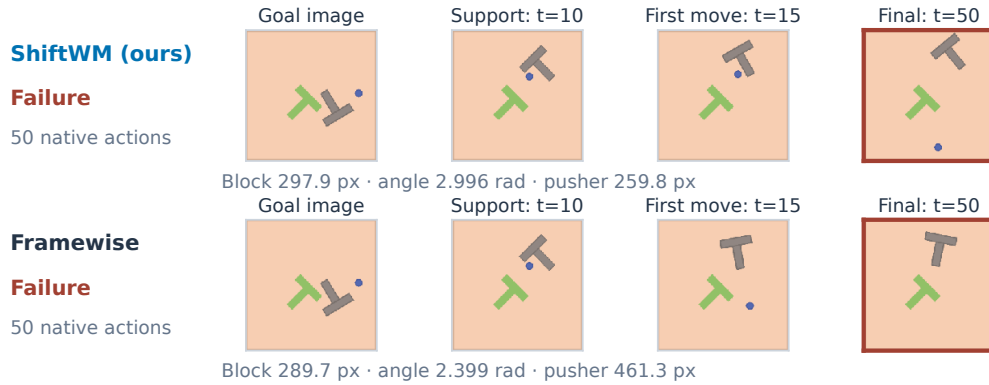


Observed simulator frames · shared start and goal · no interpolated or predicted images

Figure 11: PushT: a success and a contrasting failure. Top, episode 2031024: ShiftWM (ours) reaches the scored goal at native call 21; Framewise exhausts 50 calls. Bottom, episode 2031000: Framewise succeeds at call 48, while ShiftWM fails despite placing the block within 1.2 simulator pixels of its target. Its pusher remains 429.0 pixels away, illustrating why block placement alone does not satisfy this task. The goal is the gray block and blue pusher configuration in the goal image. The persistent green T is the upstream renderer’s reset marker, not this episode’s scored goal. Position annotations use the native 512×512 coordinate system, not the resized 224×224 image. Success requires the combined pusher/block position-error norm below 20 pixels and block-angle error below $\pi/9$. These are matched development examples, not evidence of uniform superiority.

PushT | Neither reaches the scored goal

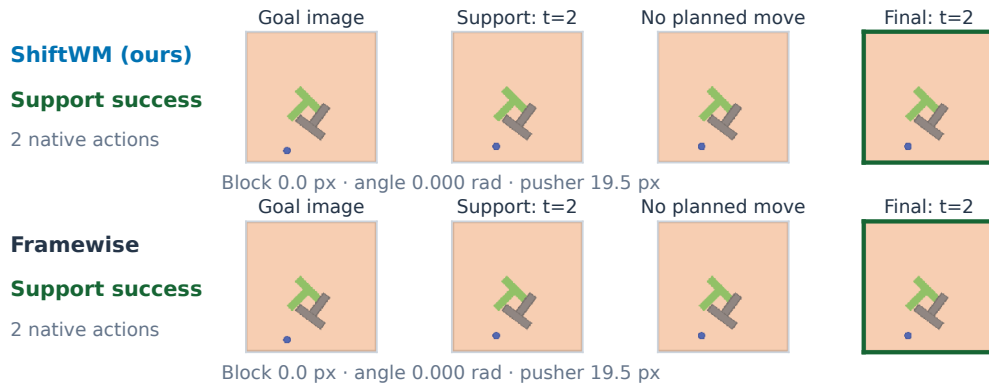
Development · episode 2031001 · training seed 0 · 50-action limit



Observed simulator frames · shared start and goal · no interpolated or predicted images

PushT | Success during shared support

Development · episode 2031026 · training seed 0 · 50-action limit



Observed simulator frames · shared start and goal · no interpolated or predicted images

Figure 12: **PushT: shared failure and success before planning.** Top, episode 2031001: neither method reaches the scored goal by call 50; terminal block and pusher errors remain large. Bottom, episode 2031026: the common support actions already satisfy the goal at call 2. No CEM plan is executed, and the repeated image is the same saved terminal observation. This task contributes to raw success but is excluded from eligible policy success; it provides no evidence of either planner’s competence. The gray block and blue pusher in the goal image define the requested configuration; the green marker remains unscored renderer context. Pixel errors use native simulator coordinates. Showing this case makes the distinction between support success and successful closed-loop control visible.

Reacher | ShiftWM succeeds; Framewise fails

Development · episode 2031004 · training seed 0 · 50-action limit

ShiftWM (ours)

Success

23 native actions



Joint-angle distance 0.048 rad

Framewise

Failure

50 native actions



Joint-angle distance 0.571 rad

Observed simulator frames · shared start and goal · no interpolated or predicted images

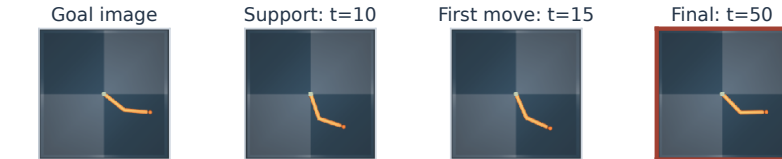
Reacher | Framewise succeeds; ShiftWM fails

Development · episode 2031001 · training seed 0 · 50-action limit

ShiftWM (ours)

Failure

50 native actions



Joint-angle distance 0.260 rad

Framewise

Success

41 native actions



Joint-angle distance 0.016 rad

Observed simulator frames · shared start and goal · no interpolated or predicted images

Figure 13: **Reacher: opposite outcomes under the same protocol.** Top, episode 2031004: ShiftWM (ours) reaches the goal joint configuration at call 23, with terminal joint-angle distance 0.048 radians; Framewise ends at 0.571 radians after 50 calls. Bottom, episode 2031001: Framewise succeeds at call 41 with distance 0.016 radians, while ShiftWM ends at 0.260 radians and fails. The target is the arm configuration shown in the available goal image. Success tests each unwrapped joint’s absolute error against 0.05 radians; the displayed joint-angle L2 distance is an endpoint diagnostic, not a 0.05 radian success threshold. Matching task, initial support, goal, and search budget exposes both improvement and regression without changing examples between methods.

Reacher | Neither reaches the scored goal

Development · episode 2031002 · training seed 0 · 50-action limit

ShiftWM (ours)

Failure

50 native actions



Joint-angle distance 0.316 rad

Framewise

Failure

50 native actions



Joint-angle distance 0.119 rad

Observed simulator frames · shared start and goal · no interpolated or predicted images

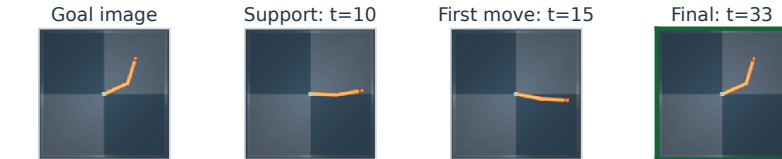
Reacher | Both reach the scored goal

Development · episode 2031000 · training seed 0 · 50-action limit

ShiftWM (ours)

Success

33 native actions

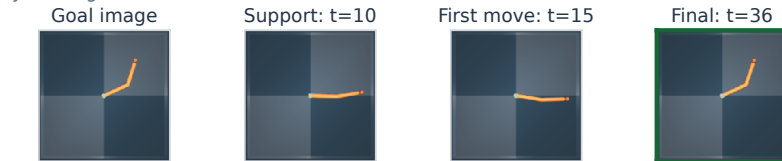


Joint-angle distance 0.023 rad

Framewise

Success

36 native actions



Joint-angle distance 0.030 rad

Observed simulator frames · shared start and goal · no interpolated or predicted images

Figure 14: **Reacher: joint failure and joint success.** Top, episode 2031002: both methods use all 50 calls and fail, with terminal joint-angle distances of 0.316 radians for ShiftWM (ours) and 0.119 radians for Framewise. A smaller diagnostic distance does not itself establish task success. Bottom, episode 2031000: both succeed, at calls 33 and 36 respectively, with terminal distances of 0.023 and 0.030 radians. These are the first tasks in the declared strata, not optimized illustrations of a method advantage. Annotations show actual stopping times; the common 50-call allowance does not mean both methods execute 50 actions. The full gallery also includes Shared context on these same tasks and retains every failure and support-only success.

G MECHANISM TESTS AND ADDITIONAL BASELINES

The remaining tests distinguish useful context information from a constant vector, and observation correction from predictor fine-tuning alone. Consistency penalties have a scale ambiguity: context magnitudes can shrink while downstream sensitivity grows. Small penalties or context variance do not establish factor separation. Interventions should measure adapter outputs, prediction changes, and planning with the same available history. Constant-context, context-shuffle, removed-pathway, and support-length controls remain incomplete. Within-episode switching is a secondary study.

Observation-only and dynamics-only *context* ablations still train the shared predictor; they are not encoder-only or dynamics-only gradient adaptation. The official AdaJEPA reproduction is complete under its separate released protocol (Appendix N.3); an input- and budget-matched comparison on our tasks remains unfinished. Matched online gradient updates and oracle contexts also remain planned controls. Their supervision, optimization, data-generation cost, parameter counts, and measured time must accompany performance.

Table 5: **Planned mechanism and resource analysis.** These targeted comparisons are incomplete; dashes are not measured zeros. Completed core forecast comparisons appear separately in Figure 4. Each future row must specify its split, support length, and measured resources.

Variant	Success $\uparrow$	Error $\downarrow$	ms/replan	Trainable M
ShiftWM (ours)	—	—	—	—
Without observation context	—	—	—	—
Without dynamics context	—	—	—	—
Unpaired contexts	—	—	—	—
Shuffled dynamics support	—	—	—	—
Shared context, matched size	—	—	—	—
Joint gradient adaptation	—	—	—	—

H DEVELOPMENT GOAL-CALIBRATION DIAGNOSTIC

After completing the prespecified development comparison, we inspect goal calibration separately from future prediction. Each diagnostic uses the same goal, support history, and support-success exclusion as planning. The corrected shifted goal and the five-block recorded-action prediction are compared with the same immutable canonical goal embedding. Recorded future actions and canonical renders are evaluator-only diagnostic inputs; they are not available to the deployed controller. There are 31 eligible PushT cases and 30 eligible Reacher cases, selected from all 32 development tasks per environment using the common support-success mask. Fresh GPU Reacher support and goal images exactly match their stored canonical counterparts in every eligible case.

For ShiftWM on Reacher, recorded-action prediction MSE is approximately 0.012 while corrected-goal MSE is 0.797. Framewise calibration yields approximately 0.009 and 0.271, respectively. This is evidence of an error mismatch, not proof that goal calibration causes the planning deficit. A separate development intervention substitutes the learned framewise goal correction while holding the ShiftWM predictor, history correction, context inference and planner fixed. This intervention is separate from the primary method comparison.

H.1 AN ELEMENTARY COST-ERROR DECOMPOSITION

This calculation explains why prediction and goal calibration require separate checks; it is not a theoretical novelty or an empirical guarantee. For a nonempty finite set of candidate action sequences $\mathcal{A}$ containing the returned sequence, let $y(a)$ and g be canonical terminal and goal latents, and let $\hat{y}(a), \hat{g}$ be their model estimates. Define $\|u\|_d = \|u\|_2/\sqrt{d}$, so squared normalized norm equals the implemented MSE. Suppose, uniformly over $\mathcal{A}$,

$$\|\hat{y}(a) - y(a)\|_d \leq e_f, \quad \|\hat{g} - g\|_d \leq e_g, \quad \|y(a) - g\|_d \leq R.$$

Writing $c(a) = \|y(a) - g\|_d^2$, $\hat{c}(a) = \|\hat{y}(a) - \hat{g}\|_d^2$ and $\varepsilon = e_f + e_g$, expansion of the squared norm gives

$$|\hat{c}(a) - c(a)| \leq 2R\varepsilon + \varepsilon^2 =: B.$$

If a returned action sequence satisfies $\hat{c}(\hat{a}) \leq \min_{a \in \mathcal{A}} \hat{c}(a) + \eta$, then

$$c(\hat{a}) - \min_{a \in \mathcal{A}} c(a) \leq 2B + \eta = 4R\varepsilon + 2\varepsilon^2 + \eta.$$

Indeed, for a true minimizer a^* , apply the cost-error bound twice: $c(\hat{a}) \leq \hat{c}(\hat{a}) + B \leq \hat{c}(a^*) + \eta + B \leq c(a^*) + 2B + \eta$. An illustrative scalar counterexample has exact predictions $y(a) = 0, y(b) = 1$, true goal $g = 0$ and estimated goal $\hat{g} = 1$: goal error alone reverses the two candidates' ranking. This is a mathematical example, not a measured simulator result. Empirical mean prediction MSE does not provide the assumed uniform norm bound; CEM does not certify η . The statement concerns latent cost on the stated candidate set, not physical success or causal attribution for the observed failures.

Development diagnostic · mean and 95% bootstrap interval

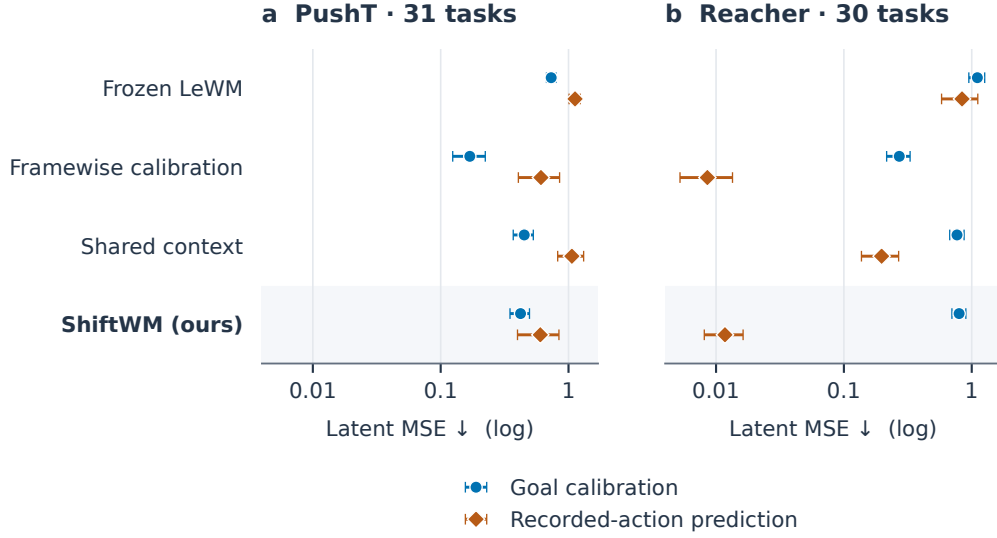


Figure 15: Prediction and goal calibration are distinct errors. Posthoc development diagnostics for frozen LeWM and three seed-0 trained models. Both errors are per-dimension MSE in pre-trained coordinates against the same canonical goal; lower is better. Blue circles denote goal calibration; orange diamonds use the five recorded future action blocks. Horizontal axes are logarithmic so small errors remain visible. Error bars are 95% trajectory-seed bootstrap intervals, excluding common support-only successes; they do not measure training-seed variation. Small recorded-action prediction error alone does not establish accurate goal calibration or successful planning.

Environment	Method	Raw	Support	Eligible
PushT	ShiftWM (ours)	2/32	1/32	1/31
PushT	Shared context	2/32	1/32	1/31
PushT	Framewise calibration	2/32	1/32	1/31
PushT	Goal-only hybrid (post hoc)	1/32	1/32	0/31
Reacher	ShiftWM (ours)	8/32	2/32	6/30
Reacher	Shared context	8/32	2/32	6/30
Reacher	Framewise calibration	13/32	2/32	11/30
Reacher	Goal-only hybrid (post hoc)	14/32	2/32	12/30

Table 6: Exploratory development results at training seed 0. The post-hoc hybrid changes only the fixed ShiftWM model’s learned goal calibration. Raw success includes support-only successes; eligible success excludes them. All methods use the same 32 development tasks and 300/30/30 CEM budget. Pending runs are not zero-valued outcomes.

Environment	Paired comparison	Wins/losses	Δ (pp)	95% interval (pp)
PushT	Hybrid — ShiftWM (ours)	0/1	-3.2	[-9.7, +0.0]
PushT	Hybrid — Framewise calibration	0/1	-3.2	[-9.7, +0.0]
Reacher	Hybrid — ShiftWM (ours)	9/3	+20.0	[+0.0, +40.0]
Reacher	Hybrid — Framewise calibration	7/6	+3.3	[-20.0, +26.7]

Table 7: Paired eligible-success differences for the post-hoc hybrid. Green bold means indicate positive differences against the named comparator, not statistical significance. Wins/losses count discordant task outcomes. Intervals resample paired development trajectories (20,000 draws, seed 1701); they do not measure variation across training seeds and are exploratory, not confirmatory tests.

I RELEASED-EVALUATOR IMPLEMENTATION CHECK

The unchanged released LeWM evaluation entry point, released checkpoints, and released datasets yield 46/50 PushT successes and 39/50 Reacher successes on the official seed-42 task selections. These runs use a separately installed, API-compatible May-2026 `stable-worldmodel` snapshot. The exact authors' training runtime was not specified and is not claimed to have been recovered. Both executions completed, and configuration, task identities, checkpoint hashes and all 50 Boolean outcomes were verified from the run records. This check supports the functioning of the released planning pathway on its released data. It does not test equivalence of that pathway to our model wrapper on identical episodes, reproduce a published aggregate score or establish equivalence with our current simulator version. Its expert-data PushT tasks and original 25-step replanning interval differ from the new intervention benchmark and its paid 10-step support phase. These scores are therefore kept separate from every adaptation comparison.

J FULL CONDITION AND DIAGNOSTIC RESULTS

The aligned methods share corrected history/goal coordinates. The *Unaligned predictor (diagnostic)* instead predicts canonical targets while shifted goals remain uncorrected. This mismatch makes it an unsuitable primary comparator for a factorization claim, although its forecast errors remain informative. Its alignment term is constant with no gradient; training curves compare prediction loss rather than different total losses. Canonical and all-condition averages retain nominal regressions that a held-out-only result could conceal. The all-condition average contains familiar factor pairs and is not a wholly unseen-composition score.

Method	Planning success (%) $\uparrow$				Forecast $\downarrow$
	Canonical	Held-out	Eligible	Extrap.	MSE@5
PushT					
Frozen LeWM	29.7	4.7	3.2	7.3	1.1854
Framewise calibration	10.9 ± 1.6	3.6 ± 0.9	2.1 ± 0.9	5.0 ± 0.8	0.5713 ± 0.0155
Shared context	14.6 ± 3.3	2.1 ± 0.9	0.5 ± 0.9	7.6 ± 1.1	0.7216 ± 0.0287
Unpaired contexts	16.1 ± 5.5	2.6 ± 0.9	1.1 ± 0.9	5.2 ± 1.4	0.5633 ± 0.0083
ShiftWM (ours)	16.7 ± 0.9	2.6 ± 0.9	1.1 ± 0.9	6.8 ± 0.5	0.5503 ± 0.0099
Unaligned predictor (diagnostic)	13.0 ± 3.9	2.1 ± 0.9	0.5 ± 0.9	6.1 ± 1.1	0.5168 ± 0.0067
Reacher					
Frozen LeWM	48.4	35.9	35.9	17.7	0.0591
Framewise calibration	42.7 ± 4.8	42.7 ± 3.3	42.7 ± 3.3	21.2 ± 5.9	0.0092 ± 0.0001
Shared context	40.1 ± 3.9	36.5 ± 3.9	36.5 ± 3.9	16.8 ± 3.3	0.0250 ± 0.0002
Unpaired contexts	38.0 ± 12.7	34.9 ± 9.0	34.9 ± 9.0	20.8 ± 2.6	0.0075 ± 0.0002
ShiftWM (ours)	43.8 ± 4.7	40.6 ± 3.1	40.6 ± 3.1	19.6 ± 1.3	0.0068 ± 0.0001
Unaligned predictor (diagnostic)	44.8 ± 4.5	41.7 ± 3.9	41.7 ± 3.9	20.0 ± 2.3	0.0072 ± 0.0000

Table 8: **Complete metric ledger.** Trained entries require three completed seeds and show mean $\pm$ sample SD; Frozen LeWM has one checkpoint. Dashes denote incomplete evidence. Canonical is (v_0, p_0) ; held-out and forecast MSE@5 use (v_2, p_2) ; extrapolation averages three prespecified conditions. Eligible success excludes held-out tasks solved during paid support acquisition; other success columns include them. MSE@5 is five-step latent prediction error in fixed reference coordinates. The unaligned predictor lacks a learned visual calibration path and is a diagnostic, not evidence for factorization.

J.1 RANDOM ACTIONS AND PRIVILEGED REPLAY

These controls retain the same task identities, goal definition, paid support, and native interaction budget. Random actions sample uniformly within the native bounds using a single fixed RNG rule; they are not estimates over three trained checkpoints or multiple independent policy seeds. ShiftWM’s three-seed raw PushT success means are 2.60% on held-out composition and 6.77% on extrapolation, compared with random-action point estimates of 7.81% and 9.38%. On Reacher the corresponding ShiftWM means are 40.63% and 19.62%, compared with 4.69% for the random control on each population. These are descriptive comparisons; no significance or policy-seed uncertainty claim is inferred from their ordering.

Privileged replay uses the recorded future actions that generated each target. Its complete success verifies reachability under this replay protocol, not fair learned-policy performance or correctness of every learned planning operation. Neither control selects actions from world-model predictions, so CEM settings in their common evaluator metadata do not describe a search performed by those policies. Both raw and support-excluded eligible counts are retained in Table 9.

Environment	Population	Control	Raw (%)	Support	Eligible (%)
PushT	Held-out	Random actions	7.81 [5/64]	1/64	6.35 [4/63]
PushT	Held-out	Privileged replay	100.00 [64/64]	1/64	100.00 [63/63]
PushT	Extrap.	Random actions	9.38 [18/192]	5/192	6.95 [13/187]
PushT	Extrap.	Privileged replay	100.00 [192/192]	5/192	100.00 [187/187]
Reacher	Held-out	Random actions	4.69 [3/64]	0/64	4.69 [3/64]
Reacher	Held-out	Privileged replay	100.00 [64/64]	0/64	100.00 [64/64]
Reacher	Extrap.	Random actions	4.69 [9/192]	0/192	4.69 [9/192]
Reacher	Extrap.	Privileged replay	100.00 [192/192]	0/192	100.00 [192/192]

Table 9: **Random-action and privileged-replay controls.** Raw and eligible columns show success percentages with successes/tasks in brackets; support counts are solved during paid history acquisition. Held-out is (v_2, p_2) (64 tasks); extrapolation pools the three prespecified conditions (192 tasks, 64 initial-state seeds). The task identities, goal index, support-success masks and recorded post-support diagnostics match the frozen-model reference. All policies share a 50-step native budget, including up to 10 support steps. Random actions use one fixed RNG rule (1701 plus initial-state seed), reused across appearances; these are not three training-seed estimates. Privileged replay uses recorded future actions and checks goal feasibility, not fair learned-policy inference. Neither control selects actions with model predictions, so CEM search metadata is unused. No independent-policy-seed uncertainty or significance claim is made.

K PAIRED PLANNING COMPARISONS

The paired reporter requires all three training seeds for both methods before releasing a contrast. It verifies the complete manifest-derived task population, source and protocol identities, support-success masks, and recorded initial distances across methods and training seeds. Initial-state clusters retain all their conditions and model outcomes during resampling. These intervals are conditional on the three observed training seeds; training-seed variability is reported separately. The main records do not contain support/goal pixel hashes or full simulator-state hashes, so those equalities are not retrospectively claimed. The reporting protocol is exploratory and does not select checkpoints or methods from test scores.

Environment	Population	Comparator	Raw Δ (pp)	Eligible Δ (pp)
PushT	Held-out	Framewise calibration	-1.0 $\pm$ 0.9	-1.1 $\pm$ 0.9
			[-3.6, +1.6]	[-3.7, +1.6]
			+0.5 $\pm$ 0.9	+0.5 $\pm$ 0.9
PushT	Held-out	Shared context	[-1.0, +2.1]	[-1.1, +2.6]
			+0.0 $\pm$ 0.0	+0.0 $\pm$ 0.0
		Unpaired contexts	[-2.1, +2.1]	[-2.1, +2.1]
PushT	Held-out		+1.7 $\pm$ 0.3	+1.8 $\pm$ 0.3
PushT	Extrap.	Framewise calibration	[+0.0, +3.6]	[+0.0, +3.8]
			-0.9 $\pm$ 1.1	-0.9 $\pm$ 1.1
		Shared context	[-3.0, +1.2]	[-3.0, +1.1]
PushT	Extrap.		+1.6 $\pm$ 0.9	+1.6 $\pm$ 0.9
		Unpaired contexts	[-0.5, +3.6]	[-0.5, +3.8]
			-2.1 $\pm$ 0.9	-2.1 $\pm$ 0.9
Reacher	Held-out	Framewise calibration	[-11.5, +7.3]	[-11.5, +7.3]
			+4.2 $\pm$ 5.9	+4.2 $\pm$ 5.9
		Shared context	[-4.2, +12.5]	[-4.2, +12.5]
Reacher	Held-out		+5.7 $\pm$ 6.5	+5.7 $\pm$ 6.5
		Unpaired contexts	[-2.6, +14.1]	[-2.6, +14.1]
			-1.6 $\pm$ 5.0	-1.6 $\pm$ 5.0
Reacher	Extrap.	Framewise calibration	[-6.2, +3.1]	[-6.2, +3.1]
			+2.8 $\pm$ 4.4	+2.8 $\pm$ 4.4
		Shared context	[-1.9, +7.6]	[-1.9, +7.6]
Reacher	Extrap.		-1.2 $\pm$ 3.3	-1.2 $\pm$ 3.3
		Unpaired contexts	[-5.7, +3.3]	[-5.7, +3.3]

Table 10: Exploratory paired planning contrasts: ShiftWM (ours) minus comparator, in percentage points. Entries are mean $\pm$ sample SD across the three fixed training seeds, with conditional 95% initial-state-cluster bootstrap intervals on the second line (20,000 draws). Green bold means indicate positive differences against the named comparator, not statistical significance. Each resampled cluster retains all conditions and all three training seeds. Eligible success excludes tasks solved during paid support acquisition. Dashes require further completed sources; no partial-seed contrasts are shown. Intervals do not capture population-level training uncertainty, are not multiplicity-adjusted, and a degenerate interval does not demonstrate equivalence.

Does better prediction translate into better planning? ShiftWM (ours) minus each matched control

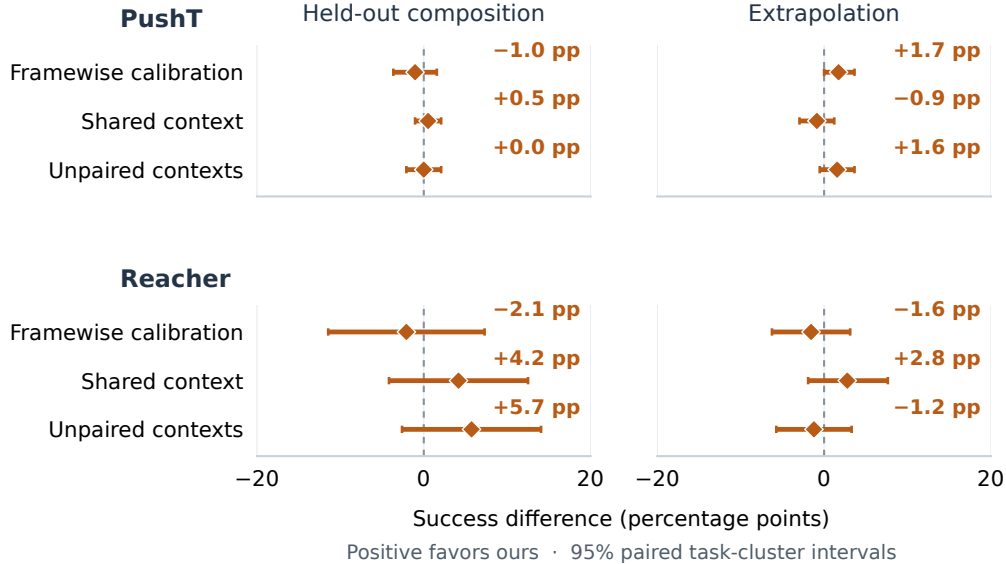


Figure 16: **Paired planning differences.** Points show mean paired raw success differences across three training seeds; positive values favor ShiftWM (ours). Bars are 95% initial-state-cluster bootstrap intervals conditional on those trained models, not training-seed standard deviations or uncertainty over the training population. Pending comparisons have no point estimate. These exploratory intervals are not multiplicity-adjusted.

L DEVELOPMENT REVISION WITH FIXED VISUAL CALIBRATION

The completed goal-only intervention motivates a separate, development-only question: whether a trained dynamics residual can improve an otherwise frozen framewise world model. This revision freezes the completed framewise visual calibrator, goal pathway, action encoder, and predictor, including normalization statistics and dropout mode. It adds a transition-context network and a zero-initialized action-embedding FiLM residual. Before optimization, the revision reproduces the donor’s predictions; only the new 218,272 parameters receive gradients. It reuses the original training/validation data, query indexing, action normalization, and 30-epoch optimizer schedule, selecting by validation prediction error. No main-test outcomes select this revision.

These extra optimization steps make this an exploratory extension, not a matched-training-budget comparison. An improvement over the frozen donor would not by itself establish that temporal conditioning is necessary: a separately trained context-free residual remains an important control. The table keeps uncompleted training or development planning missing and never merges the revision with the original three-seed study.

Environment	Model	Raw	Support	Eligible	Wins/losses
PushT	Framewise calibration	2/32	1/32	1/31	—
PushT	+ Dynamics residual (ours)	2/32	1/32	1/31	1/1
Reacher	Framewise calibration	13/32	2/32	11/30	—
Reacher	+ Dynamics residual (ours)	10/32	2/32	8/30	4/7

Table 11: **Post-hoc dynamics-residual development experiment.** Starting from the completed seed-0 framewise model, the visual calibrator and predictor stay frozen while a new dynamics-context/action-embedding residual receives 30 additional training epochs. Both policies use the same 32 development tasks and common support, with 300 CEM candidates, 30 iterations and 30 elites. Raw success includes support-only success; eligible success excludes it. Wins/losses count eligible tasks solved only by the revision/control. Dashes for the control denote no self-comparison; pending cells mean incomplete evidence. This uses extra optimization, not matched training budgets, and does not establish context necessity, physical identification, or a main-test result.

M DEVELOPMENT STUDY OF ROLLOUT OBJECTIVE AND CONTEXT DEPENDENCE

The completed dynamics-residual development experiment does not improve its frozen Framewise donor: eligible success is 1/31 for both models on PushT and 8/30 versus 11/30 on Reacher. We therefore record a new, post-hoc development protocol rather than promote that revision to the main test. All eight arms have completed training and development planning in separate environment campaigns. A 2×2 design crosses teacher-forced versus recursive prediction with inferred versus constant-input dynamics context. Each of the four variants per environment trains for 30 epochs at seed 0, freezing the same completed Framewise calibrator, goal pathway, action encoder and predictor. Both objectives supervise target frames 3–7 (indices 3 : 8), including the first prediction after the three-frame support. Teacher forcing uses available observed history for each target; recursive training feeds back predicted latents after support. Context uses support only. The constant-input control retains the identical GRU/FiLM architecture but receives all-zero support and executed-action inputs: its learned response can be nonzero, and matching nominal parameter counts does not establish equal effective capacity. All variants reuse the existing train/validation data and optimizer schedule; validation-selected checkpoints minimize the same FP32 recursive MSE averaged over all five query frames. Development planning uses the existing 32 matched tasks per environment, 300 CEM candidates, 30 iterations, 30 elites and the fixed 50-native-step budget including support. All eight prespecified arms are retained, and only validation error selects checkpoints within this follow-up protocol. Table 12 reports source-validated status and distinguishes incomplete training, trained checkpoints and completed planning; missing measurements are never substituted with zeros or partial planning outcomes. The completed teacher-forced arms give mixed evidence about context dependence. Inferred context ties the frozen donor on PushT (1/31 eligible successes) and Reacher (11/30). Constant input scores 0/31 on PushT and 15/30 on Reacher, compared with donor scores of

1/31 and 11/30, respectively. The latter is a favorable development point estimate for the constant-input control, not evidence that inferred context is necessary. The completed recursive/inferred arms tie the donor on PushT (1/31) and score 13/30 versus 11/30 on Reacher (a +6.67 percentage-point development difference). Recursive constant context scores 0/31 on PushT and 13/30 on Reacher. Thus, inferred context exceeds its constant-input control on PushT but ties the donor; on Reacher, recursive inferred and constant contexts tie. Neither objective satisfies the registered screening requirement to exceed both its constant-input control and donor in both environments. Lower recursive validation error does not establish better control, and these results do not establish a benefit from episode-dependent dynamics inference. Repeated training seeds and an untouched evaluation remain necessary to assess any subsequent method. A revised context method is not promoted without evidence of learning beyond its matched constant-input control; multiple training seeds and a fresh final holdout unused during revision development are required before a broader performance claim. These experiments remain separate from the original three-seed campaign and its negative outcomes.

Environment	Objective	Context	Status	Val. MSE	Eligible	Δ (pp)
PushT	Teacher-forced	Inferred	Complete	0.28759	1/31	+0.00
PushT	Teacher-forced	Constant	Complete	0.28791	0/31	-3.23
PushT	Recursive	Inferred	Complete	0.26388	1/31	+0.00
PushT	Recursive	Constant	Complete	0.26876	0/31	-3.23
Reacher	Teacher-forced	Inferred	Complete	0.00566	11/30	+0.00
Reacher	Teacher-forced	Constant	Complete	0.00565	15/30	+13.33
Reacher	Recursive	Inferred	Complete	0.00521	13/30	+6.67
Reacher	Recursive	Constant	Complete	0.00525	13/30	+6.67

Table 12: **Development objective/context study: protocol and completion status.** Eight seed-zero arms train for 30 extra epochs with a frozen Framewise donor. Planned means completed training is unavailable; trained means validated training is complete but planning is incomplete; complete requires all 32 matched development tasks. Validation MSE is the completed-history minimum of the common FP32 recursive criterion over targets 3–7, not a test score. Eligible success excludes support-only successes; Δ compares the same eligible tasks against the frozen Framewise donor. Bold green marks strictly positive observed differences, not statistical significance. Missing values are unavailable, never zero. Constant-input contexts can learn nonzero responses; nominal parameter matching does not establish equal effective capacity. Both objectives use the first target after support. Planning uses 300 CEM candidates, 30 iterations, 30 elites and the fixed native-action budget. This post-hoc development study does not establish a main-test, multi-seed, efficiency or context-necessity claim.

N REGISTERED DOMAIN AND PREDICTOR EXTENSIONS

The original forecasting gains, mixed planning comparisons, and failed rollout-context screen remain unchanged. A separately registered 36-run development campaign has completed training, forecasting and planning on two additional simulations and a second predictor family. A subsequent six-run observation-gain ablation is also complete. These adapted tasks are not reproductions of published drone or surgical leaderboards. Table 13 distinguishes completed development evaluation from final testing, which remains unrun. All positive and negative outcomes below remain separate from the original test results.

Stage	Drone	Tissue manipulation
Recorded trajectories	384 collected	384 collected
Full collection audit	Passed	Passed
Reachable-goal protocol	Audited	112 replay checks passed
Frozen visual feature cache	Complete	Complete
Full 30-epoch training	18/18 complete	18/18 complete
Development forecasting	18/18 complete	18/18 complete
Development planning	18/18 complete	18/18 complete
Gain ablation: train/forecast/plan	6/6 in each stage	Not registered
Final model test evaluation	Pending	Pending

Table 13: **Completed registered extension campaigns.** Each original domain has two predictor families, three methods and three training seeds. The subsequent drone ablation has two methods and three seeds. All models completed 30 epochs and both development evaluations; pending final tests are unavailable, not zero performance.

Data and physical tasks. Each domain records 384 trajectories of 200 native commands, partitioned into 192 training, 48 validation, 48 development and 96 test trajectories. Initial-state seed families are disjoint between splits; within a split, the same seeds are recorded across response gains 1, 0.75 and 1.25. Five successive two-dimensional commands form one model action block. Model inputs contain RGB observations and commanded actions. Simulator state is retained separately for collection verification and scoring. Action normalization uses only the training split. Photometric transforms and the seven training/validation appearance–gain combinations follow the original loader; development uses warm appearance with gain 0.75, while cool appearance with gain 1.25 is withheld from training.

The drone task adapts `gym-pybullet-drones` (Panerati et al., 2021), pinned at revision `7ebad1ecabd2`. A CF2X vehicle receives planar velocity requests through the upstream PID controller at fixed requested altitude, observed by a fixed external RGB camera. The intervention scales commanded response; it is not a wind or mass change. Recorded settled endpoints define goals. Success requires XY distance below 4 cm, speed below 6 cm/s, altitude error below 5 cm, and no crash or workspace escape. The complete collection audit checked all 384 trajectories, payload hashes and 77,184 native states.

The deformable-tissue task adapts LapGym’s SOFA tissue manipulation environment (Scheikl et al., 2023), pinned at revision `85bf7e05dd08`. Two-dimensional gripper commands are scaled before native clipping. Forecast collection and planning-goal verification are separate: planning goals require an independent recorded-action replay with the desired marker moved to the recorded endpoint, preserving the physical trajectory. Native XZ tissue-point distance at most 2 mm defines success. All 384 collection episodes passed the full data audit; independent replays verified all 112 eligible development/test goal references, including physical-state equality and RGB equality outside the relocated target marker. Finite workspace-clipped collection transitions are retained; planning eligibility additionally excludes unstable or invalid reference trajectories before model evaluation. This is a constrained reachable-goal distribution, not clinical validation or a claim of autonomous surgical safety.

Predictor families and matched controls. Both families reuse the frozen released PushT LeWM visual encoder and its canonical target coordinates; neither uses drone or surgical visual pretraining. The first retains the released transformer initialization. The second replaces its predictor with a two-layer GRU of hidden size 256, using the same projected state/action interface. The GRU is an in-house architecture control, not a Dreamer reproduction. Parameter counts and initialization differ across families; comparisons are primarily within each family. The three methods are *Framewise calibration*, *ShiftWM (ours)*, and *Constant dynamics context*. The last retains episode-dependent observation correction but supplies zero inputs to the dynamics-context network. Its learnable biases permit a nonzero common response; equal nominal parameter count does not imply equal effective capacity. Seeds 0, 1 and 2 give 18 runs per domain.

Every run trains for 30 epochs with AdamW, learning rate 10^{-4} , cosine minimum 10^{-6} , weight decay 0.01, gradient clipping 1, batch size 128 and stride 4. Eight-frame windows provide three observed support frames and five recursive query targets, including the first unobserved target. Training

uses BF16 where supported; selection uses the minimum full-validation recursive MSE in float32 (autocast disabled; CUDA TF32 matrix multiplication remains allowed for every arm). Alignment and context consistency weights match across the applicable methods. Source, data, cache, normalization and checkpoint identities are pinned; optimizer and RNG state support continuation. Training completion alone is not a planning result or a validated release.

Development planning and decision rule. Before model outcomes, select the first eight eligible development trajectories in seed order at gain 0.75, or report a shortfall if fewer are available. Initial-goal separation must be at least 8 cm for the drone and 4 mm for tissue manipulation; initially solved tasks are excluded and every exclusion is retained. Policy-induced invalid actions count as failures. All methods share goal images, ten recorded support commands, planner random numbers and a 200-native-command total budget including support. CEM uses horizon five blocks, 128 candidates, five iterations and 16 elites; it replans after at most five executed commands and retains the unused plan as its warm start. Search uses standardized action coordinates, with clipping after conversion to native commands. Report support-only successes separately and log partial final blocks exactly.

Random actions use the same interaction budget; privileged replay of recorded actions checks goal reachability. Replay is not a deployable baseline. We report success counts and rates, paired task and seed uncertainty, native interactions, physical errors, failures and latency. Promotion requires better closed-loop success than both within-family controls with consistent evidence across seeds and domains. Final test evaluation is separate from development, and any changed hypothesis or protocol receives a new run namespace. This campaign cannot turn the original null planning findings into positive evidence by selectively reporting a favorable domain or model.

N.1 COMPLETED DEVELOPMENT COMPARISONS

All 36 conditions have completed training, forecasting and planning. Table 14 reports every method and training seed for both domains. No task succeeds during the ten-command common support. On the drone, transformer ShiftWM achieves 20.83% success versus 16.67% for either control; with the GRU it achieves 16.67%, versus 25.00% for Framewise and 29.17% for Constant dynamics. On tissue manipulation, transformer ShiftWM achieves 4.17% versus 8.33% for either control, while GRU ShiftWM achieves 12.50% versus 8.33% for either control. These opposite signs across families and domains do not establish a general benefit of inferred dynamics context.

Table 15 retains the matched task structure when quantifying uncertainty: it jointly resamples training seeds and physical task seeds in 5,000 crossed bootstrap draws. No original interval has a strictly positive lower bound. Forecast MSE and planning success need not agree: drone GRU ShiftWM has lower horizon-five error than either control but lower closed-loop success. Final test results are not used here, and these development results do not satisfy the promotion rule above.

The fixed random-action control succeeds on 2/8 drone and 1/8 tissue tasks; privileged recorded-action replay succeeds on 8/8 in each domain. Random uses base seed 101 plus each task seed, with no independently trained models; replay is a reachability check. Original drone ShiftWM in both predictor families is below the random 25% point estimate. Neither this drone task nor the tissue simulation establishes useful real-world control or clinical performance.

Method	Seed 0	Seed 1	Seed 2	Success (%)	MSE@5 (10^{-3})
<i>Drone / Transformer</i>					
Framewise	1/8	2/8	1/8	16.67	5.650
Constant dynamics	1/8	1/8	2/8	16.67	5.718
ShiftWM (ours)	1/8	2/8	2/8	20.83	5.711
<i>Drone / GRU</i>					
Framewise	1/8	3/8	2/8	25.00	6.578
Constant dynamics	1/8	4/8	2/8	29.17	6.591
ShiftWM (ours)	1/8	2/8	1/8	16.67	6.503
<i>Tissue manipulation / Transformer</i>					
Framewise	0/8	1/8	1/8	8.33	3.588
Constant dynamics	1/8	1/8	0/8	8.33	3.563
ShiftWM (ours)	1/8	0/8	0/8	4.17	3.573
<i>Tissue manipulation / GRU</i>					
Framewise	0/8	1/8	1/8	8.33	4.405
Constant dynamics	0/8	1/8	1/8	8.33	4.417
ShiftWM (ours)	0/8	1/8	2/8	12.50	4.389

Table 14: **Complete development comparison, all 36 original models.** Within each domain, each seed uses the same eight planning tasks and a 200-native-command budget including support. Success is the mean over three training seeds; the 24 seed–task instances are not 24 independent physical tasks. MSE@5 is mean fixed-reference forecasting error across the three seeds, on identical development windows (lower is better). All controls are matched in-house implementations within each family.

Domain	Predictor	Comparator	Δ (pp)	Paired 95% interval (pp)
Drone	Transformer	Framewise	+4.17	[−25.00, +33.44]
Drone	Transformer	Constant dynamics	+4.17	[−25.00, +33.33]
Drone	GRU	Framewise	−8.33	[−29.17, +12.50]
Drone	GRU	Constant dynamics	−12.50	[−37.50, +12.50]
Tissue	Transformer	Framewise	−4.17	[−25.00, +12.50]
Tissue	Transformer	Constant dynamics	−4.17	[−20.83, +0.00]
Tissue	GRU	Framewise	+4.17	[+0.00, +25.00]
Tissue	GRU	Constant dynamics	+4.17	[+0.00, +25.00]

Table 15: **ShiftWM (ours) minus matched controls in planning success.** Differences are percentage points. **Bold green** marks positive point differences, not statistical significance. Intervals use paired crossed resampling of three training and eight physical task seeds; they are unadjusted exploratory 95% intervals. All intervals include or touch zero, and all negative comparisons remain visible.

N.2 GOAL GEOMETRY AND CONTROLLED ACTION-RANKING DIAGNOSTICS

The following diagnostics use only training/development evidence and were motivated by the unsuccessful planning comparisons. They distinguish a representational limitation from an identified cause of failure.

A fixed-context capacity bound. Across all sixteen gain-0.75 development endpoint goals (120 pairs), the warm appearance transform reduces mean pairwise latent MSE to 17.4% of the canonical value. In Figure 17, the transformer seed-0 Framewise and ShiftWM corrections retain 17.0% and 16.2%, respectively. For each observation context, all goals receive the same correction; measurements are then averaged over sixteen contexts. If $G(u, c) = g(c) \odot u + b(c)$ and each coordinate of $g(c)$ lies in $[0.9, 1.1]$, its shared translation cancels and

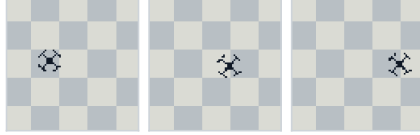
$$0.81\|u - v\|_2^2 \leq \|G(u, c) - G(v, c)\|_2^2 \leq 1.21\|u - v\|_2^2. \quad (11)$$

Thus the existing observation FiLM can expand this warm-goal mean separation to at most 21.1% of its canonical value in the fixed-context comparison. This is a capacity bound on pairwise separation,

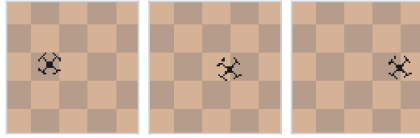
not an achieved result, an uncertainty interval, or a proof that larger distances improve control. Framewise calibration lacks this FiLM bound but also fails to recover canonical separation in this diagnostic. Canonical features themselves only imperfectly order physical distance, which is only one component of the drone success criterion.

a Same physical goals

Canonical



Warm shift



Goal 1

Goal 2

Goal 3

Recorded RGB · identical crops

b Goal separation

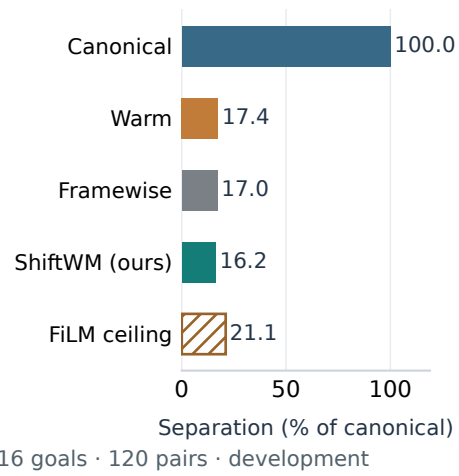


Figure 17: **Development goal geometry under the appearance shift.** Left: three recorded drone goals selected by minimum, median and maximum X position, shown with identical central 80×80 crops and the registered warm transform. Right: mean pairwise latent MSE for all sixteen goals, normalized by canonical separation; correction uses fixed support contexts and transformer seed 0. The hatched bar is the analytic $1.21 \times$ expansion ceiling for original observation FiLM applied to warm features. It is not a new-method measurement or a causal explanation of planning failure.

Four-state counterfactual action probe. Six seed-0 models score the same fixed library of 32 candidate sequences from each of four development support states before any candidate is executed. Each of the 128 branches independently resets and exactly replays the ten-command support, then executes up to 25 future commands. Two branches reach a workspace boundary before the full horizon; all branches are retained, while equal-horizon, safe branches define the within-task rank correlations and regret. All 24 model-selected winners are safe. This probe uses actual alternative action executions from a common state, not merely same-seed trajectories collected under different response gains: the latter use state-dependent feedback and do not have identical action histories.

Realized latent cost compares the canonical encoding of each executed future image with that model’s adapted goal; these future images are used only for the post-execution audit. For transformer ShiftWM, mean Spearman correlation between predicted and realized latent goal costs is 0.422; for GRU ShiftWM it is 0.278. Correlations are computed within each task and then averaged over the four tasks. Realized latent-cost versus physical-distance correlations are approximately 0.675, compared with 0.814 using frozen canonical feature-to-goal costs. The transformer ShiftWM and Constant dynamics models choose the same candidate on all four tasks, with mean terminal distance 61.7 mm and regret 40.9 mm relative to the best safe candidate. Transformer Framewise gives 66.4 mm and 45.6 mm; all three GRU variants choose the same candidates, giving 157.5 mm and 136.6 mm. The 177.4 mm uniform mean over the candidate library is a descriptive reference, not the random closed-loop controller. These $n = 4$ observations suggest both imperfect action-conditioned ranking and imperfect goal geometry; they neither isolate a unique mechanism nor establish performance on a larger task population. In particular, identical selected actions for inferred and constant contexts do not demonstrate a dynamics-inference benefit.

Completed, separately registered capacity ablation. The six-run transformer study changes only the observation gain to $g(s) = \exp\{\log(4) \tanh[0.1s / \log(4)]\}$, comparing inferred and constant dynamics context at seeds 0, 1 and 2. The attainable gain range becomes $[0.25, 4]$, while identity initialization, derivative 0.1 at zero, parameter shapes, translation, action modulation, losses and the complete 30-epoch training schedule remain matched. Higher-order curvature also changes, so optimization trajectories need not match. This is a capacity ablation, not a claim of algorithmic novelty. Its versioned checkpoints and output namespace are separate from the original study.

All six runs completed 30 epochs and both development evaluations (Table 16). Revised ShiftWM reaches 7/24 task–seed instances, versus 4/24 for the revised Constant dynamics control: +12.50 percentage points, with a paired exploratory interval $[0, 37.50]$. Each training seed gains one of eight successes, with three ShiftWM-only successes and no control-only success across the matched instances. Relative to original ShiftWM (5/24), there are three new successes and one lost success. The difference is +8.33 points, with interval $[-12.50, 29.17]$. The revised Constant dynamics control retains 4/24 in aggregate but changes which tasks succeed. The revised ShiftWM rate, 29.17%, exceeds the single random-policy reference of 25% only descriptively.

These are subsequent comparisons on the same development tasks. The forecast MSE changes little, and the fixed 16-goal/120-pair geometry measurement has not yet been repeated for the revised checkpoints. Consequently, the observed success increases do not demonstrate restored goal geometry, improved action ranking, or independent generalization. They do not justify promoting the method to final-test evaluation.

Method	Seed 0	Seed 1	Seed 2	Success (%)	MSE@5 (10^{-3})
<i>Original narrow gain</i>					
Constant dynamics	1/8	1/8	2/8	16.67	5.718
ShiftWM (ours)	1/8	2/8	2/8	20.83	5.711
<i>Revised wide gain</i>					
Constant dynamics	0/8	2/8	2/8	16.67	5.724
ShiftWM (ours)	1/8	3/8	3/8	29.17	5.713

Paired contrast	Δ (pp)	Paired 95% interval (pp)
Wide ShiftWM (ours) – wide constant	+12.50	$[+0.00, +37.50]$
Wide ShiftWM (ours) – original ShiftWM (ours)	+8.33	$[-12.50, +29.17]$
Wide constant – original constant	+0.00	$[-25.00, +33.33]$

Table 16: **Completed observation-gain ablation on drone development.** Original and revised transformer conditions share data, optimization, validation selection, eight physical tasks, support and planner budget. The six revised models use seeds 0, 1 and 2. The lower panel reports paired crossed-bootstrap success differences; all intervals include or touch zero. **Bold green** marks positive point estimates only. This follow-up ablation is exploratory, not an independent test of the geometry hypothesis.

N.3 SEPARATE OFFICIAL ADAJEPa REPRODUCTION

We additionally ran the released AdaJEPa implementation (Wang et al., 2026c), pinned at revision 51d8665b7978, with its official PushT checkpoint and evaluation goals. All four 50-episode evaluations completed (Table 17). The model observes RGB plus agent proprioception; the released goals, pretrained representation and planner budget differ from our RGB-only studies. These results therefore remain a separate reproduction, without a direct performance ranking against ShiftWM.

Official CEM uses horizon 25, 200 candidates, 30 elites and ten optimization iterations, with at most 20 replans and five grouped actions executed per replan (frameskip five). All stages use evaluation seed 100. Adaptive success exceeds frozen success by 24 percentage points on clean observations and 20 under blur. Aggregate logs and one evaluation seed do not support a paired task-level uncertainty interval. This is evidence that the released adaptive implementation runs locally under its own protocol, not a state-of-the-art or matched-baseline claim.

The checkpoint, goals, official configuration and scientific command are pinned for every stage. The first clean-frozen run began before launcher/runtime hashes were added to the record; a supplemental GPU observation documents its hardware. Later stages include those metadata fields, but current hashes cannot retroactively establish their exact values at the first stage’s launch.

Observation	Official frozen	Official adaptive	Δ (pp)
Clean	34/50 (68%)	46/50 (92%)	+24.00
Blur	29/50 (58%)	39/50 (78%)	+20.00

Table 17: **Official AdaJEPa reproduction under its separate protocol.** Each entry uses 50 released evaluation goals and seed 100, with RGB plus agent proprioception. Differences compare adaptive and frozen execution within this official protocol. **Bold green** marks positive aggregate point differences; no paired interval or significance claim is made. These inputs and budgets are not matched to ShiftWM.

O REGISTERED STUDY ON RECORDED ROBOT VIDEO

Status: complete prespecified real-data comparison. This additional study evaluates forecasting on real robot recordings from DROID (Khazatsky et al., 2024). It complements the simulator experiments; it does not replace them or establish real-robot control success. Figure 18 illustrates the acquired camera observations.

Data and splits. Before optimization, we fixed 24 of the full release’s 2,048 shards by a salted hash of their object names. This selection comprises 1,126 audited episodes and 22,017,792,821 downloaded bytes including metadata and license. Object generations, publisher checksums and local hashes identify the inputs. We assign recording sessions, defined by site plus recording date, to training, validation and test with fixed 70/15/15 hash thresholds. The resulting splits contain 851, 143 and 132 episodes from 316, 59 and 58 disjoint sessions, respectively, spanning 14 sites overall. All 327,498 native observations and 64,826 complete action blocks passed extraction and alignment checks. Unparseable session identifiers fail the audit; there is no frame-level fallback. Session separation does not guarantee distinct scenes or objects. All recorded outcomes are retained without success-based selection.

Observation and action contract. Frames at native indices 0, 5, 10, ... are paired with the intervening five seven-dimensional commands, concatenated into a 35-dimensional block. The full extraction verified every native command array against the recorded six Cartesian-position/pose components plus gripper position. Incomplete tails are counted, and 11 episodes too short for an eight-frame training window remain in the data. The inspected release has no per-step timestamp, so horizons are expressed in native transitions. Ground-truth state, rewards, success flags, language and future images are never context inputs. Training and checkpoint selection use external camera 1 only; camera 2 on held-out episodes provides a separately reported natural camera-transfer test. No synthetic appearance corruption is applied in this study.

Representation and matched controls. We freeze DINOv2-small (Oquab et al., 2023), resize complete images to 224×224 , and pool the final patch grid into 2×2 spatial cells, producing 1,536-dimensional targets. Feature and action normalization is fitted on training camera 1 only. The pinned LeWM causal predictor (Maes et al., 2026b) has width 192 and four transformer blocks; a residual output head initializes every mode to persistence in the fixed feature space. The four modes are Framewise, Constant dynamics, ShiftWM real-video variant (ours), and an Action-free control. Each uses three training seeds and the same common initialization, data and optimization: 12 registered runs, each with 30 epochs. The model observes three support frames and recursively predicts five further frames using recorded commands, without query-image teacher forcing. We select checkpoints using validation forecast loss.

Evaluation and interpretation. We register horizons of 1, 3 and 5 action blocks, plus a separately labeled 10-block extrapolation test. Comparisons include persistence and constant feature-velocity extrapolation, with raw and standardized feature MSE and cosine error. We aggregate

One recording, three camera views

DROID-100

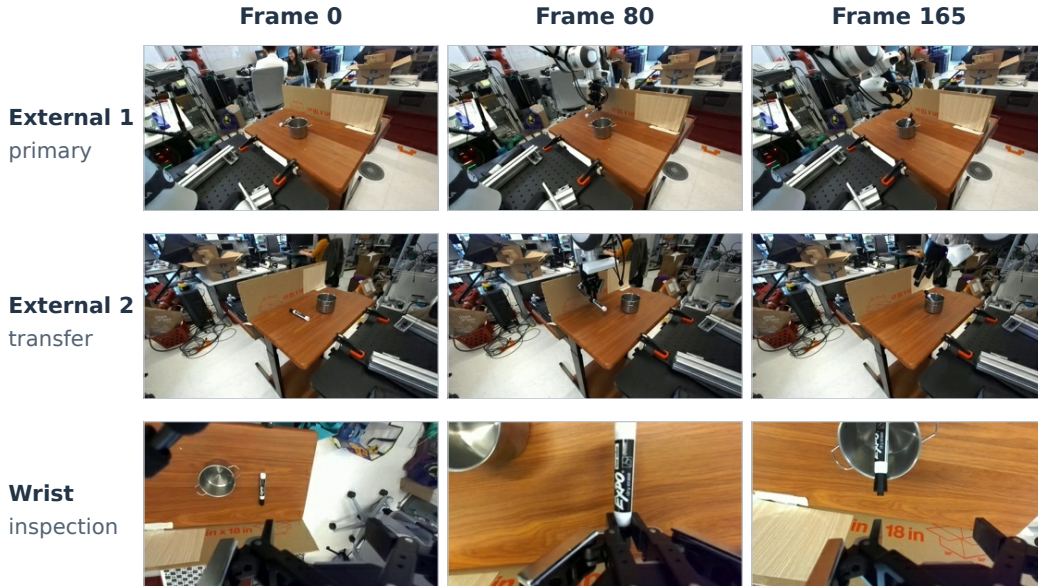


Figure 18: **Actual recorded DROID camera views, shown for data inspection.** Columns show native frame indices 0, 80 and 165 from the first decoded record of the separately downloaded DROID-100 ingestion audit; rows show its two external cameras and wrist camera. Selection uses evenly spaced frame positions, not rewards or model outcomes. All frames retain their full field of view, original colors and publisher-provided face blurring; no image is generated. The row annotations describe the registered roles of these camera streams: external camera 1 is the primary view, external camera 2 is reserved for camera-transfer evaluation, and the wrist stream is retained for inspection. The study itself uses 24 selected shards of the full DROID release, with 1,126 audited episodes, rather than this one-record ingestion example. These are observations, not predicted images, successful-policy results or an algorithm comparison. Native indices are not calibrated elapsed time.

within episodes and quantify uncertainty by source session and training seed. Camera-transfer and ordinary held-out results remain separate. Reversing recorded query actions probes model sensitivity, not the physical outcome of an unexecuted intervention. This observational study provides neither paired canonical targets nor identified physical factors; its metrics cannot establish closed-loop real-robot success.

Completed results. All 12 runs completed 30 epochs. On the primary camera, five-block standardized feature MSE is 0.138198 for ShiftWM, 0.138479 for Framewise, and 0.142392 for persistence. The 0.20% reduction versus Framewise is inconclusive: the paired 95% MSE-difference interval is $[-0.001038, 0.000413]$. The 2.94% reduction versus persistence has an interval below zero, $[-0.006714, -0.001908]$; camera transfer retains a 2.02% reduction versus persistence. At ten blocks, however, error is 1.91% and 1.24% higher than Framewise on the two cameras, respectively, with both intervals including zero. These intervals are unadjusted for multiple comparisons. Figure 19 relates recorded observations to the measured episode-level errors.

Training and action-use limitations. Every validation-selected checkpoint occurs at epoch 1 or 2; later validation deterioration despite decreasing training loss indicates overfitting. Reversing the order of future action blocks changes the final-horizon error of ShiftWM by less than 0.10% in every population. This limited sensitivity is an observational diagnostic, not proof that actions are unused or that physical interventions have no effect. The completed study supports a modest short-horizon benefit over persistence, while a substantial advantage over matched learned baselines and reliable long-horizon action-dependent prediction remain unresolved.

Real recordings, measured forecast errors

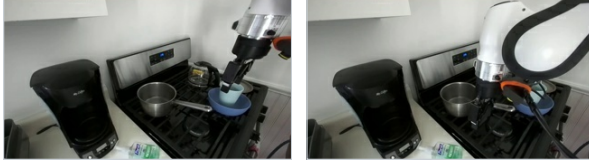
All 132 test episodes: ours lower on 73, tied on 0, higher on 59.

Mean h5 MSE reduction vs Framewise: +0.20% | 3 training seeds

— ShiftWM (ours) — Constant dynamics
— Framewise — Persistence

a Largest reduction

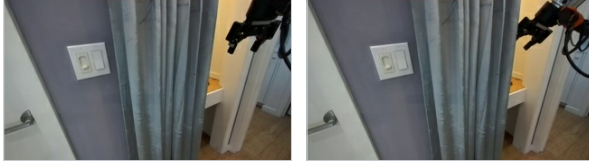
Last support · frame 10 Recorded target · frame 35



Episode averages: 7 matched windows × 3 seeds

b Largest regression

Last support · frame 10 Recorded target · frame 35



Episode averages: 7 matched windows × 3 seeds

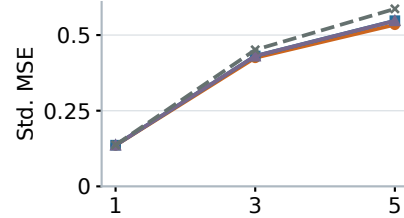
c Median case

Last support · frame 10 Recorded target · frame 35

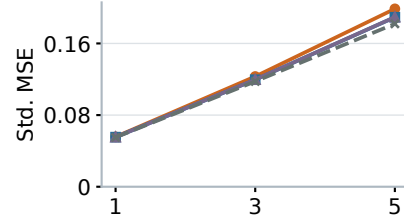


Episode averages: 8 matched windows × 3 seeds

Ours – Framewise: -0.0133 MSE



Ours – Framewise: +0.00938 MSE



Ours – Framewise: -7.75e-05 MSE

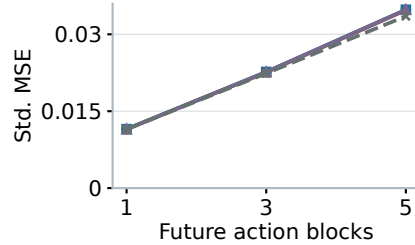


Figure 19: **Real DROID recordings with measured feature-forecast errors.** The primary held-out camera-1 population is summarized above three deterministically selected episodes: the largest reduction, the largest regression, and an episode nearest the median in ours-minus-Framewise h5 standardized MSE (ties by episode ID). Each image pair shows the last observed support frame and final recorded query target from the first evaluated window; these are actual photographs, not decoded predictions. Each curve averages all evaluated windows of that episode, then all three training seeds, at the three measured horizons. Panel y-axis ranges differ; methods within each panel share the same axes. Curves show point means; the full-test paired 95% h5 MSE-difference interval includes zero. The models observe three support frames and recorded commands; persistence repeats the last support feature and requires no training. The displayed images illustrate a recording and do not localize the aggregate error or establish why a method wins. Selection is outcome-conditioned and is not a frequency estimate: all-test counts and mean reduction are reported above. Error reductions are feature-forecast differences, not robot-control success or evidence of an identified causal mechanism.

Method	MSE@1 ↓	MSE@3 ↓	MSE@5 ↓	Gain@5 (%) ↑
Framewise	$0.04988 \pm 2.2 \times 10^{-5}$	$0.1031 \pm 4 \times 10^{-5}$	0.1385 ± 0.00015	reference
Constant dynamics	$0.04987 \pm 2.2 \times 10^{-5}$	$0.1031 \pm 5 \times 10^{-5}$	0.1384 ± 0.00016	+0.04
ShiftWM (ours)	$0.04979 \pm 4.2 \times 10^{-5}$	0.1029 ± 0.00011	0.1382 ± 0.00011	+0.20
Action-free	$0.04993 \pm 2.8 \times 10^{-5}$	0.1033 ± 0.00011	0.1389 ± 0.00013	-0.28
Persistence	0.05029	0.1053	0.1424	-2.83
Constant velocity	0.1207	0.6215	1.489	-975.22

Table 18: **Recorded DROID: primary external-camera-1 forecasting.** Train-standardized feature MSE at 1, 3 and 5 action blocks (5, 15 and 25 native transitions); lower is better. Learned entries show three-seed mean $\pm$ sample SD; support-only baselines have no training-seed SD. Gain@5 is $100(\text{Framewise} - \text{method})/\text{Framewise}$, a point reduction, not percentage points. **Bold green** marks a positive point reduction, not significance. At five blocks, ours minus the named reference in MSE with 95% paired session/seed bootstrap intervals is Framewise: $-0.0002808 [-0.001038, 0.000413]$ (includes zero); Constant dynamics: $-0.0002289 [-0.0009871, 0.0004498]$ (includes zero). Intervals are unadjusted for multiple comparisons; zero-crossing intervals are inconclusive.

Method	External camera 1		External camera 2	
	5 blocks	10 blocks	5 blocks	10 blocks
Framewise	0.1385 ± 0.00015	0.1899 ± 0.00085	$0.1543 \pm 8.9 \times 10^{-5}$	0.22 ± 0.00066
Constant dynamics	0.1384 ± 0.00016	0.1898 ± 0.00089	$0.1543 \pm 9.9 \times 10^{-5}$	0.2199 ± 0.00066
ShiftWM (ours)	0.1382 ± 0.00011	0.1935 ± 0.004	0.1542 ± 0.00032	0.2227 ± 0.0031
Action-free	0.1389 ± 0.00013	$0.1895 \pm 3.9 \times 10^{-5}$	0.1544 ± 0.00016	0.2188 ± 0.00055
Persistence	0.1424	0.1937	0.1574	0.2214
Constant velocity	1.489	5.114	1.636	5.637

Table 19: **Separate recorded-video populations.** Final-horizon standardized MSE; lower is better. External-camera-1/five-block is primary; camera 2 measures natural camera transfer, and ten blocks measure horizon extrapolation. The same session-disjoint test recordings are considered, but complete-window eligibility can differ by horizon. Three-seed means and sample SD are shown for learned methods. These feature forecasts do not measure closed-loop robot success. Raw MSE, cosine error, all paired intervals and reversed-action diagnostics are retained in the source-linked results report.

O.1 VALIDATION DEVELOPMENT: DIAGNOSING AND CALIBRATING EXCESS MOTION

After completing the registered study, we investigated overfitting using training and validation recordings only. Stronger command substitutions clarify the reversal diagnostic: replacing future commands with a sequence from another recording session increases the selected ShiftWM model’s ten-block validation error by 5.17%, averaged across three seeds, whereas the action-free control is unchanged. These are sensitivity probes against the original recorded future, not outcomes of executed counterfactual actions. Low-future-motion validation windows suffer relative to persistence, while high-motion windows benefit; these target-defined strata are descriptive and cannot determine an online adaptation rule.

We therefore evaluate a simple, matched residual-calibration control. Let $r_h = \hat{z}_h - z_0$ denote the predicted displacement from the final observed support feature and $d_h = z_h - z_0$ its recorded target, both in the frozen training-standardized coordinates. For each selected checkpoint, we fit a single scalar on training recordings:

$$\alpha = \text{clip}_{[0,1]} \frac{\mathbb{E}_{\text{train}} \langle r_h, d_h \rangle}{\mathbb{E}_{\text{train}} \|r_h\|_2^2}, \quad \tilde{z}_h = z_0 + \alpha r_h. \quad (12)$$

We set $\alpha = 0$ for zero predicted-displacement energy. The expectation averages all ten horizons and feature dimensions, then windows within each episode and equally across eligible training episodes.

Method	Validation: 5 blocks			Validation: 10 blocks		
	Original	Calibrated	Gain (%)	Original	Calibrated	Gain (%)
Framewise	0.159071	0.158942	+0.08	0.216903	0.216206	+0.32
Constant dynamics	0.158998	0.158880	+0.07	0.216746	0.216099	+0.30
ShiftWM (ours)	0.158398	0.157704	+0.44	0.219507	0.215874	+1.66
Action-free	0.159766	0.159718	+0.03	0.217435	0.217144	+0.13

Table 20: **Validation development: matched residual calibration, not a new test.** Train-standardized feature MSE, lower is better; all four methods and three seeds are included. One scalar per frozen best checkpoint is fitted on 830 eligible training episodes (7,721 windows), with no validation targets used for fitting. Each Gain column is the reduction from that row’s own uncalibrated model. **Bold green** marks positive point reductions, not significance. Against also-calibrated Framewise, ours improves 0.78% at five blocks and 0.15% at ten; the paired MSE differences with 95% session/seed bootstrap intervals are -0.001238 $[-0.002173, -0.000470]$ and -0.000332 $[-0.001395, +0.001022]$, respectively. The latter includes zero. These are exploratory validation intervals, unadjusted for multiple comparisons; the original held-out results remain unchanged.

The base recursive trajectory is unchanged; contraction is applied to its output. All four methods and three seeds receive the same fitting procedure on 830 eligible training episodes and 7,721 windows; no new neural weights are trained. The selected base checkpoints, data and scientific dependencies remain frozen, and all twelve calibration configurations pass exact offline CPU reload checks.

Table 20 reports the complete validation comparison. Calibration reduces our ten-block error by 1.66% relative to its own uncalibrated model. Against equally calibrated Framewise, the reductions are 0.78% at five blocks and 0.15% at ten; the latter paired interval includes zero. Both horizons use 141 eligible validation episodes from 59 sessions, with 1,772 and 1,631 windows, respectively. This standard least-squares control is development evidence, not a new methodological contribution or a replacement for the original test results. The separately frozen evaluation on untouched recording sessions is reported in Section P; it does not replace the original test results or change the development status of this validation comparison.

P CONFIRMATORY FORECASTING ON FRESH RECORDING SESSIONS

An untouched recording population. After the training/validation calibration study, we registered a deterministic selection of twelve additional official DROID shards, excluding all original twenty-four shards. The new download contains 556 episodes in 293 site/date sessions. Before decoding images or running any model, we excluded every episode whose session appeared anywhere in the original train, validation or test splits: 491 episodes were removed solely for session overlap. The remaining 65 episodes come from 52 sessions absent from all 433 original sessions, with no original episode-ID or serialized-record overlap. We did not resample, enlarge this selection, or filter on outcomes, motion or appearance. Session separation does not establish distinct scenes or objects.

A frozen comparison. Before decoding these observations, we separately froze the evaluation code, all twelve selected checkpoints, original training normalization, encoder, twelve train-fitted scalars, metrics and comparisons. We evaluated all four methods, three training seeds and both original and calibrated variants on each of two cameras and two horizons: 96 complete evaluations. Every method receives the same calibration procedure from Equation 12; the scalar is never fitted on fresh observations. No new neural weights are trained. The original causal three-frame support, recorded command blocks, feature extraction and stride-five windowing remain unchanged. Future observations are scoring targets only. All 65 recordings supply five-block windows. One recording lacks enough frames for ten blocks; it remains in the manifest and contributes no fabricated query.

Prespecified primary endpoint. The sole primary comparison is calibrated ShiftWM (ours) versus equally calibrated Framewise on camera 1 at the *fifth* predicted frame, measured in original training-standardized feature MSE. This is an endpoint error, not the average across five pre-

Method	External camera 1		External camera 2	
	5 blocks	10 blocks	5 blocks	10 blocks
<i>Original selected checkpoints</i>				
Framewise	0.148953	0.208396	0.185354	0.220646
Constant dynamics	0.148876	0.208241	0.185292	0.220554
ShiftWM (ours)	0.148514	0.210635	0.185000	0.222992
Action-free	0.149175	0.207603	0.185950	0.220577
<i>Train-fitted residual calibration</i>				
Framewise	0.148757	0.207555	0.185222	0.219935
Constant dynamics	0.148695	0.207461	0.185169	0.219888
ShiftWM (ours)	0.147654	0.206815	0.184255	0.219325
Action-free	0.149096	0.207252	0.185900	0.220275
Persistence	0.151979	0.210400	0.190841	0.226667
Constant velocity	1.661833	6.011535	1.753509	6.012756

Table 21: **Fresh real-DROID sessions: complete original and calibrated comparison.** Final-horizon training-standardized feature MSE ($\downarrow$), averaged equally over episodes and three training seeds. Every method uses its original validation-selected checkpoint; the second block applies its own scalar fitted on original training data only. The two support-only baselines are unchanged across variants. Five-block evaluation uses all 65 episodes from 52 new site/date sessions (866 windows); ten-block evaluation uses 64 eligible episodes from the same 52 sessions (801 windows). The one short episode remains in the audited manifest. Calibrated camera-1/five-block is the prespecified primary population; other populations are separate secondary analyses. These are feature forecasts of actual recorded observations, not physical control outcomes.

dicted frames. Table 21 reports all original and calibrated methods; Table 22 preserves their paired gains and uncertainty. Calibrated ShiftWM obtains 0.147654 versus 0.148757 for calibrated Framewise, a **0.742%** reduction. The paired MSE difference is -0.001103 , with 95% interval $[-0.002145, -0.000290]$. Thus the registered primary comparison supports a modest improvement on these fresh sessions. It does not establish a large performance margin or state-of-the-art superiority.

Secondary results and retained regressions. Camera-2/five-block transfer yields a 0.522% point reduction against calibrated Framewise, with a descriptive interval below zero. Calibrated ten-block gains are 0.356% on camera 1 and 0.277% on camera 2; both intervals include zero. The original uncalibrated model is worse than Framewise at ten blocks on both cameras: 0.210635 versus 0.208396, and 0.222992 versus 0.220646, respectively. These are relative error increases of 1.075% and 1.063%, and are retained rather than replaced by the calibrated outcomes. Confidence intervals use 10,000 paired session-cluster draws crossed with training-seed resampling, after averaging windows within each episode and weighting eligible episodes equally. Only the stated primary comparison is confirmatory; secondary intervals have no multiplicity adjustment.

Scope of the evidence. This separate test supports the reproducibility of a small short-horizon forecasting gain. Residual contraction is a standard least-squares control, not a novel architectural or theoretical contribution. Long-horizon superiority over learned controls remains inconclusive. All input, checkpoint and result hashes were checked, and endpoint arithmetic was independently recomputed from every recorded episode. The evidence concerns observational forecasts of real robot videos; it does not measure closed-loop physical control.

What changes in individual recordings. Figure 20 connects the calibration rule to recorded inputs and measured forecast errors. It shows the largest episode improvement, the median episode and the largest regression, alongside the complete distribution: 46 of 65 episodes improve and 19 regress. Episode-average gains need not hold in every window; the fixed first window of the median episode is one such counterexample. The images are recorded observations, while prediction quality is measured in the frozen feature space.

A fixed residual scale, tested on real recordings

a Same forecast; contracted displacement

$$\tilde{z}_h = z_0 + \alpha (\hat{z}_h - z_0)$$

Train-fitted α · fixed at inference

Standard calibration control



— Ours, original — Ours, calibrated — Framewise, calibrated

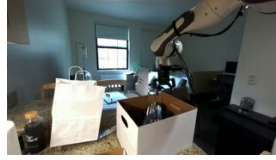
Actual observations from the first eligible window

b Largest episode gain

Episode Δ : +0.0069



Last support · 10



Recorded future · 35

c Median episode

Episode Δ : +0.0009



Last support · 10



Recorded future · 35

d Largest regression

Episode Δ : -0.0040



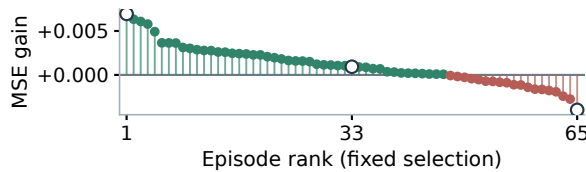
Last support · 10



Recorded future · 35

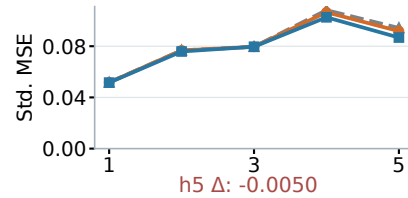
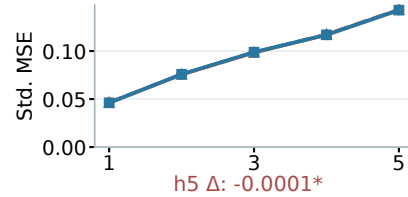
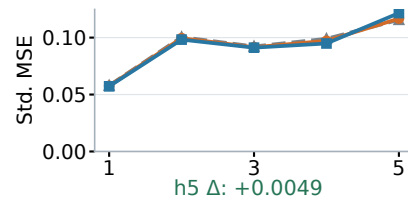
e All 65 episodes

46 gains · 19 regressions



That window's error

Future blocks 1-5



Population mean + 95% CI



Figure 20: **Residual calibration on actual DROID recordings.** (a) One train-fitted scalar contracts each forecast displacement about the last observation; the schematic uses seed 0's $\alpha = 0.908$. The base rollout is unchanged. (b–d) Episodes are selected by their three-seed, episode-average h5 gains: largest improvement, median and largest regression among all 65 fresh recordings. Photographs are recorded support/target frames from each first eligible window (native indices 10 and 35). Curves show that window's five measured feature errors, averaged over three seeds; vertical ranges differ between cases. Positive Δ favors calibrated ShiftWM over equally calibrated Framewise. *The median episode improves overall but regresses in the displayed window. (e) All episode gains and regressions are shown. The separate population interval is the original paired session/seed bootstrap: MSE gain 0.001103, 95% CI [0.000290, 0.002145], or 0.742% lower mean h5 error. This measures observational forecasting, not physical control.

Population	Gain (%) $\uparrow$	Paired MSE difference [95% CI]
<i>Original selected checkpoints</i>		
Camera 1, 5 blocks	+0.294	-0.000438 [-0.001373, +0.000366]
Camera 1, 10 blocks	-1.075	+0.002240 [-0.000480, +0.005574]
Camera 2, 5 blocks	+0.191	-0.000353 [-0.001285, +0.000483]
Camera 2, 10 blocks	-1.063	+0.002346 [-0.000464, +0.005829]
<i>Train-fitted residual calibration</i>		
Camera 1, 5 blocks (primary)	+0.742	-0.001103 [-0.002145, -0.000290]
Camera 1, 10 blocks	+0.356	-0.000740 [-0.002391, +0.000907]
Camera 2, 5 blocks	+0.522	-0.000967 [-0.001971, -0.000174]
Camera 2, 10 blocks	+0.277	-0.000609 [-0.002464, +0.001283]

Table 22: **Fresh-session gains and uncertainty versus matched Framewise.** ShiftWM (ours) is compared with Framewise within the same original or calibrated variant. Gains are relative error reductions, not percentage points; **bold green** denotes a positive point estimate, not statistical significance. Intervals use 10,000 paired session-cluster bootstrap draws crossed with training-seed resampling. Only the calibrated camera-1/five-block comparison is confirmatory. Its interval excludes zero; both calibrated ten-block intervals cross zero. All secondary intervals are descriptive and unadjusted for multiplicity. The uncalibrated ten-block regressions are retained explicitly.

Q MATCHED GENERALIZATION DEVELOPMENT ON RECORDED VIDEO

This registered development study investigates the early validation optimum in the original recorded-video campaign. Three fixed interventions change either the learning rate (Slow: 3×10^{-5}), weight decay (Decay: 0.1), or the predictor/context capacity (Compact: width 96, depth 2, context dimensions 16/64). Each arm trains Framewise, Constant dynamics, ShiftWM (ours), and Action-free at three seeds for all 30 epochs. The Compact intervention changes these capacity settings jointly and does not isolate their separate effects.

All inputs, normalization and model selection use the original training and validation partitions. Neither the original test nor the separately acquired fresh holdout selects these revisions. Each checkpoint uses the common all-five-query validation objective. Subsequent five- and ten-block forecasts are summarized by averaging windows within episodes and then equally averaging episodes and seeds. Development improvements, if present, require another independent evaluation for a new confirmatory claim; ordinary optimization and capacity controls are not presented as additional method novelty.

Table 23 reports every matched arm and method. Tables 24 and 25 provide the session/seed paired intervals, including regressions and intervals crossing zero. Table 26 records the selected epochs, parameter counts and measured invocation times. All 36 selected packages pass strict weights-only CPU reload parity on original-validation causal inputs.

Arm	Method	MSE@5 ↓	MSE@10 ↓	Gain@5 (%)	Gain@10 (%)
Slow	Framewise	0.15848	0.21626	reference	reference
Slow	Constant dynamics	0.15841	0.21605	+0.04	+0.09
Slow	ShiftWM (ours)	0.15802	0.21556	+0.29	+0.32
Slow	Action-free	0.15902	0.21564	-0.34	+0.28
Decay	Framewise	0.15907	0.21689	reference	reference
Decay	Constant dynamics	0.15899	0.21673	+0.05	+0.07
Decay	ShiftWM (ours)	0.15839	0.21948	+0.43	-1.20
Decay	Action-free	0.15976	0.21742	-0.44	-0.25
Compact	Framewise	0.15964	0.21925	reference	reference
Compact	Constant dynamics	0.15963	0.21923	+0.01	+0.01
Compact	ShiftWM (ours)	0.15888	0.22054	+0.48	-0.59
Compact	Action-free	0.16019	0.21877	-0.34	+0.22

Table 23: **Complete matched generalization development study.** All 36 runs completed 30 epochs on original training data. Entries are three-seed, equal-episode validation means; MSE uses fixed train-standardized feature coordinates. Five blocks is the primary development horizon; ten blocks is extrapolation. Slow reduces learning rate, Decay increases weight decay, and Compact jointly reduces predictor/context capacity. Gains compare each method to Framewise within the same arm. **Bold green** identifies positive point reductions, not statistical significance. Every arm and method is retained; these are validation results, not an additional held-out test.

Arm	Method	Δ MSE	Gain (%)	Paired 95% interval
Slow	Constant dynamics	-0.000065	+0.04	[-0.000111, -0.000015]
Slow	ShiftWM (ours)	-0.000458	+0.29	[-0.000671, -0.000283]
Slow	Action-free	+0.000539	-0.34	[-0.000062, +0.001162]
Decay	Constant dynamics	-0.000073	+0.05	[-0.000105, -0.000045]
Decay	ShiftWM (ours)	-0.000676	+0.43	[-0.001535, -0.000053]
Decay	Action-free	+0.000696	-0.44	[+0.000208, +0.001192]
Compact	Constant dynamics	-0.000017	+0.01	[-0.000072, +0.000039]
Compact	ShiftWM (ours)	-0.000759	+0.48	[-0.001353, -0.000278]
Compact	Action-free	+0.000543	-0.34	[+0.000029, +0.001149]

Table 24: **Matched validation uncertainty at 5 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are $100(\text{Framewise} - \text{method})/\text{Framewise}$. Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

Arm	Method	Δ MSE	Gain (%)	Paired 95% interval
Slow	Constant dynamics	-0.000201	+0.09	[-0.000322, -0.000086]
Slow	ShiftWM (ours)	-0.000698	+0.32	[-0.001144, -0.000339]
Slow	Action-free	-0.000610	+0.28	[-0.002089, +0.000888]
Decay	Constant dynamics	-0.000157	+0.07	[-0.000241, -0.000079]
Decay	ShiftWM (ours)	+0.002593	-1.20	[-0.000748, +0.005568]
Decay	Action-free	+0.000532	-0.25	[-0.000728, +0.001798]
Compact	Constant dynamics	-0.000017	+0.01	[-0.000194, +0.000153]
Compact	ShiftWM (ours)	+0.001291	-0.59	[-0.000945, +0.003425]
Compact	Action-free	-0.000483	+0.22	[-0.002460, +0.001390]

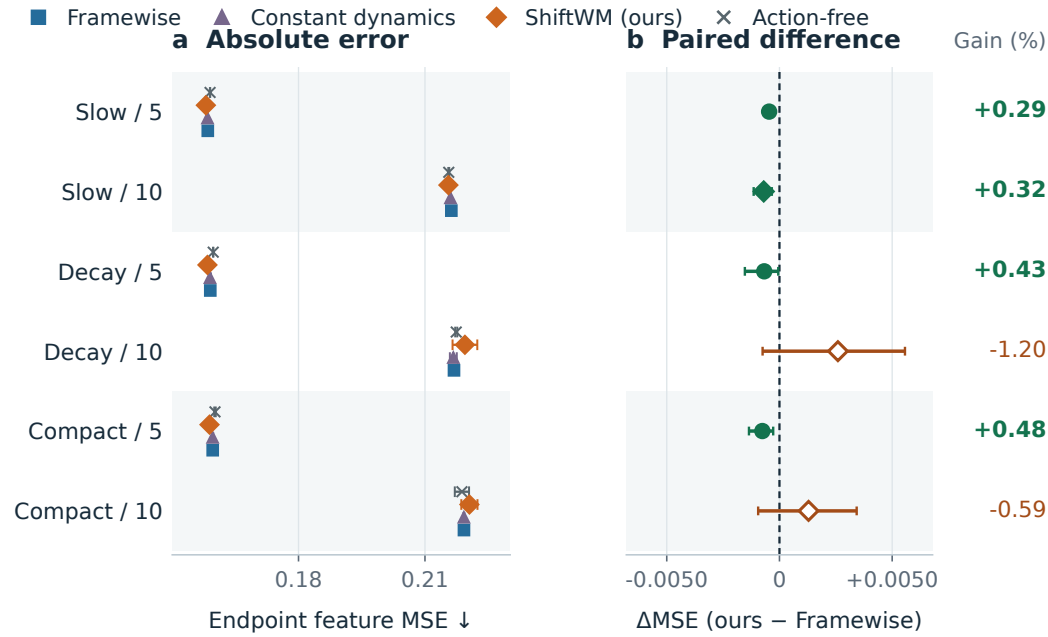
Table 25: **Matched validation uncertainty at 10 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are $100(\text{Framewise} - \text{method})/\text{Framewise}$. Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

Arm	Method	Best epochs	Total (M)	Trainable (M)	Seconds/run
Slow	Framewise	3/3/3	3.763	3.469	151 [149, 219]
Slow	Constant dynamics	3/3/3	3.763	3.714	159 [159, 234]
Slow	ShiftWM (ours)	3/3/3	3.763	3.714	161 [161, 236]
Slow	Action-free	3/3/3	3.763	3.312	141 [140, 213]
Decay	Framewise	1/1/1	3.763	3.469	148 [146, 218]
Decay	Constant dynamics	1/1/1	3.763	3.714	158 [156, 238]
Decay	ShiftWM (ours)	2/2/1	3.763	3.714	161 [158, 240]
Decay	Action-free	1/1/1	3.763	3.312	140 [138, 213]
Compact	Framewise	2/2/2	0.841	0.763	100 [99, 159]
Compact	Constant dynamics	2/2/2	0.841	0.828	110 [110, 175]
Compact	ShiftWM (ours)	2/3/3	0.841	0.828	112 [111, 173]
Compact	Action-free	2/1/2	0.841	0.722	93 [93, 149]

Table 26: **Reusable checkpoints and computation.** Best epochs are ordered by seeds 0/1/2 and selected by the shared all-five-query validation objective after full 30-epoch training. Parameter counts include frozen parameters; trainable counts exclude inactive modules. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, validation and CPU reload verification; prior interrupted work, if any, is excluded. Different GPU partitions are used, so these times are operational evidence, not a hardware-matched speed comparison. All 36 selected packages pass strict CPU reload parity.

Matched controls for DROID forecasting

Original validation only · 36 runs · three seeds per method



Whiskers: seed SD (a); paired 95% interval (b).

Open marks: interval includes zero. Green: favorable point estimate.

Figure 21: Matched optimization and capacity controls on original validation. All 36 registered runs complete 30 epochs; selected checkpoints use the common all-five-query validation objective. Rows identify arm and forecast horizon in action blocks. Slow lowers learning rate, Decay increases weight decay, and Compact jointly reduces predictor/context capacity. (a) Equal-episode three-seed mean endpoint feature MSE with sample seed standard deviation; all four methods are shown. (b) ShiftWM (ours) minus matched Framewise with 95% intervals from 10,000 paired session/seed bootstrap draws. Green positive reductions and open/filled interval markers distinguish point direction from interval exclusion of zero. The adjacent gain column is the relative MSE reduction, not the interval unit. Five and ten blocks include 141 and 141 eligible validation episodes, respectively. These exploratory development intervals are unadjusted for multiple comparisons; no additional held-out-test claim is made.

R TESTING CAUSAL SUPPORT RELIABILITY

We tested whether errors within a longer observed prefix predict when the factorized forecast should be trusted over Framewise. This is a calibration and ensembling diagnostic, not a new architectural contribution. The protocol, code, donor checkpoints and decision rule were frozen before execution; only original training and validation recordings were used.

Causal prefix backtesting. Both frozen donors received their respective original training-fitted residual calibrations. Within thirteen observed frames, each donor forecast the last ten from the first three and recorded commands, without incorporating any of those ten targets into its context or rollout. Let d denote the factorized-minus- Framewise forecast and e the observed-minus-Framewise residual, both in fixed training-standardized coordinates. The candidate sets

$$\alpha_s = \text{clip}_{[0,1]} \frac{\langle e, d \rangle_s + \lambda \alpha_0}{\langle d, d \rangle_s + \lambda}, \quad (13)$$

where each inner product averages prefix horizons and feature dimensions. The zero-denominator fallback is α_0 . The prior and shrinkage scale were fitted using original training only; three-fold training-session cross-validation selected from four fixed shrinkage values. All three seeds selected $\alpha_0 = 1$ and $\kappa = 100$, where $\lambda = \kappa$ times training-average donor-disagreement energy. The base donors had already trained on those sessions, so this procedure cross-validates only gate fitting, not end-to-end model training.

For the actual query, both donors restarted from the final three observed prefix frames. Their forecasts were blended with fixed α_s , using recorded future commands and no future images. All gate packages were locked before decoding validation features. The longer-prefix population contains 758 training episodes in 295 sessions and 132 validation episodes in 57 sessions. Respectively 93 and 11 recordings were too short for the common 23-frame window and remain in the audit. This observation budget differs from the main three-frame-support experiment. The candidate requires both donors and two additional prefix rollouts; it is not a single-model improvement.

Failed promotion and complete controls. Table 27 retains all eleven arms, including simple averaging, training-fitted constant blends, unshrunk, one-step and session-shuffled gates. The candidate’s h5 reduction versus the global prior is 0.007%, with paired MSE difference -0.00001134 and 95% interval $[-0.00002378, +0.00000256]$. Its h10 reduction is 0.022%, with interval $[-0.00008776, +0.00000306]$ for the paired MSE difference. It fails the prespecified 1% gain, interval and all-seed conditions. Ordinary equal averaging has lower mean error than the adaptive candidate; this does not establish that prefix reliability transfers to the next query. We therefore do not promote this candidate or test it on the previously inspected holdouts. Intervals resample sessions and training seeds and remain exploratory. Table errors use FP64 sufficient-statistic arithmetic on frozen FP32 donor predictions. The exported FP32 wrapper has exact reload parity; this is not a claim of bitwise parity with FP64 score arithmetic.

Method	Endpoint MSE ↓		Gain vs. global prior ↑	
	h5	h10	h5 (%)	h10 (%)
Framewise, calibrated	0.152472	0.213742	-0.464	-0.055
ShiftWM (ours), calibrated	0.151768	0.213624	+0.000	+0.000
Equal blend ($\alpha = 0.5$)	0.151513	0.211785	+0.168	+0.861
Global-prior blend (reference)	0.151768	0.213624	+0.000	+0.000
Train-query constant blend	0.151768	0.213624	+0.000	+0.000
Unshrunk support gate	0.151606	0.212266	+0.107	+0.636
Shrunk support gate (candidate)	0.151757	0.213577	+0.007	+0.022
One-step support gate	0.151761	0.213595	+0.005	+0.013
Shuffled support gate	0.151761	0.213590	+0.004	+0.016
Persistence	0.155705	0.215908	-2.594	-1.069
Constant velocity	1.916075	6.879813	-1162.502	-3120.525

Table 27: **A support-reliability candidate that did not pass its development gate.** Original camera-1 validation only: 132 episodes, 57 sessions and 1,351 common windows, equally weighted over episodes and three seeds. All arms have thirteen observed prefix frames and identical ten-step query windows, whereas the main forecasting protocol uses three support frames. Gains are relative to the *global-prior blend*, which equals calibrated ShiftWM because all training-fitted priors are one. **Bold green** marks positive point estimates, not significance or a passed decision rule. The candidate improves h5 by only 0.007%; its paired difference is -0.00001134 , with 95% interval $[-0.00002378, +0.00000256]$. Ordinary equal averaging has the lowest mean errors. The candidate fails the prespecified 1% improvement, interval and all-seed criteria; it is retained as a negative development result. No new neural weights are trained.

S MATCHED TEN-STEP TRAINING ON RECORDED VIDEO

This registered control tests whether training at the evaluation horizon improves long-horizon prediction. It reuses the unchanged width-192, depth-4 LeWM-based predictor with Framewise, Constant dynamics, ShiftWM (ours), and Action-free variants. All twelve runs train for 30 epochs at seeds 0/1/2, using three observed support frames and ten recursively predicted query targets. The optimizer, training-only normalization, action blocks, camera, training stride, and validation stride match the original five-step campaign.

No query image conditions forecasts or inferred contexts. Model selection uses FP32 standardized MSE averaged over all ten query steps and windows, preserving the original window weighting. An equal-episode metric is logged only as an auxiliary diagnostic. All evaluation tables equally weight episodes and then training seeds, as in the original evaluation. Training and checkpoint-selection horizons change together; this ordinary horizon-mismatch control is not a new architecture or evidence of additional method novelty.

Tables 28 and 29 compare all h10-trained methods on each standard validation population. Table 30 qualifies ours–Framewise differences using paired seed/session intervals. Tables 31 and 32 compare original and new checkpoints on identical h10-eligible query starts, including their five-step prefixes. The primary diagnostic is mean error across all ten query predictions; endpoint and five-step effects are secondary. Positive point gains, regressions and intervals including zero are all retained. These exploratory validation results require independent evaluation before any new confirmatory claim. Neither original nor fresh test payloads enter this study.

Table 33 documents the selected reusable checkpoints and measured invocation times. The accompanying evidence ledger records every configuration, checkpoint, metric source and independent numerical check.

Method	Endpoint MSE@5	Endpoint MSE@10	Gain@5 (%)	Gain@10 (%)
Framewise	0.15866	0.21429	reference	reference
Constant dynamics	0.15863	0.21423	+0.02	+0.03
ShiftWM (ours)	0.15830	0.21342	+0.23	+0.40
Action-free	0.15936	0.21496	-0.44	-0.31

Table 28: **Matched h10-trained methods: endpoint mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

Method	Mean query MSE@5	Mean query MSE@10	Gain@5 (%)	Gain@10 (%)
Framewise	0.11286	0.15261	reference	reference
Constant dynamics	0.11284	0.15258	+0.01	+0.02
ShiftWM (ours)	0.11271	0.15222	+0.13	+0.25
Action-free	0.11323	0.15322	-0.33	-0.40

Table 29: **Matched h10-trained methods: mean query mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

Scoring	Ours	Framewise	Gain (%)	Paired 95% interval
Endpoint 5	0.15830	0.15866	+0.23	[-0.00052, -0.00022]
Mean 1–5	0.11271	0.11286	+0.13	[-0.00022, -0.00008]
Endpoint 10	0.21342	0.21429	+0.40	[-0.00133, -0.00049]
Mean 1–10	0.15222	0.15261	+0.25	[-0.00059, -0.00022]

Table 30: **ShiftWM (ours) versus equally h10-trained Framewise.** Intervals are for ours minus Framewise in standardized feature MSE, so negative favors ours. The primary control metric averages all ten query predictions. Endpoint metrics and mean-five errors are secondary. Intervals jointly resample the three training seeds and recording-session clusters (10,000 draws); they are exploratory and unadjusted for multiple comparisons. † marks intervals including zero. Green bold point gains alone do not establish a difference.

Method	h	Original h5	Trained h10	Gain (%)	Paired 95% interval
Framewise	5	0.10972	0.10962	+0.09[†]	[-0.00028, +0.00007]
Framewise	10	0.15346	0.15261	+0.56	[-0.00135, -0.00039]
Constant dynamics	5	0.10968	0.10961	+0.07[†]	[-0.00025, +0.00009]
Constant dynamics	10	0.15338	0.15258	+0.52	[-0.00130, -0.00034]
ShiftWM (ours)	5	0.10938	0.10951	-0.12 [†]	[-0.00022, +0.00059]
ShiftWM (ours)	10	0.15385	0.15222	+1.06	[-0.00272, -0.00080]
Action-free	5	0.11011	0.11002	+0.08[†]	[-0.00022, +0.00005]
Action-free	10	0.15403	0.15322	+0.52	[-0.00114, -0.00046]

Table 31: **Matched training-horizon control: mean errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows average all query steps. Gains are $100(\text{original} - \text{h10})/\text{original}$; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. [†] marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

Method	h	Original h5	Trained h10	Gain (%)	Paired 95% interval
Framewise	5	0.15671	0.15630	+0.26	[-0.00081, -0.00002]
Framewise	10	0.21690	0.21429	+1.21	[-0.00383, -0.00148]
Constant dynamics	5	0.15664	0.15627	+0.23[†]	[-0.00076, +0.00002]
Constant dynamics	10	0.21675	0.21423	+1.16	[-0.00375, -0.00138]
ShiftWM (ours)	5	0.15623	0.15600	+0.15[†]	[-0.00090, +0.00061]
ShiftWM (ours)	10	0.21951	0.21342	+2.77	[-0.00936, -0.00309]
Action-free	5	0.15749	0.15709	+0.25	[-0.00067, -0.00010]
Action-free	10	0.21743	0.21496	+1.14	[-0.00323, -0.00168]

Table 32: **Matched training-horizon control: endpoint errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows score the final query step. Gains are $100(\text{original} - \text{h10})/\text{original}$; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. [†] marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

Method	Best epochs	Total (M)	Trained (M)	Seconds/run
Framewise	1/1/1	3.763	3.469	249 [242, 336]
Constant dynamics	1/1/1	3.763	3.714	262 [256, 368]
ShiftWM (ours)	1/1/1	3.763	3.714	264 [258, 368]
Action-free	1/1/1	3.763	3.312	234 [229, 321]

Table 33: **Horizon-ten checkpoints and computation.** Best epochs follow seeds 0/1/2, selected by window-weighted FP32 validation MSE over all ten recursive predictions after complete 30-epoch training. Total parameters include inactive frozen modules; trained parameters exclude them. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, five evaluation passes and CPU reload checks, excluding prior interrupted work. Hardware partitions differ, so invocation times are operational records, not a matched speed comparison. All 12 selected packages pass registered and independently repeated weights-only CPU reload checks.

Horizon matching on real DROID recordings

Validation only · 12 full 30-epoch runs · selected epoch 1 for every model

a Match the training horizon

h10-trained – h5-trained · mean of steps 1–10

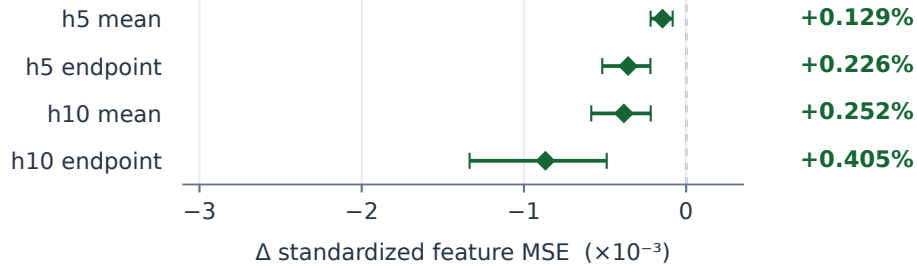
Reduction



b Compare equally h10-trained models

ShiftWM (ours) – Framewise

Reduction



Δ standardized feature MSE ($\times 10^{-3}$)

← Lower error · bars: paired 95% CI

Secondary: ours, matched h5 mean -0.116% (CI includes zero)

Figure 22: Horizon matching on real DROID recordings (original validation only). (a) For each unchanged mode, h10-trained minus original h5-trained standardized feature MSE, averaged over steps 1–10 on identical h10-eligible windows. Training and checkpoint-selection horizons change together. (b) ShiftWM (ours; factorized) minus equally h10-trained Framewise, for all-query mean and endpoint errors at h5 and h10. Dots and bars show signed MSE differences and paired 95% recording-session $\times$ training-seed bootstrap intervals (10,000 draws; seed 5198010). Negative differences favor the first model. Bold green percentages are relative point reductions, not confidence intervals. Both panels use the same absolute-error axis. There are 141 eligible episodes from 59 sessions and 3 model seeds; standard h5 has 1,772 windows, while h10 and matched h5 prefixes have 1,631. Windows are averaged within episodes, then episodes and seeds equally. All 12 runs completed 30 epochs; all selected epoch 1 by the fixed window-weighted validation criterion. The unfavorable secondary matched h5-prefix mean for factorized is -0.116% relative reduction, a small increase in error; new-minus-original MSE difference +0.0001264, paired 95% CI [-0.0002152, +0.0005941]. All 20 registered contrasts are retained in the companion evidence ledger. These are exploratory, multiplicity-unadjusted validation comparisons, not independent test or state-of-the-art claims.

T OBSERVATION-ANCHORED SPATIAL DEVELOPMENT STUDY

This registered architecture study addresses recursive prediction drift while preserving the locations of visual features. It is a new spatial variant; it does not preserve the original separate observation/dynamics factorization. The frozen DINOv2-small encoder produces a 4×4 grid of 384-dimensional features. Training-only channel statistics are shared across all sixteen positions, so mixtures remain in common feature coordinates. Three observed grids pass through a shared spatial encoder and the pinned LeWM temporal predictor. A causal action-prefix GRU supplies

commands only through the requested horizon, and a support-only transition context conditions the trunk through FiLM. Context remains fixed throughout the forecast.

For horizon h , let Z_0 be the last actual observed grid, H_h the horizon state, and E_0 the spatial encoding of Z_0 before context FiLM. The transport decoder computes

$$T_h = \text{softmax}_{\text{row}} \left(Q(H_h) K(E_0)^\top / \sqrt{96} + 4I \right), \quad (14)$$

$$\hat{Z}_h = (1 - g_h) \odot Z_0 + g_h \odot (T_h Z_0) + \tanh(R(H_h)), \quad g_h = \sigma(W_g H_h + b_g). \quad (15)$$

Every forecast retains the actual anchor instead of recursively replacing it with predictions. The row-stochastic mixture and unit-bounded innovation imply the componentwise bound $|\hat{Z}_{h,p,c}| \leq \max_q |Z_{0,q,c}| + 1$ in shared normalized coordinates. This is an output invariant, not an accuracy or stability guarantee. Transport mixes semantic features; it is not a validated physical image warp or RGB generator.

The five registered arms share the spatial inputs and trunk: Autoregressive recursively updates the feature history; Anchored additive keeps the actual observation anchor; Spatial transport (ours) adds mixing and bounded innovation; Context-off and Action-free disable the corresponding information paths. Transport begins near persistence (identity score bias 4 and gate bias -3), whereas the zero-initialized additive heads begin at exact persistence. Active parameter counts differ slightly. These arms test the package-level decoder effect; they do not separately identify mixing versus bounded innovation.

All fifteen models complete 30 epochs at seeds 0/1/2 with identical AdamW settings, batch size 128 and ten query steps. Window-weighted all-ten validation MSE selects checkpoints; final metrics average windows within episodes, then weight episodes and seeds equally. The evaluation contains 141 episodes from 59 recording sessions and 1631 windows. Neither original nor fresh held-out test observations enter this development study.

Tables 34 and 35 report every arm in native and original pooled coordinates. Each coordinate system has its own normalization; comparisons are made within columns. Tables 36 and 37 preserve all sixteen registered comparisons, including regressions and intervals containing zero. These intervals are exploratory and unadjusted for multiple comparisons; stronger claims require an independently frozen evaluation. Table 38 records every selected checkpoint and the nonidentical active parameter counts.

T.1 WHAT THE SPATIAL CONTROLS ESTABLISH

At the ten-step endpoint, Spatial transport (ours) reduces native feature MSE by 5.30% relative to the matched autoregressive predictor (0.193141 versus 0.203950) and by 3.55% relative to Anchored additive (0.200244). Both paired intervals exclude zero. These are relative reductions in feature error, not percentage-point improvements in task success. On the original pooled coordinates, the reduction is 2.54% against autoregression, while the 0.47% difference against anchoring has an interval containing zero. The stronger native-grid result therefore does not establish an equally strong advantage at every spatial resolution.

Removing the command-conditioning path increases native ten-step error by the equivalent of a 4.50% relative reduction for the full model. In contrast, the separate support-context path adds only a 0.10% native endpoint point gain; all four context comparisons have intervals containing zero, including two small unfavorable pooled-coordinate differences. These controls provide evidence for the action-conditioned decoder package, but do not establish an additional benefit from its learned support context. The action-free and context-off controls also remove active parameters: context-off has 878,497 versus 1,026,305 for the full model (14.40% fewer). They are information-path ablations, not exact capacity matches.

Across sixteen registered contrasts, fourteen point estimates favor transport, nine intervals exclude zero in that direction, and seven intervals contain zero. Two of the latter have unfavorable point estimates. Because all comparisons use development validation data and unadjusted intervals, this study supports further architectural investigation rather than a new confirmatory superiority claim.

Separating mixing and bounding. A registered follow-up completes the two-by-two decoder comparison with six additional 30-epoch models: bounded additive (Ours-3) and unbounded trans-

Method	$h = 5$ MSE	$h = 10$ MSE	$h = 10$ gain
Persistence	0.164028	0.226129	—
Autoregressive	0.149573 ± 0.000095	0.203950 ± 0.000504	—
Anchored additive	0.149606 ± 0.000224	0.200244 ± 0.000111	+1.817%
Spatial transport (ours)	0.145074 ± 0.000987	0.193141 ± 0.000991	+5.300%
Context-off	0.145143 ± 0.000954	0.193330 ± 0.001182	+5.208%
Action-free	0.149480 ± 0.000478	0.202249 ± 0.000865	+0.834%

Table 34: **Native 4×4 features.** All five arms share the 4×4 representation and support/query horizons; Context-off and Action-free explicitly remove information paths. Values are equal-seed means $\pm$ sample seed SD after within-episode window averaging; persistence is deterministic. Signed gain compares each arm with Autoregressive in the same column; green bold marks positive point estimates only. Paired uncertainty is reported separately. Native scores use shared-channel training normalization. All results use development validation.

Method	$h = 5$ MSE	$h = 10$ MSE	$h = 10$ gain
Persistence	0.163156	0.225622	—
Autoregressive	0.148577 ± 0.000135	0.202916 ± 0.000239	—
Anchored additive	0.148705 ± 0.000252	0.198704 ± 0.000131	+2.076%
Spatial transport (ours)	0.147921 ± 0.001069	0.197766 ± 0.001302	+2.538%
Context-off	0.147635 ± 0.000949	0.197553 ± 0.001423	+2.643%
Action-free	0.152414 ± 0.000690	0.206188 ± 0.000937	-1.613%

Table 35: **Original 2×2 coordinates.** All five arms share the 4×4 representation and support/query horizons; Context-off and Action-free explicitly remove information paths. Values are equal-seed means $\pm$ sample seed SD after within-episode window averaging; persistence is deterministic. Signed gain compares each arm with Autoregressive in the same column; green bold marks positive point estimates only. Paired uncertainty is reported separately. Predicted raw grids are pooled, then scored against exact original cached targets with original normalization. These values cannot be compared numerically with the native table. All results use development validation.

Transport comparator	h	Gain	Difference [paired 95% interval]
Autoregressive	5	+3.008%	-0.004499 [-0.006132, -0.002491]
Autoregressive	10	+5.300%	-0.010809 [-0.012461, -0.009028]
Anchored additive	5	+3.029%	-0.004532 [-0.005930, -0.002795]
Anchored additive	10	+3.547%	-0.007102 [-0.008831, -0.005019]
Context-off	5	+0.048% [†]	-0.000069 [-0.000633, +0.000478]
Context-off	10	+0.097% [†]	-0.000188 [-0.000943, +0.000485]
Action-free	5	+2.948%	-0.004406 [-0.006551, -0.002149]
Action-free	10	+4.503%	-0.009107 [-0.011896, -0.006061]

Table 36: **Transport comparisons: Native 4×4 features.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). [†] marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

Transport comparator	h	Gain	Difference [paired 95% interval]
Autoregressive	5	+0.441% †	-0.000656 [-0.002499, +0.001918]
Autoregressive	10	+2.538%	-0.005150 [-0.007196, -0.002727]
Anchored additive	5	+0.527% †	-0.000784 [-0.002281, +0.001370]
Anchored additive	10	+0.472% †	-0.000938 [-0.002897, +0.001784]
Context-off	5	-0.194%†	+0.000286 [-0.000386, +0.001016]
Context-off	10	-0.108%†	+0.000213 [-0.000644, +0.001075]
Action-free	5	+2.948%	-0.004493 [-0.007086, -0.001778]
Action-free	10	+4.085%	-0.008422 [-0.011892, -0.004691]

Table 37: **Transport comparisons: Original 2×2 coordinates.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). † marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

Method	Selected epochs	Total params	Active params
Autoregressive	16/25/22	1,026,305	1,007,776
Anchored additive	15/25/16	1,026,305	1,007,776
Spatial transport (ours)	14/19/11	1,026,305	1,026,305
Context-off	10/17/14	1,026,305	878,497
Action-free	6/5/4	1,026,305	988,001

Table 38: **All fifteen completed spatial models.** Every model completed 30 epochs; selected epochs follow seeds 0/1/2 and minimize window-weighted all-ten validation error. Active counts exclude frozen unused heads. The shared trunk does not imply identical active parameter counts; Context-off and Action-free remove inputs and parameters. All selected models passed relocated CPU forecast parity.

ID	Mechanism	Status
Ours-1	Original context-based ShiftWM	Completed; different protocol
Ours-2	Observed anchor + additive innovation	Completed
Ours-3	Observed anchor + bounded innovation	Completed; follow-up
Ours-4	Gated mixing + unbounded innovation	Completed; follow-up
Ours-5	Gated mixing + bounded innovation	Completed

Table 39: Stable mechanism identifiers, not a performance ranking. Ours-1 uses the original context-based protocol and is excluded from the spatial comparison axis. Context-off and action-free are Ours-5 ablations; autoregression and persistence retain baseline labels. Ours-3/4 completed their registered follow-up after the original controls were revealed; all six new models trained 30 epochs.

port (Ours-4), each with three seeds. Together with the six previously revealed anchored-additive and bounded-transport controls, this isolates conditional component comparisons under the same training recipe. At the native ten-step endpoint, adding the mixing package reduces error by 3.48% without bounding and 3.43% with bounding; both paired intervals exclude zero. In contrast, the incremental native-coordinate effect of bounding is inconclusive at both reported horizons, and all four mixing-by-bounding interaction intervals contain zero. Two small pooled-coordinate bounding effects without mixing have favorable intervals; the complete table retains them alongside every null and unfavorable point estimate.

These results support the action-conditioned mixing package as the main measured improvement over the additive anchor in this development study. They do not establish an accuracy advantage from the bounded correction itself. The package includes its learned gate and identity bias and adds approximately 1.84% active parameters, so its comparison does not isolate mixing from those accompanying changes. Ours-4 and Ours-5 trade small point differences across metrics; version numbers are mechanism identifiers, not a monotonic performance ranking. The six new inference packages pass exact relocated CPU reload checks.

T.2 COMPARING ARCHITECTURAL VERSIONS

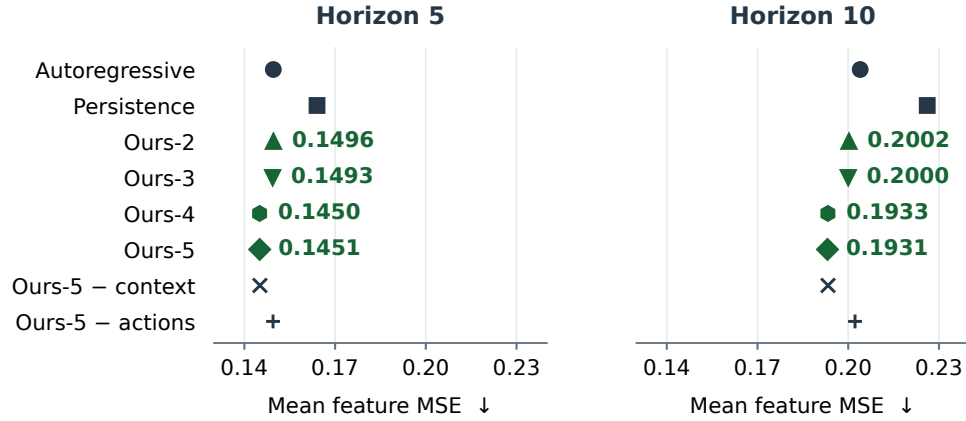
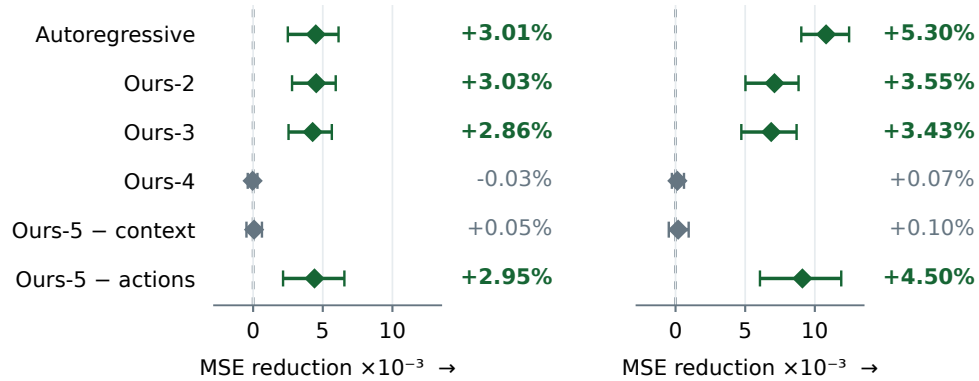
The mechanism key and comparison display use stable identifiers for our variants. Their numbers do not imply a ranking. All spatial versions now have complete three-seed results; paired uncertainty accompanies their comparisons.

Metric	Comparator	h	Relative gain	MSE reduction $\times 10^3$ [95% CI]
Native	Autoregressive	5	+3.008%	+4.499 [+2.491, +6.132]
Native	Autoregressive	10	+5.300%	+10.809 [+9.028, +12.461]
Native	Ours-2	5	+3.029%	+4.532 [+2.795, +5.930]
Native	Ours-2	10	+3.547%	+7.102 [+5.019, +8.831]
Native	Ours-5 — context	5	+0.048%	+0.069 [-0.478, +0.633]
Native	Ours-5 — context	10	+0.097%	+0.188 [-0.485, +0.943]
Native	Ours-5 — actions	5	+2.948%	+4.406 [+2.149, +6.551]
Native	Ours-5 — actions	10	+4.503%	+9.107 [+6.061, +11.896]
Original 2×2	Autoregressive	5	+0.441%	+0.656 [-1.918, +2.499]
Original 2×2	Autoregressive	10	+2.538%	+5.150 [+2.727, +7.196]
Original 2×2	Ours-2	5	+0.527%	+0.784 [-1.370, +2.281]
Original 2×2	Ours-2	10	+0.472%	+0.938 [-1.784, +2.897]
Original 2×2	Ours-5 — context	5	-0.194%	-0.286 [-1.016, +0.386]
Original 2×2	Ours-5 — context	10	-0.108%	-0.213 [-1.075, +0.644]
Original 2×2	Ours-5 — actions	5	+2.948%	+4.493 [+1.778, +7.086]
Original 2×2	Ours-5 — actions	10	+4.085%	+8.422 [+4.691, +11.892]

Table 40: All 16 completed spatial contrasts: Ours-5 versus each comparator. Positive reductions favor Ours-5; bold green percentages indicate positive point estimates, not statistical significance. Paired 95% recording-session and seed bootstrap intervals use 10,000 draws (seed173), without multiplicity correction. Both unfavorable coarse-coordinate context-ablation point estimates remain included. Native and original 2×2 coordinate errors are different metrics and cannot be pooled. Original validation only.

Comparison	Metric	h	Relative gain	Effect $\times 10^3$ [95% CI]
Ours-3 vs Ours-2	Native	5	+0.173%	+0.259 [-0.017, +0.535]
Ours-3 vs Ours-2	Native	10	+0.117%	+0.234 [-0.160, +0.649]
Ours-3 vs Ours-2	2×2	5	+0.185%	+0.275 [+0.009, +0.533]
Ours-3 vs Ours-2	2×2	10	+0.223%	+0.443 [+0.009, +0.907]
Ours-5 vs Ours-4	Native	5	-0.025%	-0.037 [-0.387, +0.309]
Ours-5 vs Ours-4	Native	10	+0.069%	+0.133 [-0.263, +0.606]
Ours-5 vs Ours-4	2×2	5	-0.074%	-0.109 [-0.559, +0.287]
Ours-5 vs Ours-4	2×2	10	-0.030%	-0.060 [-0.652, +0.449]
Ours-4 vs Ours-2	Native	5	+3.054%	+4.569 [+2.917, +5.870]
Ours-4 vs Ours-2	Native	10	+3.481%	+6.970 [+5.013, +8.666]
Ours-4 vs Ours-2	2×2	5	+0.601%	+0.893 [-1.162, +2.359]
Ours-4 vs Ours-2	2×2	10	+0.502%	+0.998 [-1.653, +2.923]
Ours-5 vs Ours-3	Native	5	+2.861%	+4.273 [+2.542, +5.650]
Ours-5 vs Ours-3	Native	10	+3.434%	+6.868 [+4.723, +8.690]
Ours-5 vs Ours-3	2×2	5	+0.343%	+0.509 [-1.618, +1.935]
Ours-5 vs Ours-3	2×2	10	+0.250%	+0.495 [-2.224, +2.513]
Interaction D	Native	5	—	+0.296 [-0.115, +0.778]
Interaction D	Native	10	—	+0.101 [-0.471, +0.710]
Interaction D	2×2	5	—	+0.384 [-0.061, +0.938]
Interaction D	2×2	10	—	+0.503 [-0.175, +1.344]

Table 41: All 20 predeclared component effects from the original-validation follow-up: six new models plus six reused controls, each trained for 30 epochs. The first four blocks report error reductions (second-model MSE minus first-model MSE): positive favors the first model. Bounding is isolated within the Ours-3/Ours-2 and Ours-5/Ours-4 pairs; mixing is compared within Ours-4/Ours-2 and Ours-5/Ours-3. The last block retains the separately signed interaction $D = (\text{MSE}_5 - \text{MSE}_4) - (\text{MSE}_3 - \text{MSE}_2)$: negative means bounding helps more with mixing; no percentage is defined for D . All four interaction intervals include zero. Bold green denotes positive pairwise point estimates, not significance. Paired session/seed 95% bootstrap intervals use 10,000 draws, seed 173, without multiplicity correction. This follow-up was designed after existing controls were revealed; mixing includes the gate, identity bias and approximately 1.84% extra active parameters.

(a) Native spatial error**(b) Paired gains for Ours-5**

Bars: 95% paired CI. Labels: relative point gains.

Figure 23: Completed spatial versions on original DROID validation. (a) Absolute native 4×4 standardized feature MSE at endpoints 5 and 10; green marks identify our mechanism variants, not a winner. (b) Ours-5 is the reference for paired reductions against six comparators; right of zero favors Ours-5. Only registered matching contrasts are used. Ours-4 and Ours-5 are close: both native comparison intervals include zero, and Ours-4 has the lower h5 point estimate. Bars show 95% paired recording-session $\times$ training-seed bootstrap intervals (10,000 draws; seed 173); percentages are relative point reductions. Gray intervals include zero; favorable green intervals exclude zero without multiplicity adjustment. All 21 spatial models completed 30 epochs: 15 original models plus 6 new component models; the four-arm factorial subset has 12 models including reused controls. Ours-3/4 were added in a registered follow-up after the original controls were revealed. Evaluation uses 1,631 matched windows, 141 episodes, 59 sessions and 3 seeds. Windows are averaged within episodes, then episodes and seeds equally. Ours-1 has a different protocol and remains off this axis. Companion tables preserve the original 16 comparisons and all 20 component effects, including four signed interactions and every unfavorable or inconclusive result. These are development comparisons, not independent-test or state-of-the-art claims.

What does the world model predict?

Future visual features, given past images and robot commands

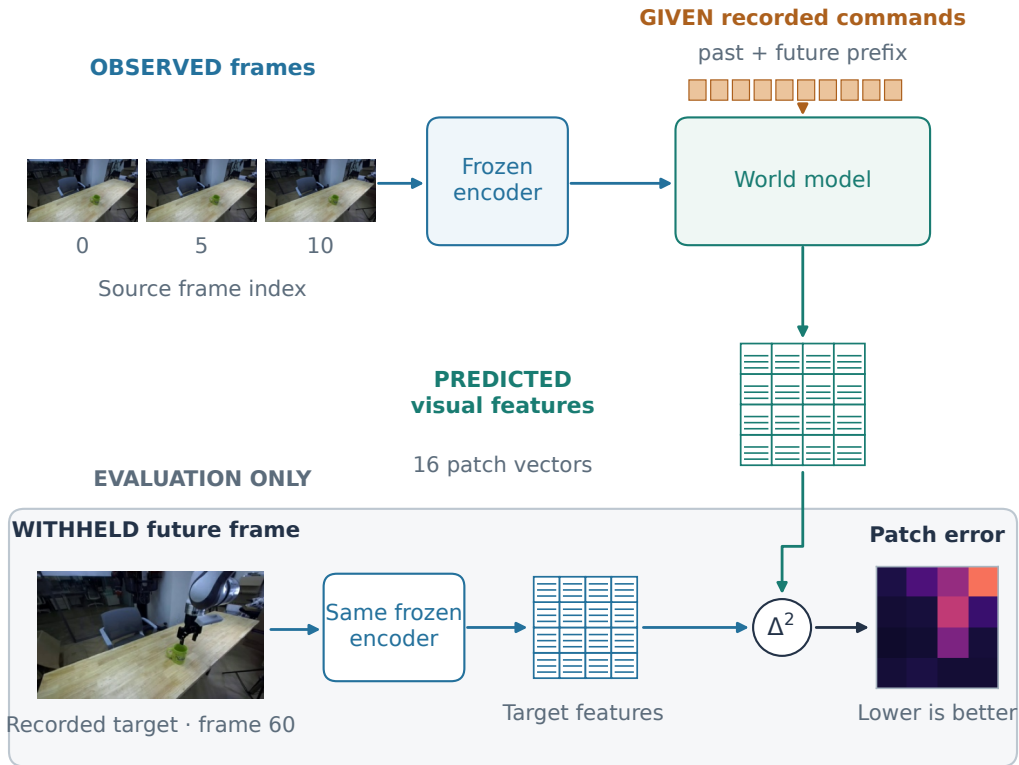
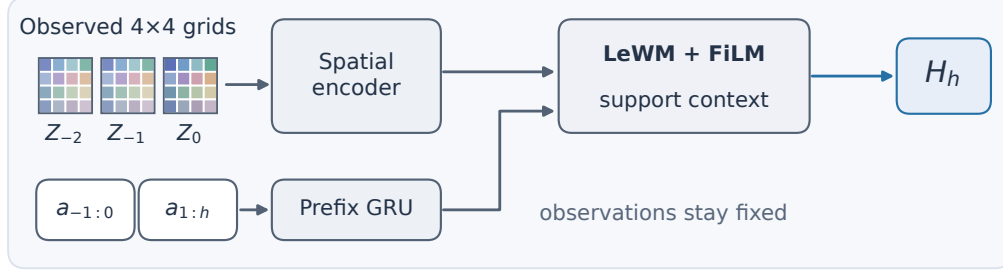
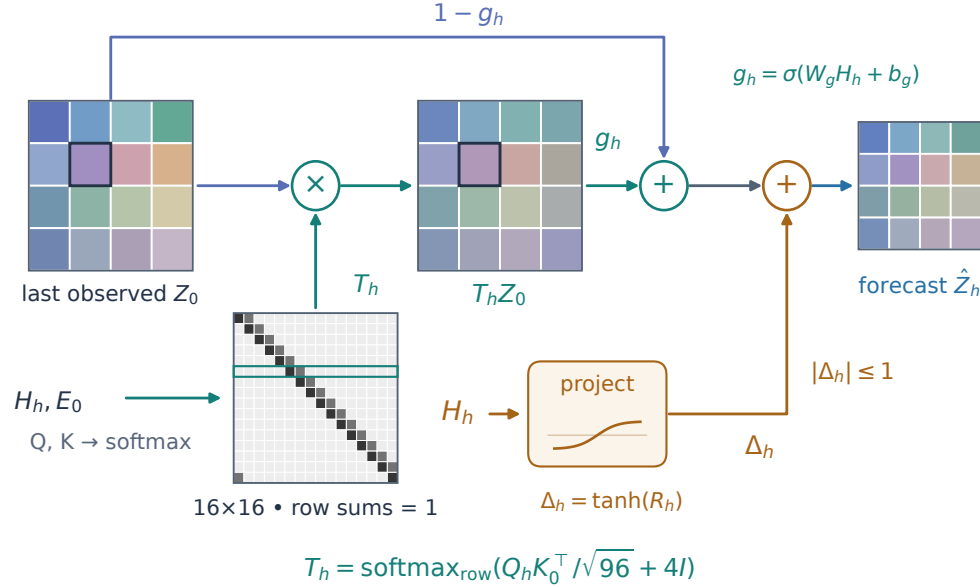


Figure 24: **The recorded-video prediction task.** The model receives three observed images and recorded robot commands, then forecasts visual-feature vectors for a future time. The withheld future RGB frame is used only for evaluation: the same frozen DINO encoder supplies its target features, which are compared with the forecast. The model outputs features, not a generated RGB frame or a chosen robot action. The 16 vector glyphs denote feature vectors schematically; the final heatmap is the actual three-seed $h = 10$ per-patch standardized feature error for the prespecified median episode’s first window (mean MSE 0.1727; one shared scale 0–0.787 across the companion gallery). Bright patches have larger feature error; this is not a pixel-reconstruction error map. The example’s episode-average gain is +5.358%, but this fixed displayed window has a -0.305% gain against the autoregressive baseline. Recorded images are unchanged DROID observations (CC BY 4.0); the target frame never enters the forecasting model.

a Build a causal horizon state



b Anchor, mix, then correct



Fixed anchor at every horizon • schematic latent features

Figure 25: Observation-anchored spatial candidate (ours), transport arm. (a) Three observed feature grids, support-only context and a causal action prefix produce horizon state H_h . (b) The last observed grid Z_0 stays fixed at every horizon. Queries project H_h ; keys project E_0 , the last observed grid’s spatial encoding *before* context FiLM. Row-stochastic weights mix semantic patch features, a sigmoid gate with explicit affine bias b_g blends them with the anchor, and a bounded innovation adds a correction. Operations use training-standardized, shared-channel coordinates; final features are subsequently unnormalized. Colors and matrix shading are schematic, not measured outputs or learned weights. This spatial candidate differs from the original two-context ShiftWM architecture and makes no accuracy, RGB-generation or physical-warp claim.

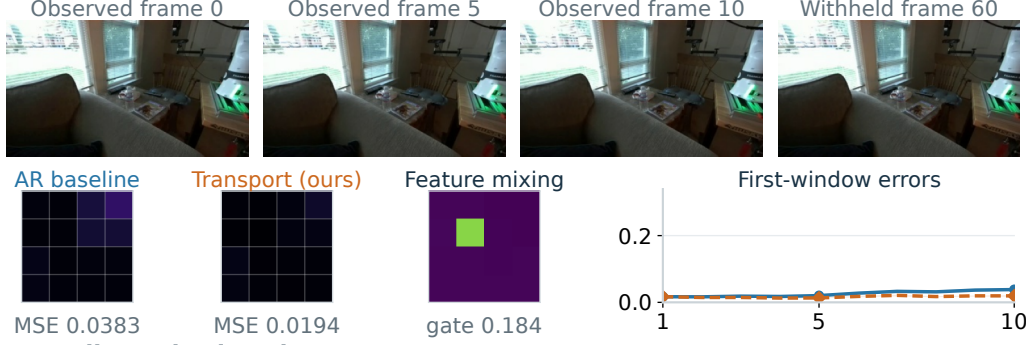
Recorded scenes, measured feature forecasts

141 episodes · 136 gains / 5 regressions · native 4×4 +5.30%

A Largest episode gain

episode +44.9%

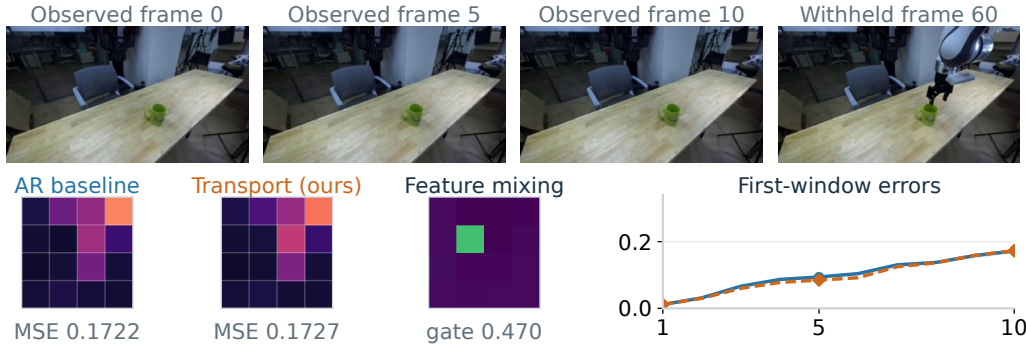
shown +49.3%



B Median episode gain

episode +5.4%

shown -0.3%



C Largest regression

episode -9.6%

shown -14.5%

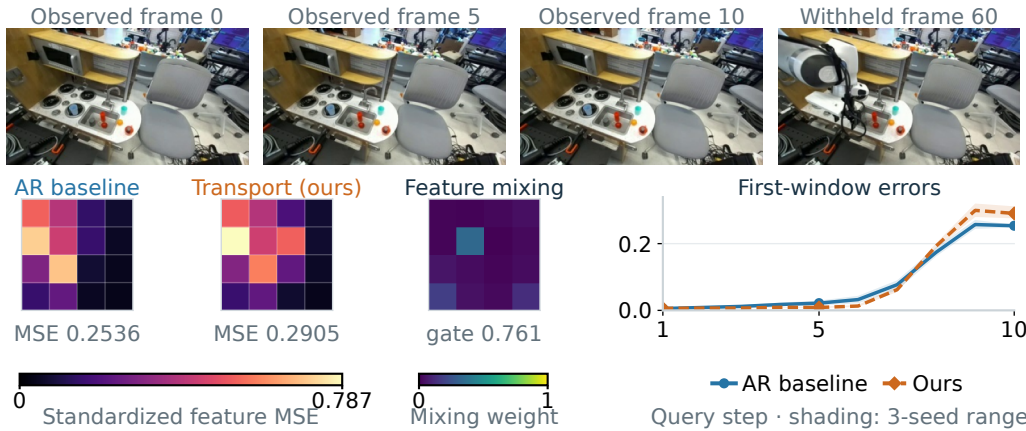


Figure 26: **Recorded-video examples and spatial feature errors.** Best, median and worst cases use prespecified three-seed episode-level native h10 gain versus the matched 4 × 4 autoregressive baseline. The first eligible window is always shown: the median episode gains +5.358%, but its displayed window regresses by -0.305%. Withheld future frames enter only the evaluator. Images: DROID (CC BY 4.0). Error maps average squared standardized errors over 384 channels and three seeds, with one unclipped scale for all six maps. Curves show mean error and three-seed ranges (not confidence intervals): baseline, solid circles; ours, dashed diamonds. Mixing grids show actual h10 anchor-source weights for target cell (row 2, column 2), averaged across seeds; gates appear below. These are semantic feature mixtures, not physical flow or causal attribution. Population counts and episode- and seed-averaged native-grid endpoint gain appear above. This gain-conditioned validation gallery is not an independent test.

U ARTIFACT PROVENANCE

The earlier TTA draft is retained under `paper/archive_tta`. This manuscript uses the unmodified official ICLR 2027 style files. Architecture and split diagrams have editable TikZ sources, a figure brief, and a review record. Numerical plots are populated only from recorded evidence. Every plotted value and uncertainty interval must link to a run record; unavailable values remain missing.

U.1 REUSABLE MODELS AND SCOPE

The release target is a full weights-only model, preprocessing and action normalization, history/reset rules, base revision and hash, and a planning example for each validated environment family. Adapter-only weights require the corresponding base hash. Current packages preserve every loaded tensor; the simulation release stores identical tensors once across models. Active-path counts do not measure archive size or runtime memory. Portable local frozen exports and validation-selected checkpoints from the completed 30-epoch runs have been load-verified. The existing local demo uses recorded held-out simulator clips and measured latent errors. The public real-DROID release contains twelve validation-selected predictors, the shared frozen encoder, and twelve training-fitted calibration wrappers, with model cards, source, licenses and checksums. All base models and calibrated variants pass exact offline prediction parity after relocation. The separate generalization development release adds all 36 predictors from Appendix Q, including every comparison arm, with the shared encoder, source and checksums. Each passes an isolated offline CPU reload check. These packages remain labeled as validation-selected development models; they do not replace the original test checkpoints. A separate simulator development release provides 42 predictors: the 36 original extension models and six drone geometry revisions. All completed 30 epochs; their selected states reproduce source predictions exactly after offline relocation on both cached-feature and image inputs. Together, these five releases provide 117 trained predictors, including all mixed outcomes. The fourth release contains the twelve ten-step-trained predictors from Appendix S; all selected epoch one after full 30-epoch runs and passed exact relocated CPU prediction parity. The fifth release contains all fifteen spatial predictors from Appendix T. Each completed 30 epochs and passed exact relocated CPU prediction parity; all public archive payload hashes were independently verified against its release manifest. These models forecast visual features; evaluation status remains separate from artifact availability.

U.2 VISUAL AND DIAGNOSTIC INTERPRETATION

The introduction’s generated scene is an illustration, not simulator output; the evaluated appearance changes are pointwise RGB transforms, not viewpoint changes. Method feature strips depict correction, not reconstructed images or guaranteed canonical recovery. Candidate branches are schematic latents, not predicted videos or physical trajectories. Split response glyphs are not measured trajectories or ranked physical responses. Numerical plots use source-linked run records, not generated artwork.

The nominal simulator, collection policy, and interaction protocol differ from the released-evaluator check, so their performance gap cannot be attributed solely to appearance/dynamics shifts. The goal-only hybrid was chosen after inspecting development failures; its task-bootstrap intervals are exploratory summaries conditional on that selection. Any further revision motivated by these diagnostics remains separate from the fixed primary campaign and cannot be tuned on its test outcomes.

V TRAINING CAMPAIGN SNAPSHOT

Environment	Mode	Seed	Epochs	State
reacher	ShiftWM (ours)	0	30/30	complete
reacher	Unaligned predictor (diagnostic)	0	30/30	complete
reacher	Shared context	0	30/30	complete
pusht	ShiftWM (ours)	0	30/30	complete
pusht	Unaligned predictor (diagnostic)	0	30/30	complete
pusht	Shared context	0	30/30	complete

	Environment	Mode	Seed	Epochs	State
3564					
3565					
3566	reacher	ShiftWM (ours)	1	30/30	complete
3567	reacher	Unaligned predictor (diagnostic)	1	30/30	complete
3568	reacher	Shared context	1	30/30	complete
3569	pusht	ShiftWM (ours)	1	30/30	complete
3570	pusht	Unaligned predictor (diagnostic)	1	30/30	complete
3571	pusht	Shared context	1	30/30	complete
3572	reacher	ShiftWM (ours)	2	30/30	complete
3573	reacher	Unaligned predictor (diagnostic)	2	30/30	complete
3574	reacher	Shared context	2	30/30	complete
3575	pusht	ShiftWM (ours)	2	30/30	complete
3576	pusht	Unaligned predictor (diagnostic)	2	30/30	complete
3577	pusht	Shared context	2	30/30	complete
3578	reacher	Framewise calibration	0	30/30	complete
3579	reacher	Unpaired contexts	0	30/30	complete
3580	pusht	Framewise calibration	0	30/30	complete
3581	pusht	Unpaired contexts	0	30/30	complete
3582	reacher	Framewise calibration	1	30/30	complete
3583	reacher	Unpaired contexts	1	30/30	complete
3584	pusht	Framewise calibration	1	30/30	complete
3585	pusht	Unpaired contexts	1	30/30	complete
3586	reacher	Framewise calibration	2	30/30	complete
3587	reacher	Unpaired contexts	2	30/30	complete
3588	pusht	Framewise calibration	2	30/30	complete
3589	pusht	Unpaired contexts	2	30/30	complete
3590					
3591					
3592					
3593					
3594					
3595					
3596					
3597					
3598					
3599					
3600					
3601					
3602					
3603					
3604					
3605					
3606					
3607					
3608					
3609					
3610					
3611					
3612					
3613					
3614					
3615					
3616					
3617					